The Breakfast Paradox

Why two thirds of the hostel breakfasts at IIIT Hyderabad go uneaten, and what a student team can do about it

PROJECT CASE STUDY · SYSTEMS THINKING AND DESIGN (PD2.405) · IIIT HYDERABAD · SEPTEMBER TO OCTOBER 2026

Team	Team Invictus: Sricharan Peri, Prathyusha Kalluri and Neha Susan Manoj. All three worked across every phase. Faculty: Prof. Raman Saxena.
Time	About six weeks, from 3 September 2026 to early October 2026
Scope	Hostel breakfast in four dining halls, from registration and billing to cooking, serving and disposal. Lunch and dinner are the comparison.
Evidence	Six student voices, visits to three kitchens, an interview and kitchen walkthrough with the head of facilities, Yuktāhār's photographed registers, and a month of Kadamba scan records (32,162 registrations)
Outputs	Fifteen deliverables across four phases and ten activities, a scenario model, and a recommendation with a pilot plan
Status	A recommendation tested on paper and against real records. Nothing has been piloted on campus yet.

In short

Every hostel student at IIIT Hyderabad is registered for breakfast automatically, and the charge follows the registration. About two thirds of those breakfasts are never eaten, yet the kitchens still cook for most of them.

We went in expecting a student problem. We found a system problem. The gap is recorded every day, in the scan records and in the kitchens' own books, and the record never reaches the people who set the rules. So nothing changes.

We recommend three steps that run together. Build the missing route for that record. Give students a real way to skip Sunday breakfast. Tune the kitchens in the meantime, and call that what it is: a quick fix. On Kadamba's April records, the kitchen change cuts surplus food by about 75%. Only the rule change reduces what students pay.

63%	72%	362 to 91	₹10.6 lakh
of Kadamba breakfast registrations unused in April 2026	of Yuktāhār breakfast registrations unused in September 2026	surplus portions a day at Kadamba, today and with the kitchen fix (projected)	charged to Kadamba students for uneaten breakfasts in April (estimate)

This case study is a shorter version of the project document. It follows the same order, from the brief to the recommendation, and keeps every figure. Figures carry the same numbers as in the project document.

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The project

Background and the brief

Hostel students at IIIT Hyderabad are registered and billed for breakfast automatically, and many do not eat it. The food is cooked anyway, and the surplus leaves by routes that no office tracks.

The course in Systems Thinking and Design offered four campus puzzles, and we chose the Breakfast Paradox. Its brief follows a student whose alarm has gone off four times at 7:45 after a late night. He picks sleep, and by 11:15 he is hungry and cannot concentrate. Kadamba serves breakfast from 7:30 to 9:30, classes start at 8:30, and barely a third of the hostel shows up. The rest of the food is thrown away.

The brief blames one villain, the clock. We treated the story as a set of claims to test, and tied each claim to an interview question or a record that could check it.

The problem we set out to solve

The puzzle asks why students skip a meal they registered for. Phase 1 showed that question to be too narrow. Registration is a default that students did not choose, and the kitchens do not cook to the registration count; each applies its own rule of thumb. We restated the problem.

> Breakfast registration at IIIT Hyderabad does not reflect who will eat, by roughly two thirds, while lunch under the same rules mostly does. Each kitchen estimates its own conversion from registrations to cooked food, the records that could calibrate that estimate are already kept, and none of those records reaches the people who set the registration and billing rules. The project asks why this arrangement persists and what a small group of students could propose to change it.

Aim and research objectives

Our aim was to explain the persistence of the gap between breakfast registration and breakfast attendance as a property of the whole system, and to recommend an intervention that can be tested on the campus. Nine research objectives followed from it.

Objective ID	What it asks	Activity
O1	Fix the boundary; separate visible symptoms from underlying issues	1
O2	Identify the people and bodies in the system and what each needs	2
O3	Establish who holds authority, power and leverage over each decision	3
O4	Trace how a registration becomes cooked food	4
O5	Measure behaviour over a morning, a week and a month	5
O6	Build a model and test which feedback mechanisms close	6
O7	Trace events to structures and beliefs; reframe the problem	7 and 8
O8	Find where change could act and what any change must protect	9
O9	Design, test and evaluate interventions	10

Scope and boundary

Inside the boundary is breakfast in the four dining halls (Kadamba, Palash, Bakul and Yuktāhār) as it is registered, billed, cooked, served and disposed of. Lunch and dinner sit outside and serve as the comparison, since the same portal, rules and caterer apply. The class timetable is outside too, but stays on the maps because it shapes the morning.

Kadamba has a month of scan records, Yuktāhār photographed attendance registers. Bakul rests on its manager's interview and no records (single-sourced), and Palash is a candidate kitchen sharing Bakul's off-site kitchen.

Team and timeline

Team Invictus is Sricharan Peri, Prathyusha Kalluri and Neha Susan Manoj. The work ran from the first interview on 3 September 2026 to the rebuild of the deliverables in early October. First drafts of Phases 1 and 2 were written between 3

and 14 September. Field evidence arrived later, in three batches on 26, 27 and 28 September, and each forced a revision of earlier work (Figure 1.1).

- 26 September: kitchen visits, recordings, photographed registers and Kadamba's April scan export. A working figure of about 30% of registrations unused gave way to measured figures of 63 and 72%.
- 27 September: a team member's first-hand account, through which an observer confirmed Kadamba's 70% rule.
- 28 September: the Bakul interview summary. Three operator regimes replaced two, and a statement that payment follows served plates became an open conflict, with dependent levers held.

Phases 3 and 4 were then written on the revised evidence.

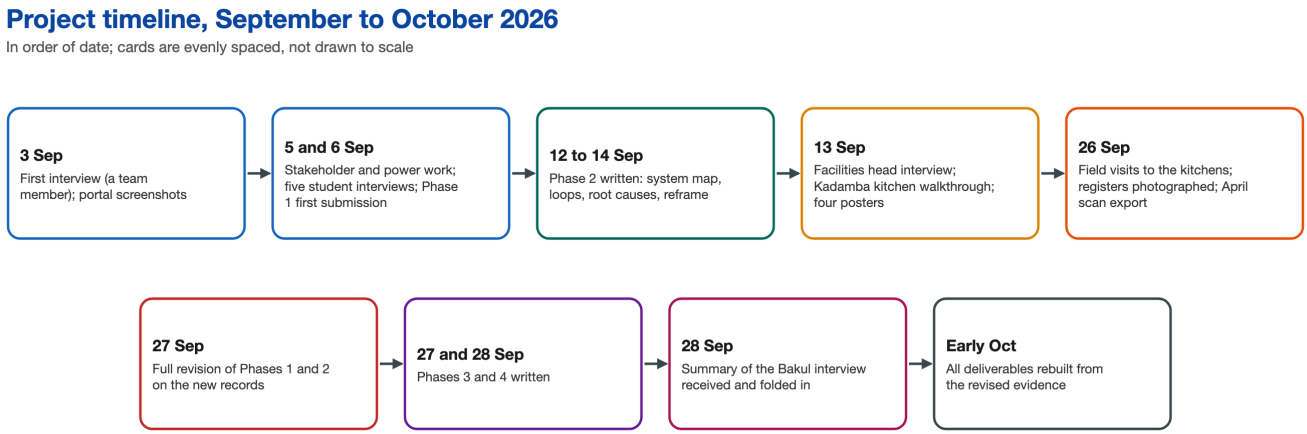

Figure 1.1. Project timeline, September to October 2026. Source: team record of activities.

How we approached it

The course framework

The course framework divides the work into four phases, ten activities and fifteen deliverables, and our headings keep its wording and order.

Phase	Activity	Deliverable
1: Understanding the System	1. Understanding the System Context	System Context Brief
	2. Identifying and Mapping Stakeholders	Stakeholder Map
	3. Analysing Power, Interests and Leverage	Stakeholder Power-Interest and Leverage Map; Power Analysis Brief
	4. Tracing the Existing Process	Current-State Process Map and Process Trace
	5. Analysing System Behaviour over Time	System Timeline and Behaviour-Over-Time Map
2: Modelling the System	6. Developing the System Map and Causal Loop Analysis	System Map; Causal Loop and Feedback Analysis
	7. Identifying Systemic Patterns and Root Causes	Systemic Problem Analysis
	8. Reframing the Problem	Reframed Systemic Problem Statement; Design Opportunity Statement
3: Identifying Leverage Points and Design Principles	9. Defining Design Principles and Intervention Opportunities	Leverage-Point and System Intervention Map; Design Principles
4: Designing, Testing and Evaluating Interventions	10. Designing, Testing and Evaluating Interventions	Intervention Concepts and Prototypes; Scenario Model and Impact Evaluation; Final System Design Recommendation

Plan and working method

Three revision passes on 27 and 28 September went back through every earlier activity.

Activity	Worked	Revision passes
1 System context	3 to 6 Sep	13 Sep; all three passes
2 Stakeholders	5 and 6 Sep	6 and 13 Sep; all three passes
3 Power, interest, leverage	6 Sep	6 Sep audit; all three passes (payment basis kept open)
4 Existing process	6 Sep	13 Sep; all three passes
5 Behaviour over time	6 Sep	All three passes (records replaced recall)
6 System map and loops	12 and 13 Sep	13 Sep; all three passes (one scheme of 18 mechanisms)
7 Patterns and root causes	12 and 13 Sep	13 Sep (iceberg corrected); all three passes
8 Reframe	14 Sep	Split in two on 14 Sep; all three passes
9 Principles and intervention points	27 Sep	28 Sep (LP11, LP12, P7); Bakul pass
10 Design, test, evaluate	27 Sep	28 Sep (Concept H, gate G4, stress tests); Bakul pass

Three habits, each adopted after a week went wrong without it, governed the work. We revised in dependency order (Phase 1, Phase 2, Activity 9, Activity 10), began every revision with a gap audit, and wrote each batch up as one dated evidence summary, the only source of facts for later revisions. Decisions were logged with date and reason, and a changed decision stayed on record beside its successor.

Each deliverable stood alone, followed the framework's headings and closed with the evidence still outstanding.

Concepts we drew on

We drew on established frameworks from systems thinking, service design and research practice, without a systematic literature search.

Concept	How we used it
System boundary	Line drawn around the interactions that matter (Activities 1 and 9)
Mental models and the iceberg	Each event traced down to the belief that sustains it (Activities 7 and 8)
Causal loops and behaviour over time	Records charted first; a loop accepted only if every link is evidenced (Activities 5 and 6)
Archetypes and traps	Graded present, candidate, risk or not established (Activities 7 and 9)
Power-interest grid	Authority judged decision by decision; power kept apart from leverage (Activity 3)
Leverage points (Meadows, 1999)	Levers ranked by depth on the twelve-place scale, then by feasibility (Activity 9)
Service blueprint	What the student sees, split from what the kitchen does (Activities 4 and 6)
Reframing and design thinking	From why students do not eat to what holds the gap in place (Activities 8 and 10)
Payoffs of the status quo	Convenience, revenue, avoided workload and safety; a design that ignores them meets resistance (Activity 9)
Testing without a pilot	Backtests on recorded days; stress tests with assumed shocks (Activity 10)

No source we had covered standing registration, billing at registration and kitchen planning in an institutional canteen, or set a standard of evidence for a student team with partial records. We adapted the methods in seven ways, among them the loop acceptance rule, graded archetypes, pass-or-fail gates before scoring, and conflicts kept open.

Research methods

Research design

The project is a single-case study of one institutional system. Interviews, observation and visits explained how it works and why; records measured it. Interviews can show a mechanism but not how common it is, so prevalence claims rest on records. We chose interviews and records over a survey: the population was small and reachable, the mechanism unfamiliar, and several students did not know the rules a survey would ask about.

Participants and sources

Sampling was purposive, driven by access.

Source	What it gave	Main limit
Team member's self-account, 3 Sep	The system and their own mornings	One person, also a stakeholder
Nine portal screens	Registration dialog, capacities, rates, charges	Each shows one day
Five student interviews, 6 Sep	Two daily eaters, three skippers	Four to five distinct people
Head of facilities, Kadamba kitchen and four posters, 13 Sep	Governance, menu rotation, cadence, stock, waste, food safety	One respondent; posters state intent
Operator talks, 26 Sep, and a first-hand account, 27 Sep	Ordering, registers, rules of thumb, surplus routes	Self-presentation; Bakul single-sourced
Yuktāhār books and registers, 26 Sep	26 photographs, 22 tables	Some breakfast rows hidden by a hand
Kadamba scan export, April 2026	Every breakfast registration with its scan time	One hall, one meal, one month

We did not interview the head of the dining services body (two different people are on record for that post), the Mess Committee, the Warden's office or the Student Parliament's mess seats.

Instruments and the bias review

Our first questionnaire (five roles, 39 questions) was criticised as very direct and likely to produce leading answers. We rewrote it to standing rules: the fact before the reason, every question checked for four kinds of bias and tied to a named objective, and respondents described as roles. Three guides followed, for students, operators and governance, through six versions on 5 and 6 September.

A wording review then found four questions that invited the answer they expected, such as whether Skip Meal "actually changes anything or just feels symbolic". All four were rewritten before any real interview.

Fieldwork

- The five student interviews followed a short written list of 23 questions. Two team members transcribed them independently and found a labelling gap, so we count four to five distinct people and relabelled nothing.
- On 13 September we interviewed the head of the facilities department and walked through Kadamba's kitchen. One conflict stayed open: the poster gives Kadamba's capacity as 1,200, the interview 420 seats and 1,200 diners.
- On 26 September we visited three kitchens, recorded eight audio files (about two hours and fifteen minutes) and photographed 26 Yuktāhār pages. At Kadamba we observed the display plate and the tasting.
- Machine transcription, with translation from Telugu, gave a reliable Yuktāhār record and the gist for Kadamba and Bakul. Where an observer who understood the talk disagreed with the machine, the observer won.

Records

Kadamba's export has one row for each breakfast registration from 1 to 30 April 2026: 32,162 rows giving the meal date, the line (vegetarian or non-vegetarian) and the scan time, blank when the person did not eat. We built five analysis tables from it, down to arrivals per minute from 9:00 to 9:45. The 26 Yuktāhār photographs became 22 tables, each row marked for how confidently it could be read.

The portal sits behind single sign-on, so we used only screens a student captured while logged in, and no one's credentials. The dialog's Skip Meal toggle warns that the student will still be charged, and its tabs give each hall's capacity

and rate (Figure 4.1).

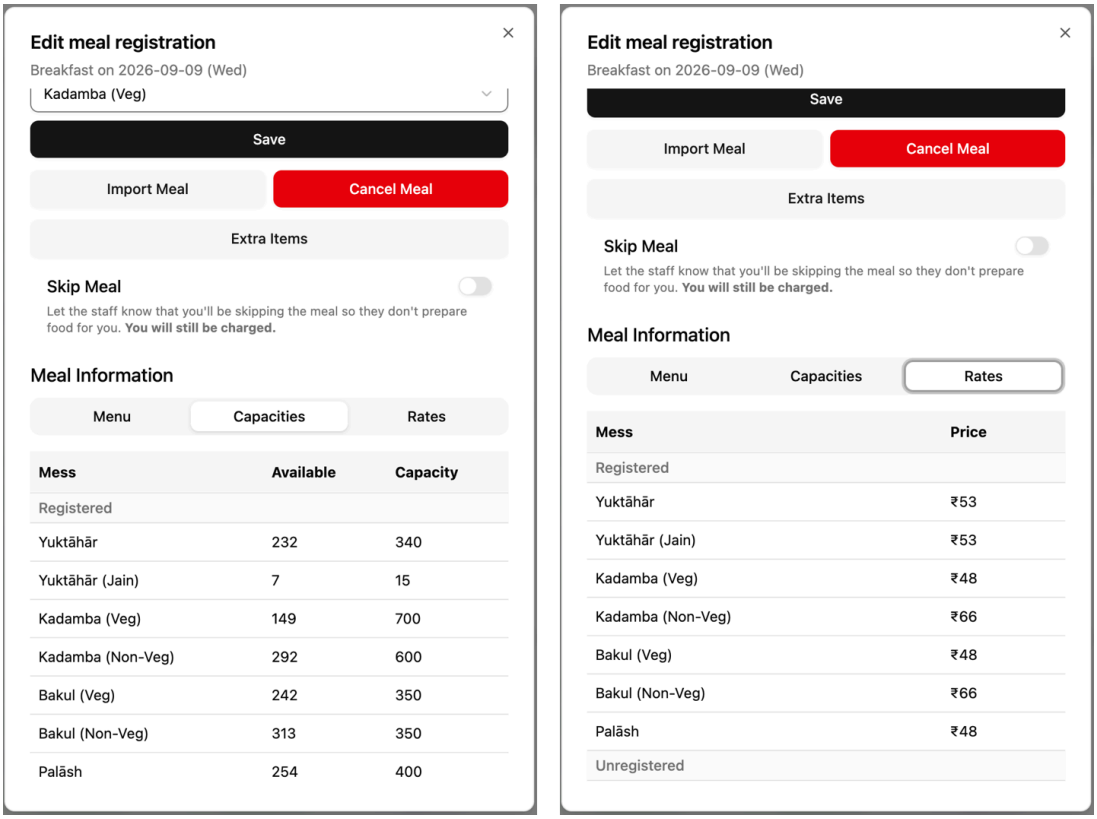

Figure 4.1. Portal screens used as evidence: the registration dialog with capacities by hall (left) and its rates tab (right), breakfast on 9 September 2026. Source: screens of the dining portal captured by a student team member, cropped to the dialog.

Analysis methods

Turnout is scans divided by registrations, with lunch from the same register as a control. A causal loop counts only if its last link causes its first and every link is evidenced; other chains keep a dashed closing link. The scenario model replays proposed kitchen rules over the 23 recorded days from 8 to 30 April, forecasting only from earlier days of the same weekday.

Evidence labels and source precedence

Every claim carries one of seven labels, kept through every later use.

Label	What it means
Confirmed	Read off a record, seen on site, or stated alike by independent sources
Single-sourced	Stated by one person or document and not checked against a second
Candidate	An inference combining sources, stated nowhere directly
Assumed	A judgement made to continue the analysis
Projected	Scenario-model arithmetic on real records under stated assumptions
Simulated	A reasoned expectation of a group’s response, never their feedback
Planned	Designed and not carried out

When sources conflicted, the Bakul summary of 28 September ruled on Bakul, the first-hand account of 27 September ruled on Kadamba and Yuktāhār, and the first operations summary kept authority over record-level numbers. Machine-translated claims stayed marked *(unclear)* until an observer confirmed them.

Revision protocol

Each affected deliverable was audited against the batch summary, revised and checked, and each phase was then read for consistency. An earlier deliverable was patched whenever a later one contradicted it, and evidence was frozen at a stated date for each writing, with the gaps listed.

Ethics and data handling

The interviews were part of a course assignment, and the project record holds no signed consent forms. Interviewees and staff are named by role. The scan export has no student identifier, and only aggregates are reported. No kitchen's records are used in a way that could be turned against the person who kept them.

Limits of the method

- Six student voices: evidence of mechanism, never of how many.
- One visit per kitchen and real self-presentation bias: one kitchen head put breakfast turnout at about 50%, where his own register shows about 28%.
- Records for two halls in two different months; none for Bakul or Palash.
- Two operator accounts received only as syntheses.
- No direct observation of a live breakfast, of disposal or of a student's morning.
- Nothing tried on campus: stakeholder responses are simulated, co-design tests planned, and the model's skip uptake, adoption pace and shocks assumed.

Phase 1: Understanding the System

Phase 1 looked at the breakfast system from five sides: its edges, its stakeholders, its power, its process and its behaviour over time. It found that registration records a default nobody chooses, and that each kitchen discounts it by its own rule, every rule still above recorded turnout.

Activity 1: Understanding the System Context

Activity 1 asked for the system's edges, its purpose set against its actual mechanism, and its visible faults. Its deliverable is the System Context Brief.

Boundary and context

The system is breakfast as registered, billed, cooked and served in four halls: Kadamba, Palash, Bakul and Yuktāhār. Three layers sit inside the boundary: registration and billing (the portal and the counter QR scan), the kitchens (one outside caterer per hall) and governance (Figure 5.1).

The boundary ring holds people any change must plan for: at Kadamba 30 to 50 outsiders and about half the operator's staff eat from the same pots, so cutting surplus also cuts food they rely on (single-sourced; revising an earlier figure of 20 to 30 staff). The class schedule lies outside but overlaps the meal.

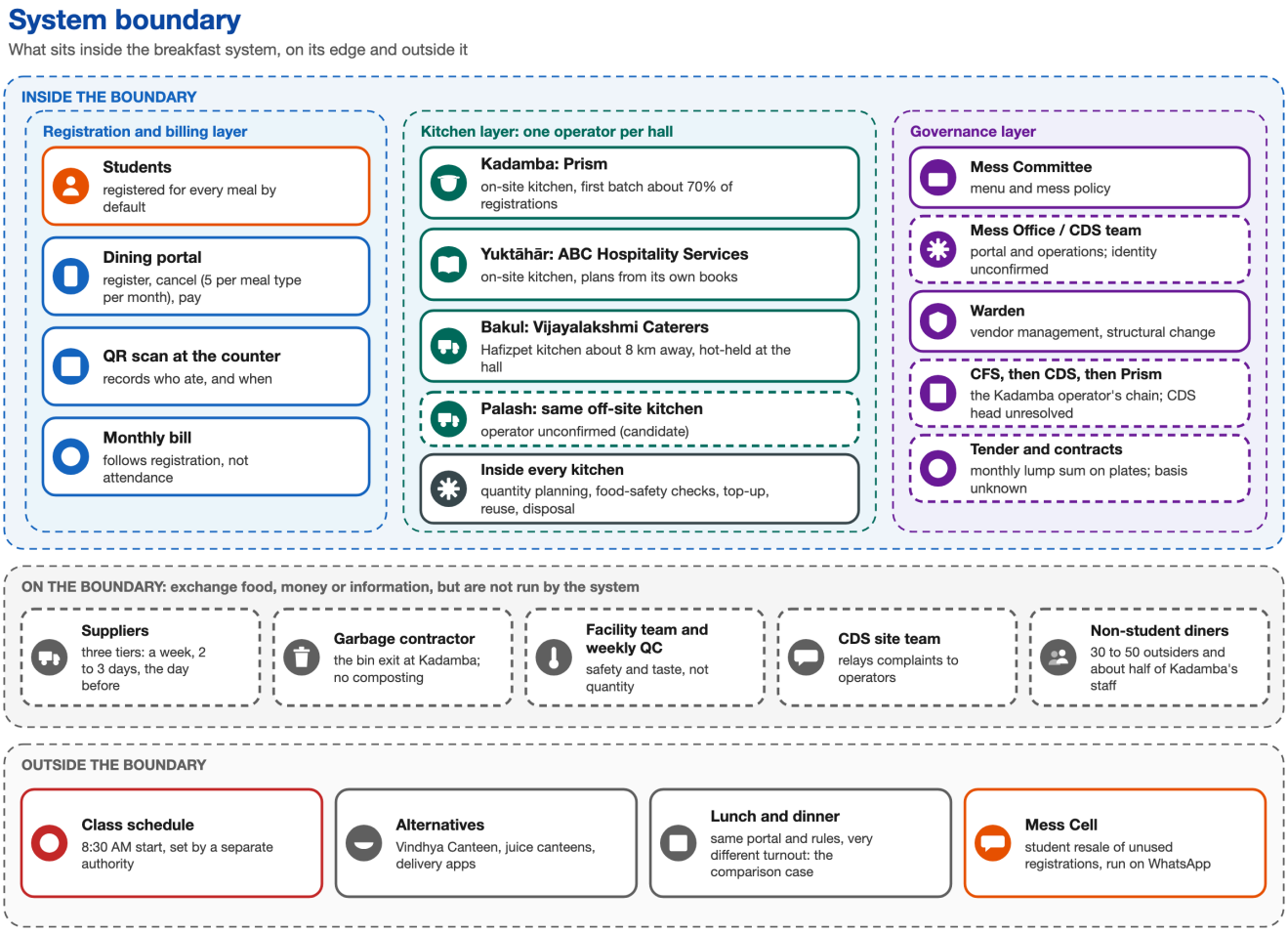

Figure 5.1. System boundary: three layers inside, boundary actors on the ring, class schedule and alternatives outside. Source: authors' analysis of the portal, published rules and operator accounts.

Registration is automatic: every student is registered for every meal, holidays included. The only exit is a cancellation, five per meal type per month and at least four days ahead (confirmed). Billing follows registration, Skip Meal leaves the charge, and a walk-in pays about 70% more (confirmed, posted rates).

The context map puts the registration count at the centre, with governance above it, the three operators' ratios beside it and the surplus exits below (Figure 5.2). Notice the "no formal channel" between the academic office, which sets the 8:30 class start, and the Mess Office.

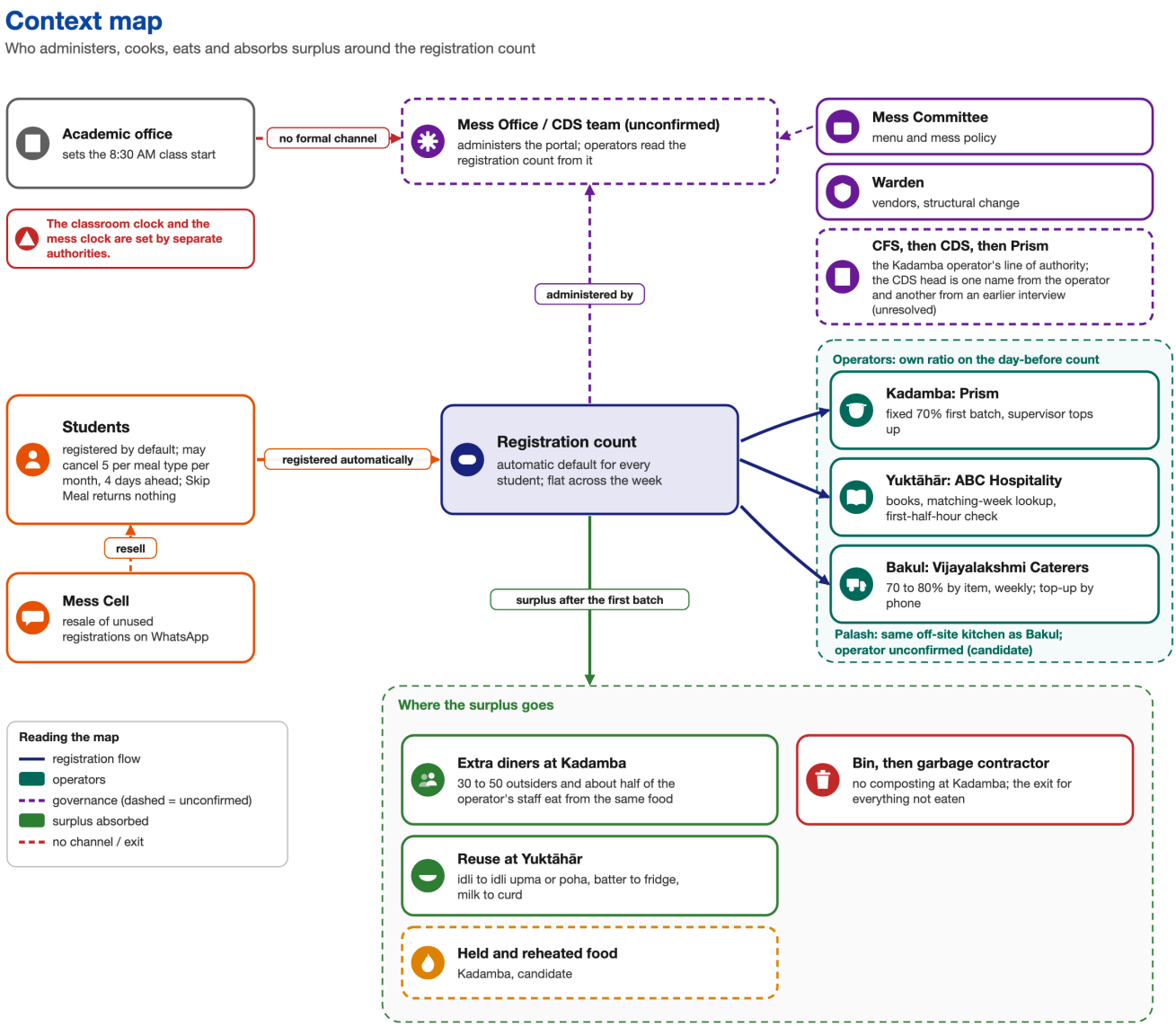

Figure 5.2. Context map: who administers, cooks, eats and absorbs surplus around the registration count. Source: authors' analysis.

Purpose against mechanism

The institution's stated objective includes "minimizing food wastage through advance meal planning". That works only if registration expresses an intent to eat.

Step	Stated design	Actual mechanism
What registration records	Demand, estimated in advance	A default students do not choose; flat across the week
How kitchens plan	From the advance count	Each operator applies its own conversion to the day-before count
Where surplus goes	Minimised by planning	Reuse, outside diners and staff, held food (candidate), the bin

Registrations overstate breakfast turnout about 2.7 times at Kadamba and 3.5 times at Yuktāhār (confirmed, records). The gap is already measured, yet nothing carries it into registration or billing rules. In April 2026, 20,221 registered Kadamba breakfasts were billed and not eaten (confirmed).

Symptoms and underlying issues

Visible symptom	Underlying issue
About two thirds of breakfast registrations unused yet charged; Yuktāhār lunch far better used	The default records no decision on the morning; the cause is specific to breakfast
Kadamba registrations flat while turnout varies by weekday	An exit closing four days ahead cannot follow daily variation

Visible symptom	Underlying issue
Arrivals pile up at the close	Kitchens cook just in case and still run short at the close
No student can name feedback that changed registration, billing or hours	No channel has a confirmed instance of changing those rules

Actors, institutions and resources

Formal governance is the Mess Committee, the Mess Office, the Warden, the tender and per-plate payment, and the halls post its rules and rates (Figure 5.3). The informal institution that matters as much is Yuktāhār's paper books, which record per dish what was issued, cooked, left and wasted (confirmed).

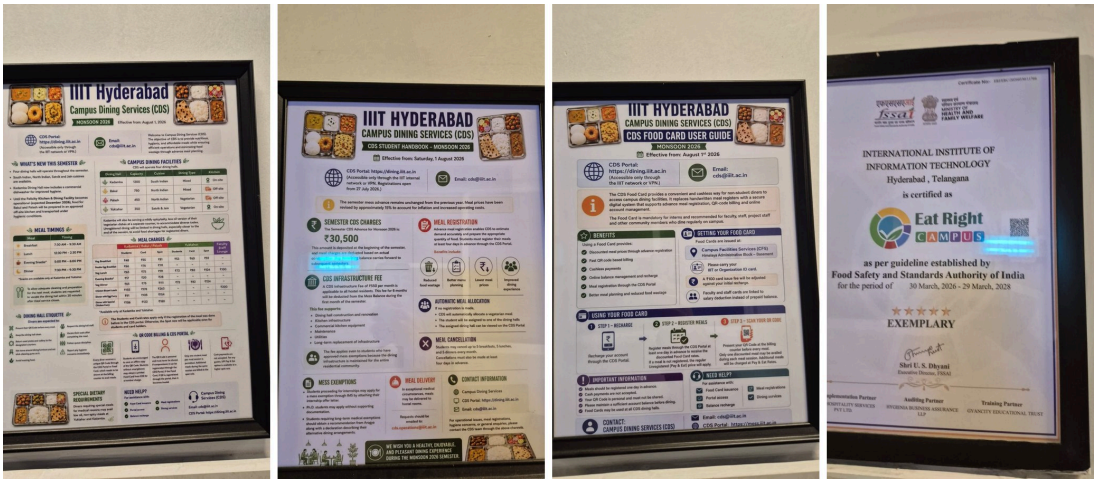

Figure 5.3. What the dining halls post: overview poster, student handbook, food-card guide and food-safety certificate. Source: field photographs, 13 September 2026.

Two resources already measure the gap: the institution's scan record of who ate and when (confirmed for Kadamba breakfast), and the operators' own records, laid out in Yuktāhār's kitchen on our visit (Figure 5.4). Operators control quantities but never registration, billing or the menu.

Figure 5.4. Yuktāhār's kitchen on the day of the visit: the registers laid out, the cold room, and staff at the weighing station. Source: field photographs, 26 September 2026.

Initial problem framing

About 63% of Kadamba's breakfast registrations went unused in April 2026 and about 72% of Yuktāhār's in September. The context brief asked why registration misses who will eat by roughly two thirds when it works for Yuktāhār's lunch, and why each kitchen guesses its own conversion when the data to calibrate it is already recorded.

Decisions and revisions

- We scoped the study to breakfast and kept lunch and dinner outside as a control, to test whether the cause is the meal or the system.
- We treated the opening story, a late night against an early class, as hypotheses; only its first link had testimony behind it.
- On 5 September we split purpose into purpose, mechanism and gap, because a registration that rewards staying booked explained more than the class clash alone. When field evidence later contradicted the brief, we changed the statement and marked the revision.

Activity 2: Identifying and Mapping Stakeholders

Activity 2 asked who takes part in the system, influences it, benefits from it or bears its effects. Its deliverable is the Stakeholder Map.

Who the stakeholders are

Seventeen lenses produced 32 candidates, and 24 survived a test against direct evidence. Seven appeared only once we saw operator practice, among them suppliers and the garbage contractor.

Tier	Count	Examples
Primary	7	Students, the three operators, site management, serving and kitchen staff
Secondary	5	Mess Committee, Mess Office (candidate: CDS), CDS site team, Warden, Parliament mess seats
Indirect	12	Academic administration, suppliers, garbage contractor, non-student diners, Mess Cell, Ping, canteens

Five governance identities stay unconfirmed and apart, including who leads CDS. Each operator also runs an organisation the institution does not see; at Yuktāhār it is a small team who keep the books (Figure 5.5).

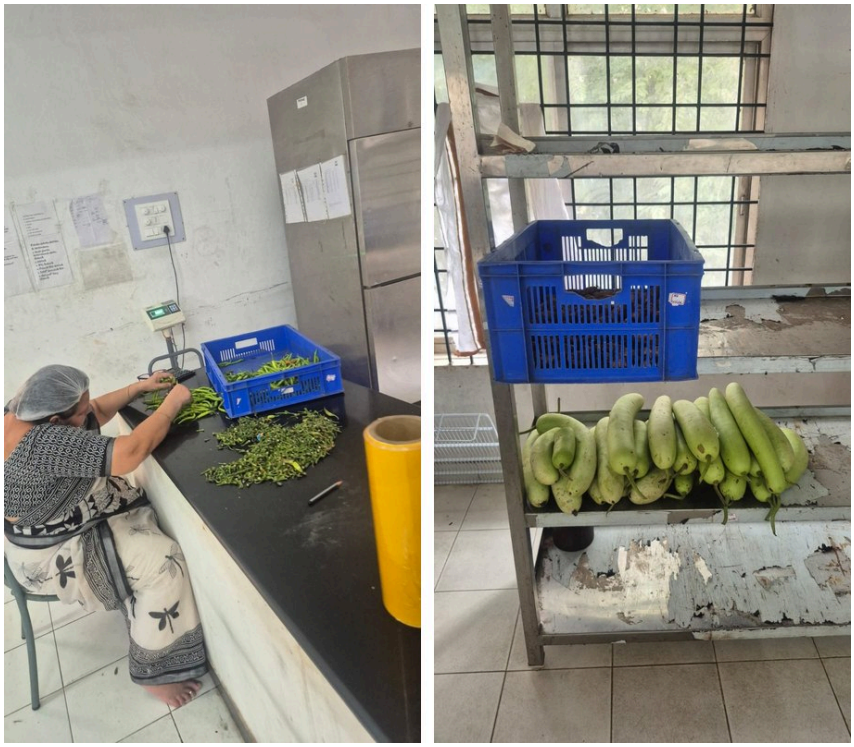

Figure 5.5. Staff at Yuktāhār's weighing station and the vegetable store, the people behind the books. Source: field photographs, 26 September 2026.

Interests and roles

The operators differ most. Prism treats the count as an obligation it is paid for; Vijayalakshmi Caterers holds that "there should not be a shortage of food, even if some wastage occurs" (single-sourced). Kadamba is a ladder of command; Yuktāhār has no hierarchy, with two project managers and about three site staff (single-sourced; revised from five); Bakul splits across two sites.

Informal actors do real jobs without standing: Mess Cell is a resale market, Ping a public-pressure channel, and Yuktāhār's twice-daily meeting the real forecasting institution. Peer influence sits inside the Students node, so friends are not a stakeholder.

Three maps of the same people

The four-ring map places Students as the customer, governance, operators and staff as internal, and outside actors as external (Figure 5.6). In the public ring, Mess Cell, the tool developers and Ping have no standing in the system.

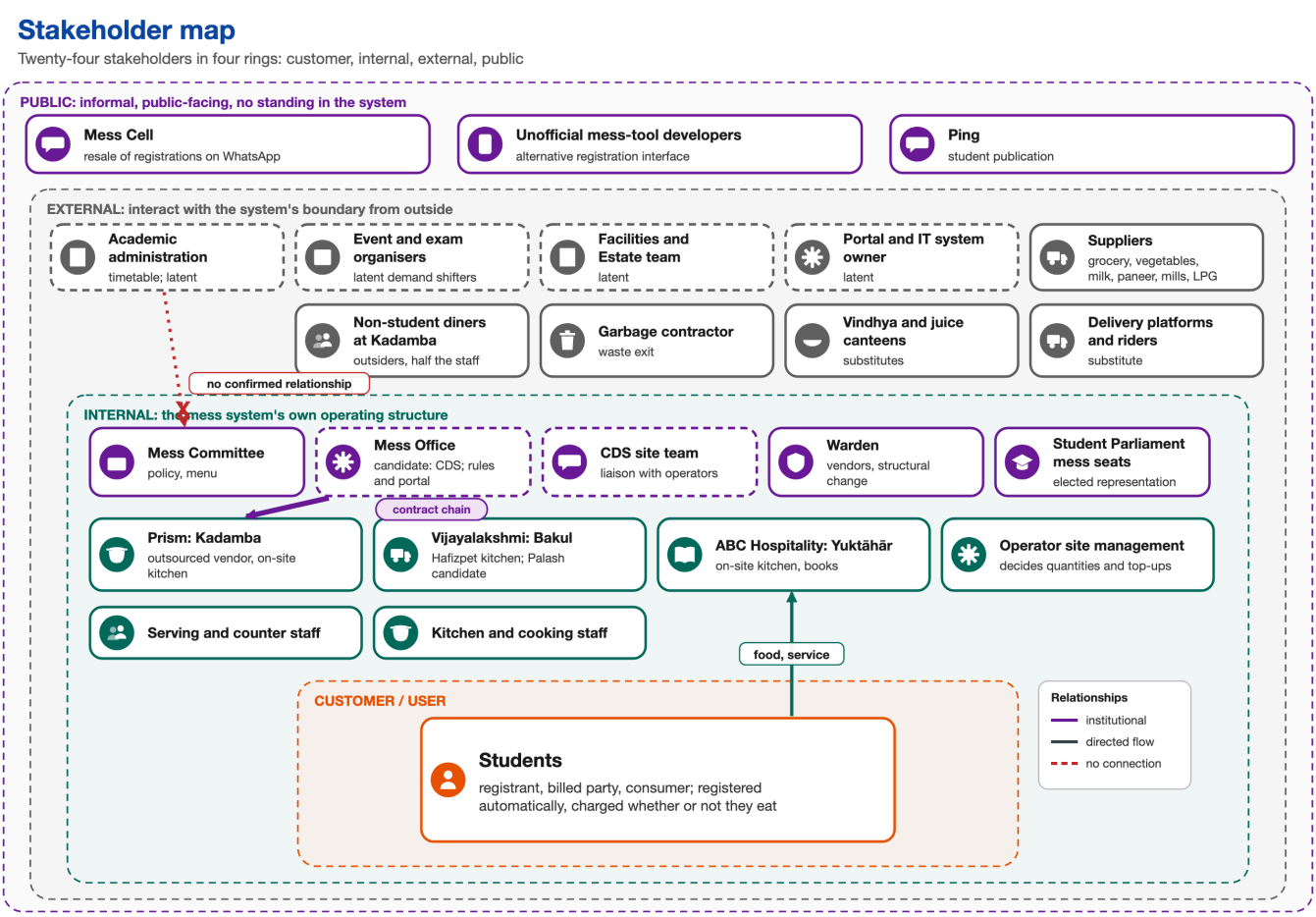

Figure 5.6. Stakeholder map in four rings: customer, internal, external and public. Source: authors' analysis, using the course's four-ring stakeholder-map template.

The relationship network regroups the same actors into five clusters (Figure 5.7). Notice the operator interior: a ladder at Kadamba, a flat ring of books and keepers at Yuktāhār, and two sites with one manager at Bakul.

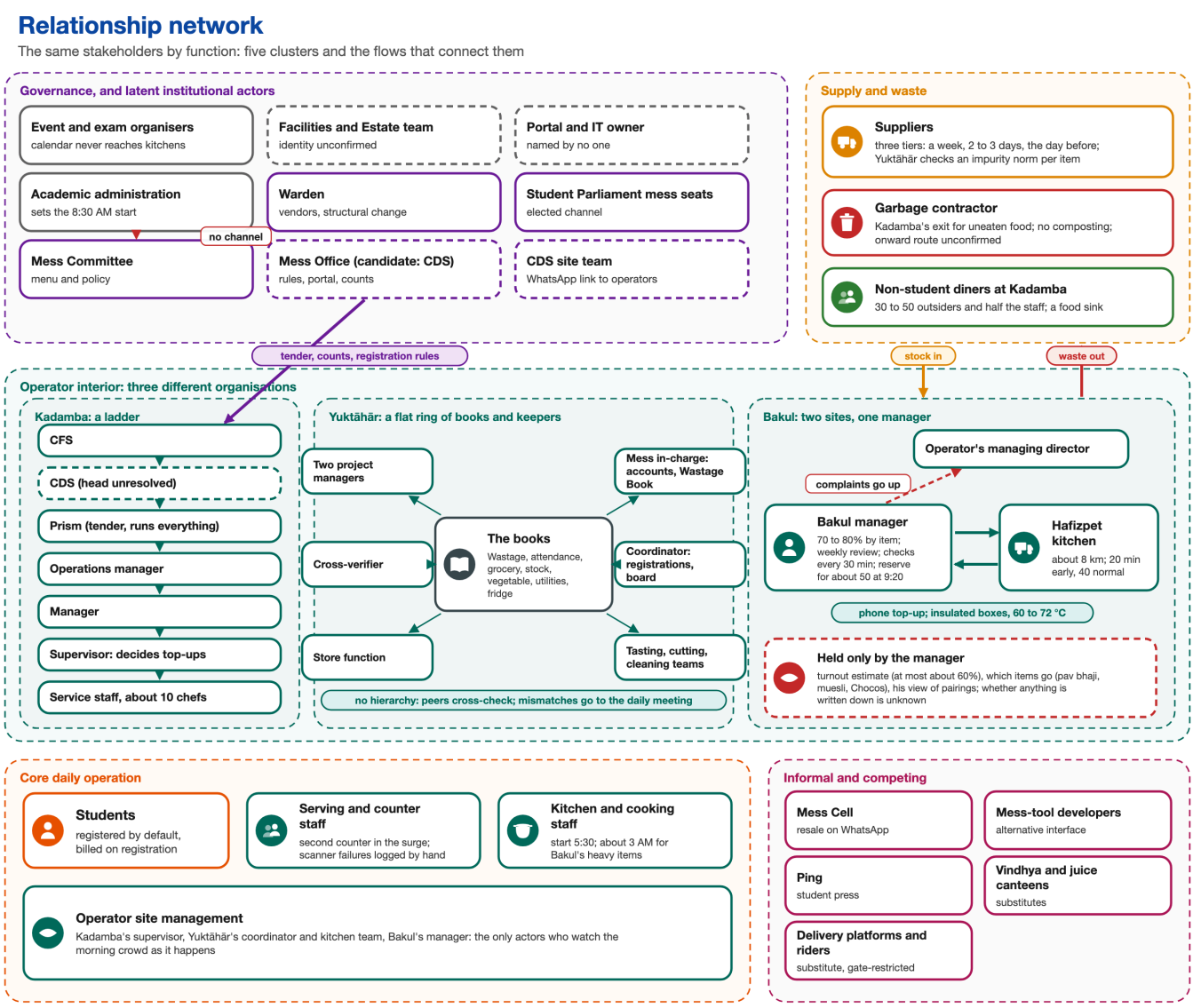

Figure 5.7. Relationship network: five clusters, with the flows of authority, money, food, information and complaints. Source: authors' analysis.

By closeness to a student's morning, academic administration sits in the outermost ring with no adjacent relationship, while Mess Cell and Ping sit beside students (Figure 5.8). Counter staff and the in-service deciders also sit close; they alone see the morning crowd as it happens.

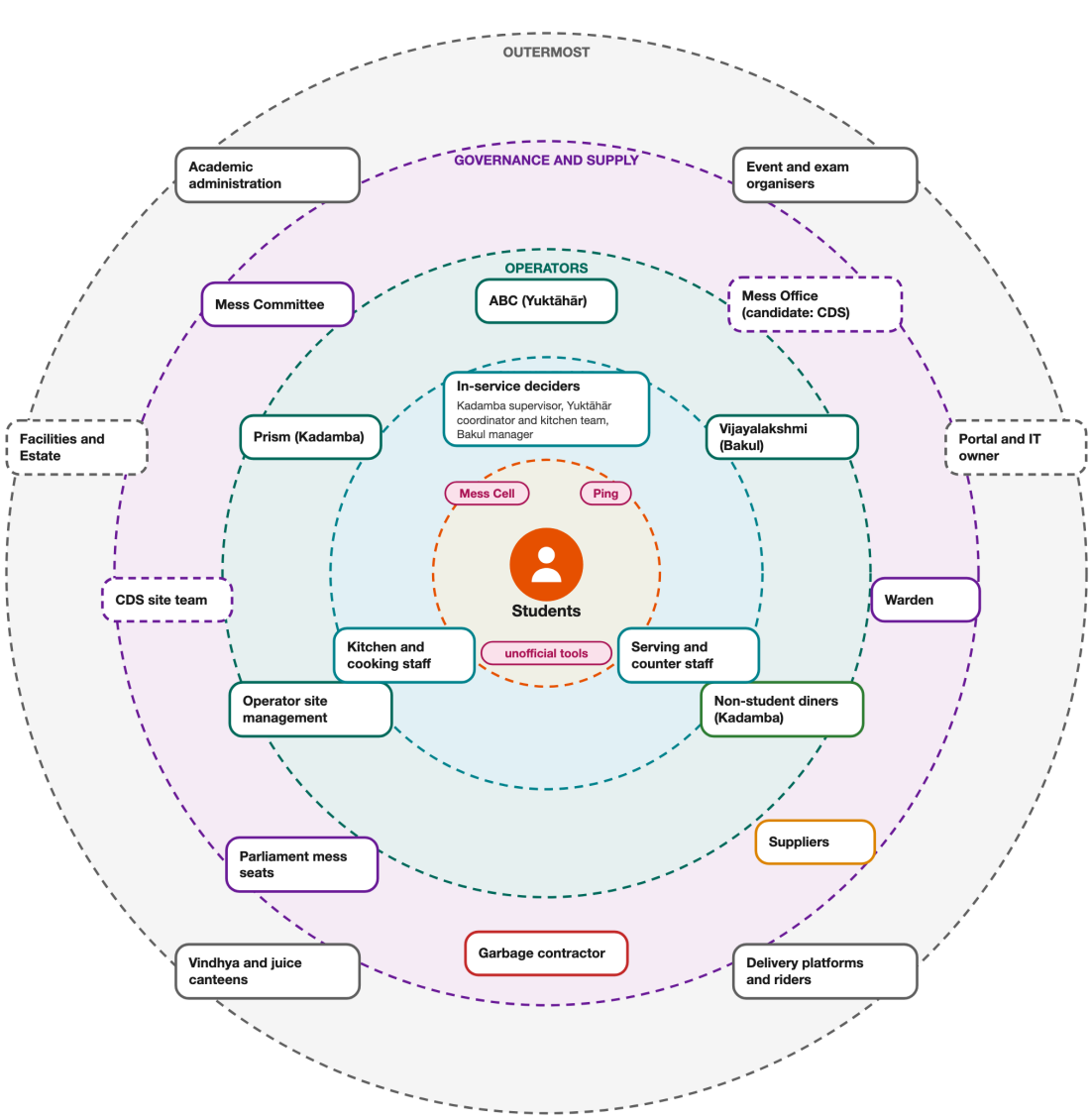

Figure 5.8. Proximity map: how close each stakeholder is to a student's actual morning. Source: authors' analysis.

Dependencies and conflicts

Bakul depends on the road, about 20 minutes early and 40 in normal traffic, which revises a first version with a 40-minute leg and 30-minute top-ups. The contract dependency stays open: paid on registered plates, an operator keeps the margin on every no-show; paid on served plates, it bears the over-cooking (candidate). The sharpest conflicts set billing certainty against uncertain attendance, menu ownership against waste visibility, and waste cuts against the outside diners and staff who eat the surplus.

Decisions and revisions

- We cut the list from 32 to 24 on direct evidence and kept the five minor and three excluded candidates on record with reasons.
- The first map treated the vendors as one actor. Asking who decides how much food appears revealed three firms with three regimes, so they became their own cluster.
- We split the Mess Committee from the Mess Office on 5 September and kept every unconfirmed governance identity apart.
- When the operator visits of 26 and 27 September contradicted first-version findings on vendors, lead time and who orders food, we replaced them everywhere. The first version had also denied that vendors and students meet at the counter.

Activity 3: Analysing Power, Interests and Leverage

Activity 3 asked how authority, resources, precedent and bargaining shape the system, and who can and cannot influence it. It produced the Stakeholder Power-Interest/Leverage Map and the Power Analysis Brief.

Who decides what

Authority is split into pieces that do not touch. Kadamba's line runs from CFS to CDS to Prism and down Prism's ladder to the service staff (Figure 5.9). Beside the chain sit the facility team's side channel and Yuktāhār's flat structure. Who heads CDS stays open, because two sources name different people.

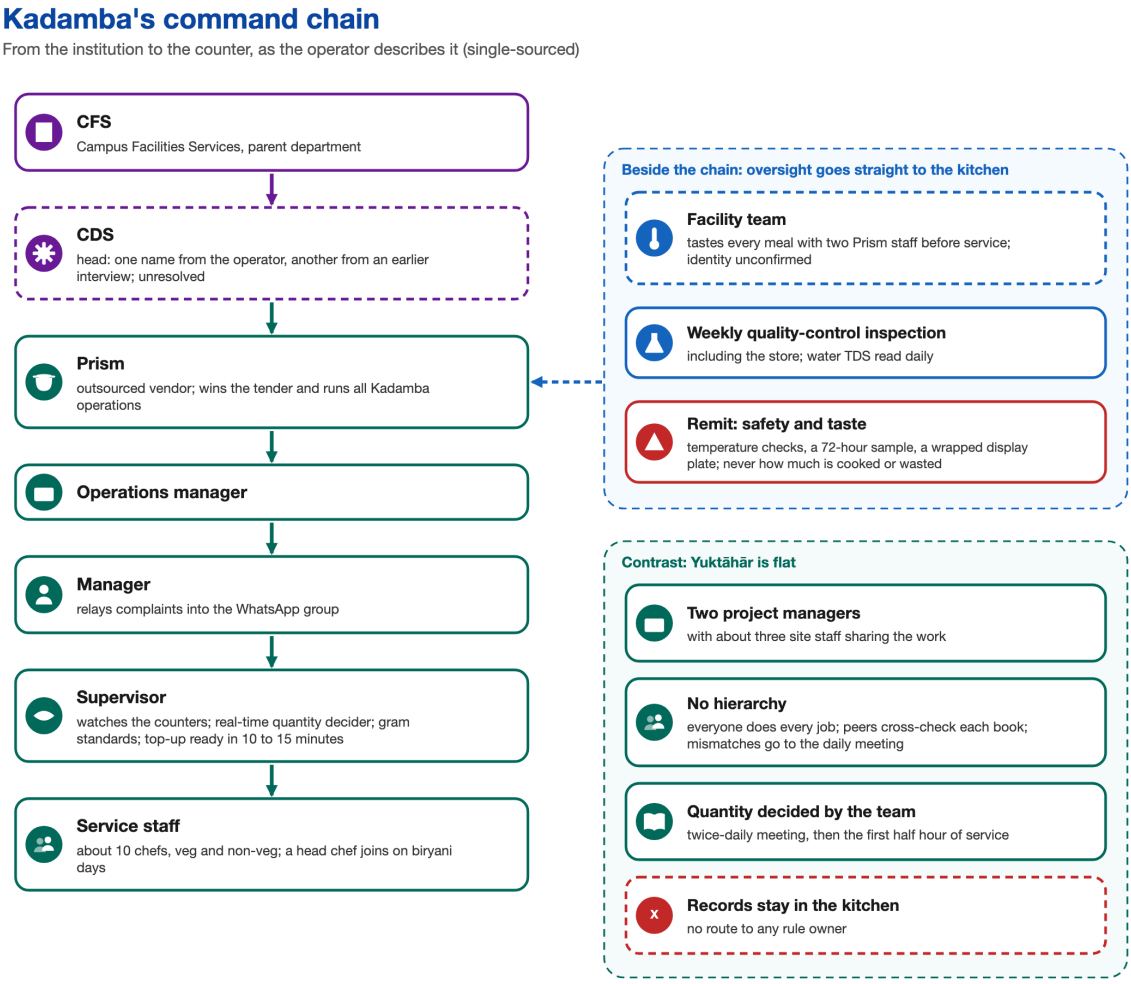

Figure 5.9. Kadamba's command chain, with the facility team's side channel and Yuktāhār's flat contrast. Source: the Kadamba manager's account and the team's first-hand account (single-sourced).

Decision	Who holds it	Evidence
Class timing	Academic administration	Confirmed; no link to mess governance
Registration cut-off and cancellation	Mess Office; Committee sign-off unclear	One case, December 2024 (candidate)
Menu	Student committee, from operator options	Confirmed in outline
Quantity, first batch, reuse	The operator	Confirmed at Yuktāhār; single-sourced elsewhere
Breakfast timing	Unknown	About 37 scans a day at or after 9:30 (confirmed)
Food safety	Facility team; weekly quality control	Nothing covers quantity

The operator holds high local authority over quantity and none over the rules that decide how many students register. Nobody has raised the 8:30 overlap with academic administration.

Incentives

Nothing rewards registering only to eat. Skip Meal returns nothing and resale returns value: of six respondents none used Skip Meal for a return and four used Mess Cell (confirmed). The Mess Office's workload motive produced tighter

cancellation rules in December 2024, a candidate fix that fails. At Kadamba a 70% first batch against 37.1% turnout implies about 1.9 times over-preparation (candidate).

Who can and cannot influence

Students carry high consequence and low power. The operator has the best data and no authority: Yuktāhār's books record registered, eaten and wasted per item and weekday (confirmed), and no route takes them out of the kitchen. Even its own site head believes breakfast turnout is about 50%, while his September register shows 17.7 to 40.2%, mean about 28 (confirmed). The record that would correct him is in his own book (Figure 5.10).

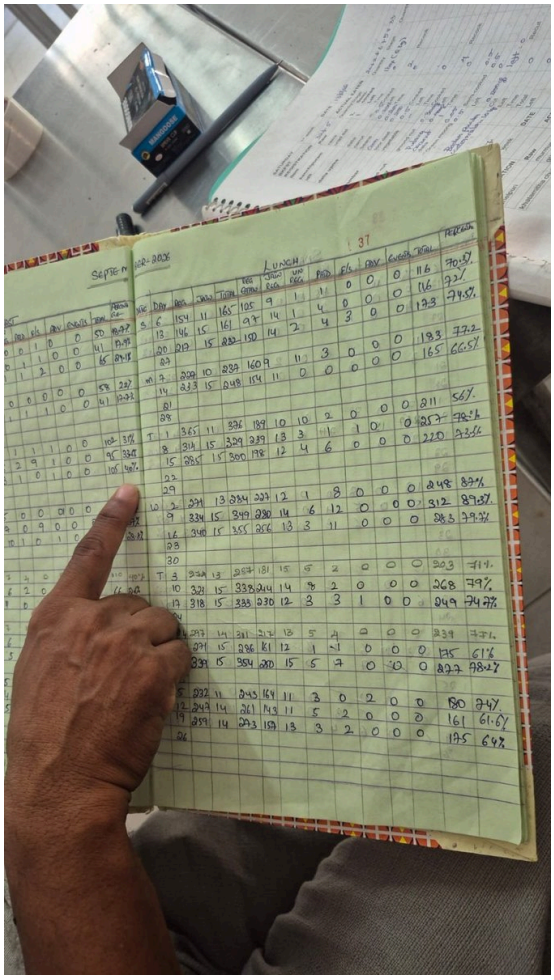

Figure 5.10. Yuktāhār's September attendance book, lunch on the right and the edge of the breakfast page on the left. Source: field photograph, 26 September 2026.

The power maps

Placement on the maps is ordinal. On power against interest, operators have high interest and local power but no rule-making power, which revises a first placement at moderate power and low interest (Figure 5.11).

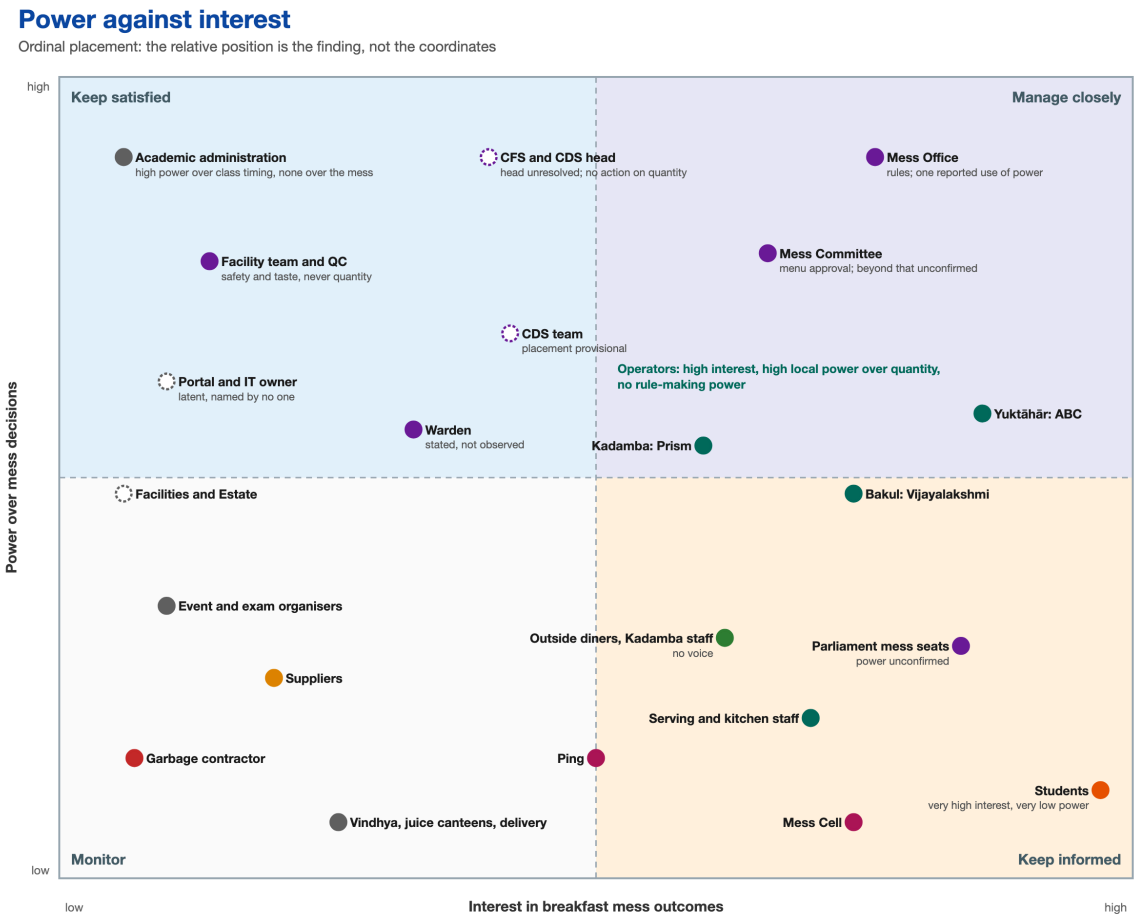

Figure 5.11. Power against interest. Placement is ordinal. Source: authors' analysis.

Separating leverage from power shows two different levers: operators over waste, and the Mess Office over student flexibility rules (Figure 5.12). Tightening operator parameters alone, such as the first-batch ratio, hides the gap further upstream.

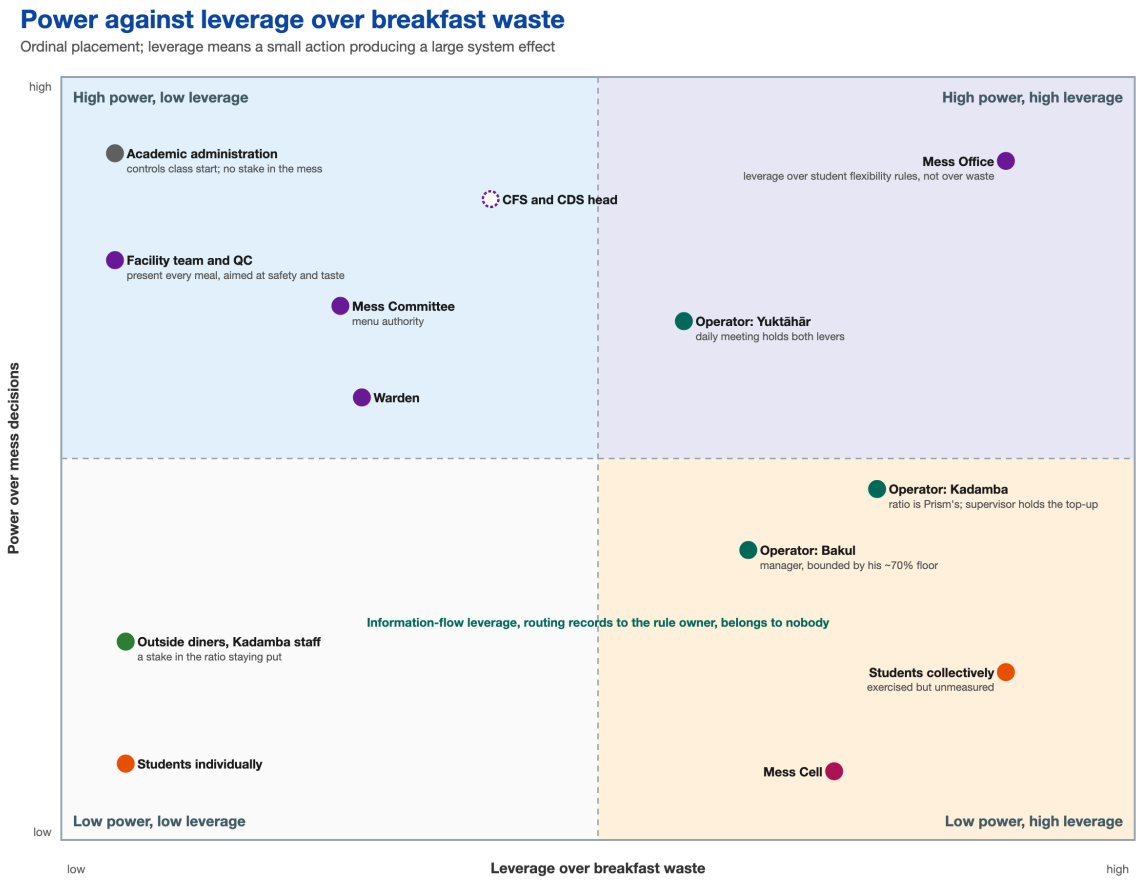

Figure 5.12. Power against leverage. Placement is ordinal. Source: authors' analysis.

The flow map marks confirmed absences in red: academic administration to every mess-governance actor, the operators' records to every rule-setter, and the calendar to the kitchen (Figure 5.13). A run-out is visible and routes a complaint; a surplus is silent (candidate).

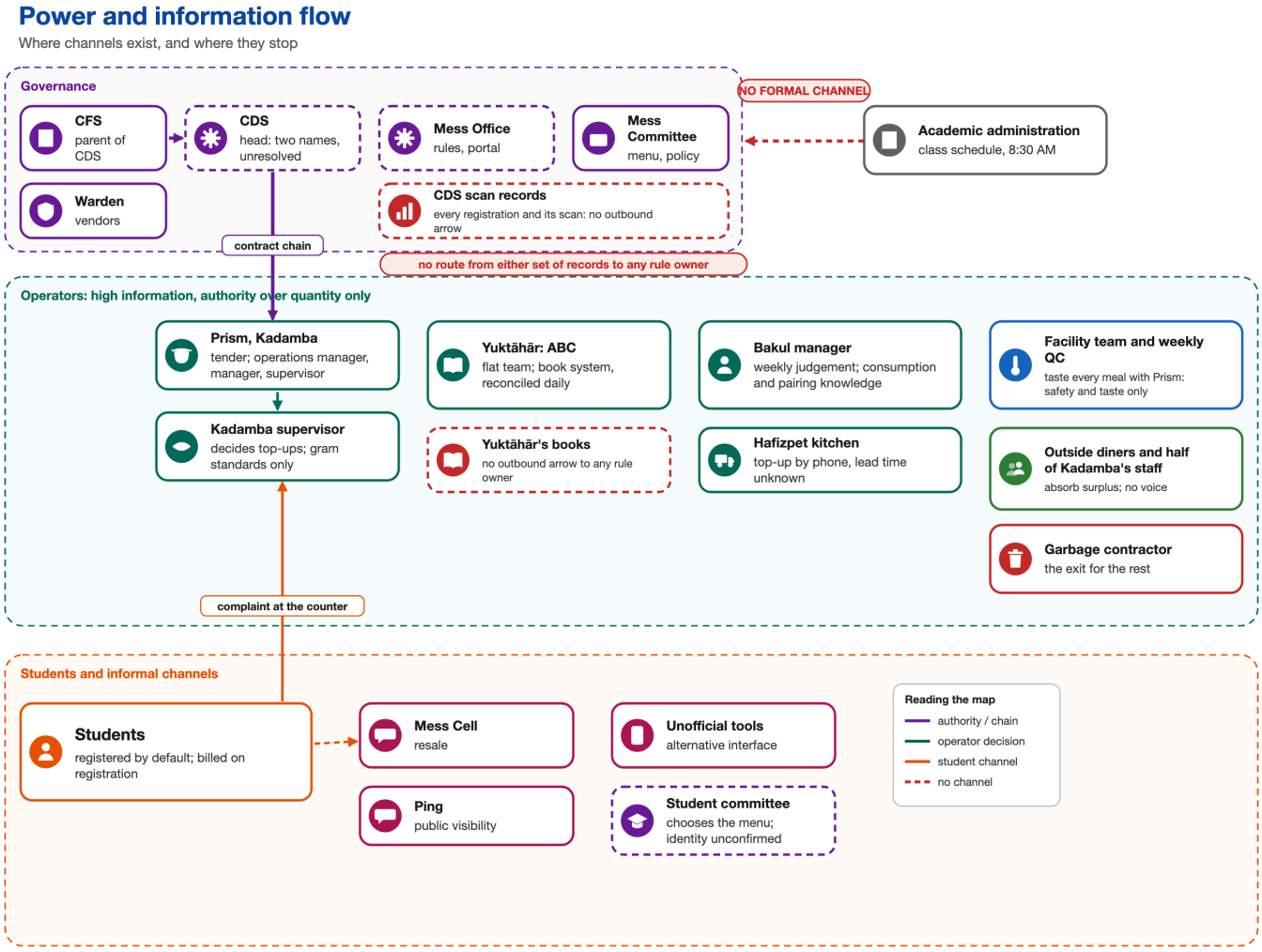

Figure 5.13. Power and information flow: where channels exist and where they stop. Source: authors' analysis.

Decisions and revisions

- We assessed power one decision at a time and rejected a single score per actor, since the same actor is decisive on one question and absent on the next.
- We separated power from leverage. Combining them, as the first version did, hid that the actor with most leverage over waste has no rule-making power.
- We kept unverified body names apart and did not treat the "Mess Council" as an actor. Collective leverage, first called sleeping, became exercised but unmeasured.
- We left the payment basis unresolved and labelled dependent conclusions candidate, because one Kadamba remark and one Bakul report pull apart.

Activity 4: Tracing the Existing Process

Activity 4 asked how the system actually operates: its decisions, rules, escalation routes, exceptions and workarounds. Its deliverable is the Current-State Process Map / Process Trace.

The student side

In practice registration is a monthly standing booking: the portal locks weekly and students set the booking once. In the monthly cycle the student's only active step is to opt out (Figure 5.14).

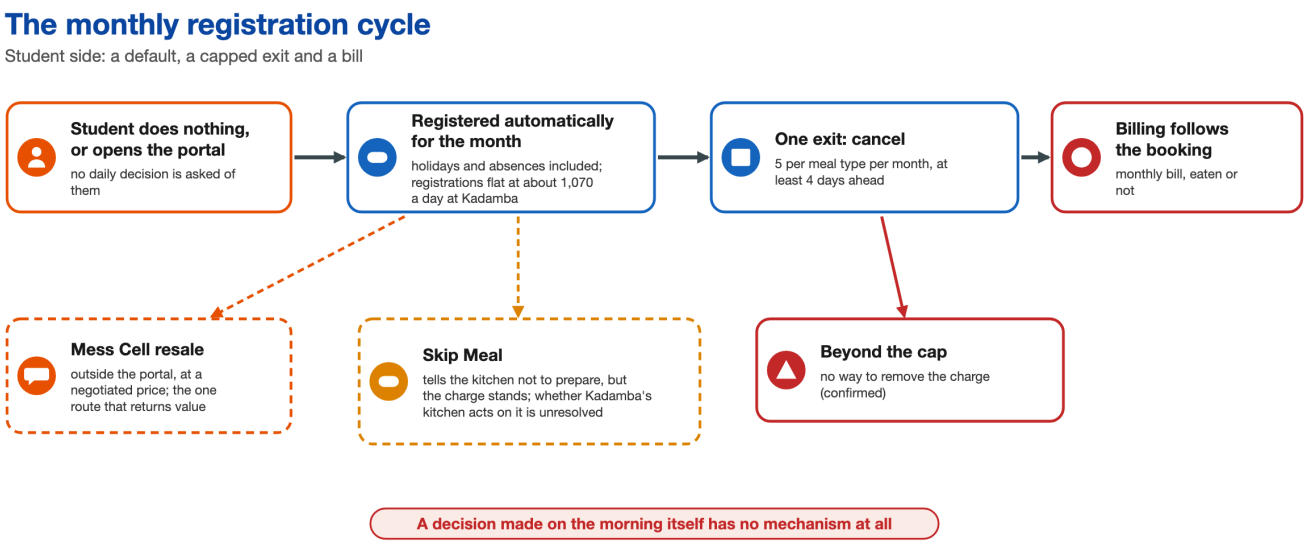

Figure 5.14. The monthly registration cycle, from the automatic default to the bill. Source: portal rules, student interviews and the team's own experience as a registered student.

The daily decision, go or skip, is made on waking, with the 8:30 class pressing on it (Figure 5.15). Four of six respondents have used Mess Cell, and none who addressed Skip Meal has used it. One early hypothesis fell here: friends do not drive attendance.

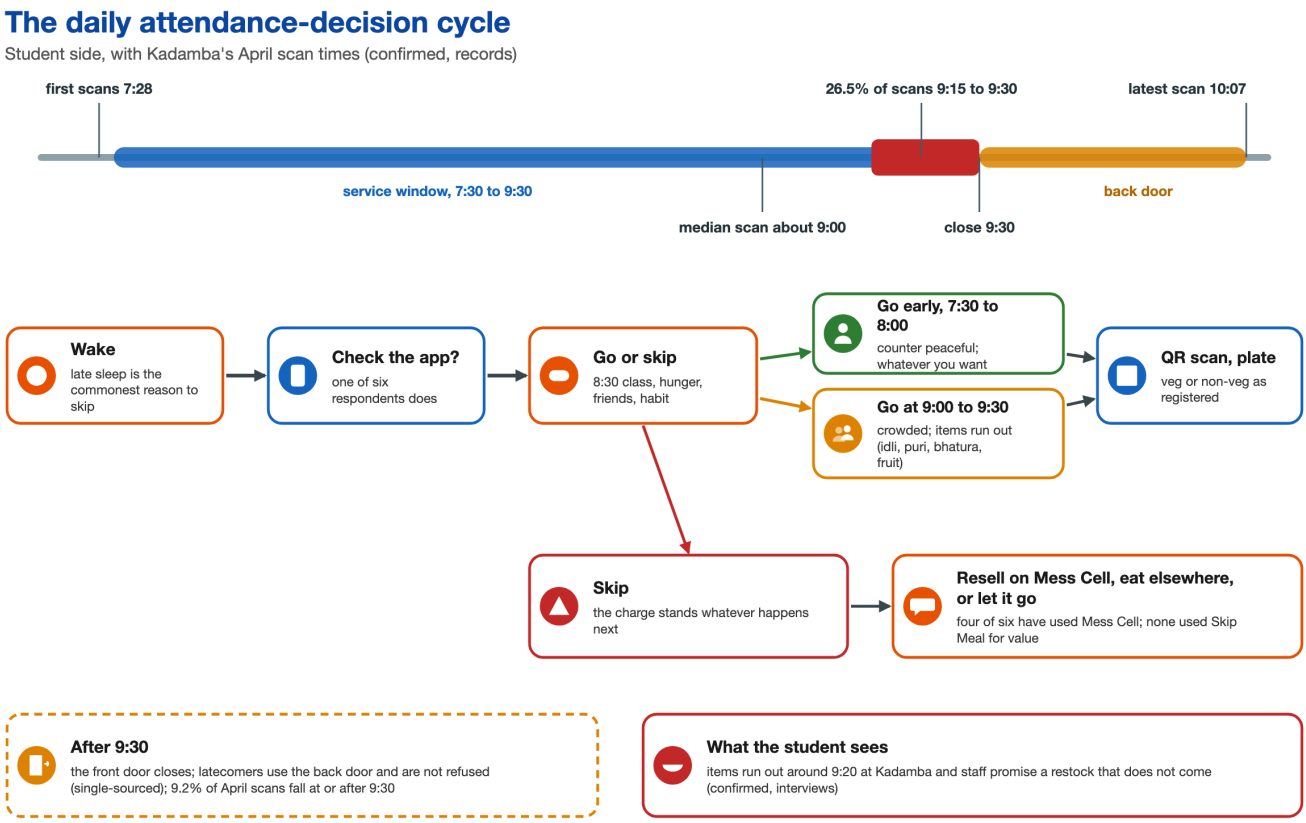

Figure 5.15. The daily attendance-decision cycle on the student side. Source: student interviews; scan times from Kadamba's April 2026 records (confirmed).

Three kitchens, three logics

No kitchen cooks to the count. Each reads the closed count the day before and converts it by its own rule.

	Kadamba (Prism)	Yuktāhār (ABC Hospitality Services)	Bakul (Vijayalakshmi Caterers)
First batch	Fixed at about 70% of registrations, by gram standard (single-sourced)	No fixed ratio: the matching past week's Wastage Book page, weekday register and events (confirmed)	70 to 80% by item, reset weekly, floor of about 70% (single-sourced)
Who decides	Prism management; a supervisor for top-ups	Storekeeper, coordinator and kitchen staff in a daily meeting	The manager alone; cooking at Hafizpet, about 8 km away
Correction in service	Top-up in 10 to 15 minutes	First-half-hour projection; re-cook on early run-out	Phone call, cooking, then a 20 to 40 minute drive; total lead time unstated (single-sourced)
Stance	The count is an obligation it is paid for	Waste is a cost to cut	Shortage is the failure to avoid

Every Bakul number is the manager's estimate (Figure 5.16). Notice that his first batch and weekend floor both exceed the highest turnout he expects, at most about 60%, so the batch exceeds turnout by design (candidate). An earlier reading from a noisy recording, a fixed 70% like Kadamba's, is withdrawn.

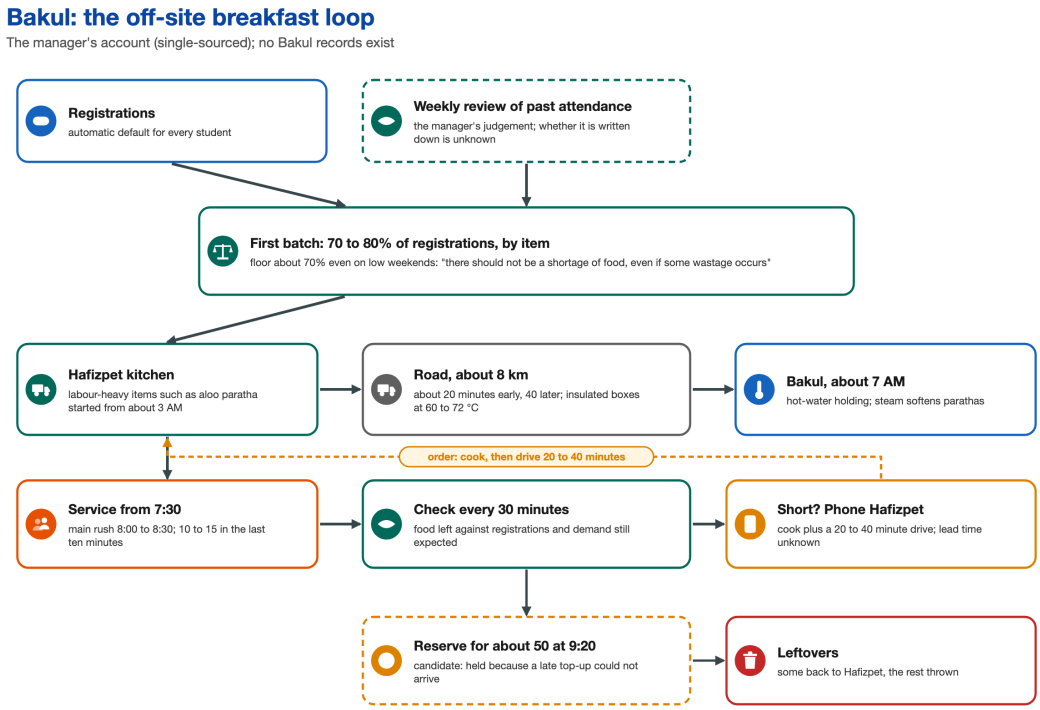

Figure 5.16. Bakul's breakfast loop from the weekly review to the late reserve. Source: the Bakul manager's account (single-sourced); no Bakul records exist.

Yuktāhār's books

Yuktāhār runs on paper books, each cross-checked by a peer. The production and waste sheets put what was made and what came back on one page, down to run-out time and plate waste (Figure 5.17).

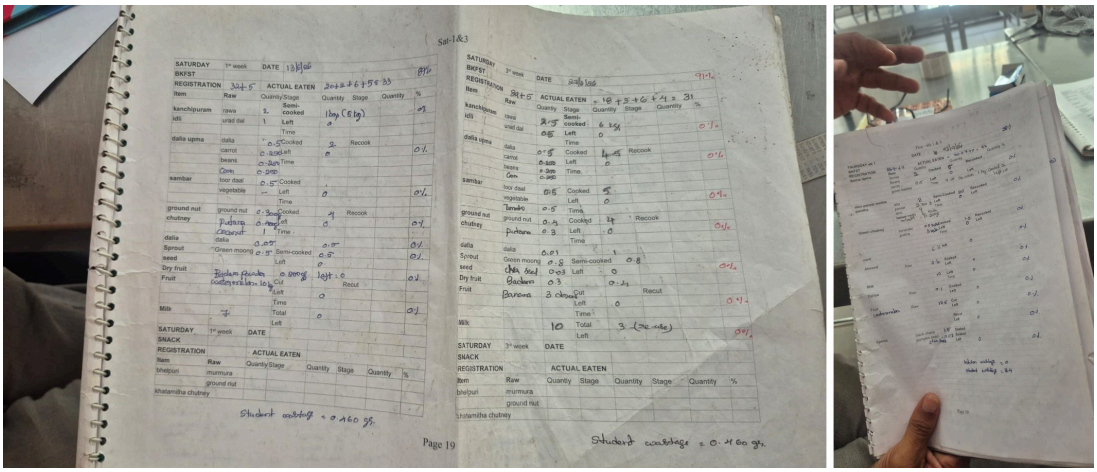

Figure 5.17. Yuktāhār's breakfast production and waste sheets (a Saturday in June, a Thursday in September). Source: field photograph, 26 September 2026.

The Grocery Book header carries the day's registrations, such as breakfast 265 plus 14 on 18 September, beside the stock registers (Figure 5.18). Its issue list is what the store pulls and cleans.

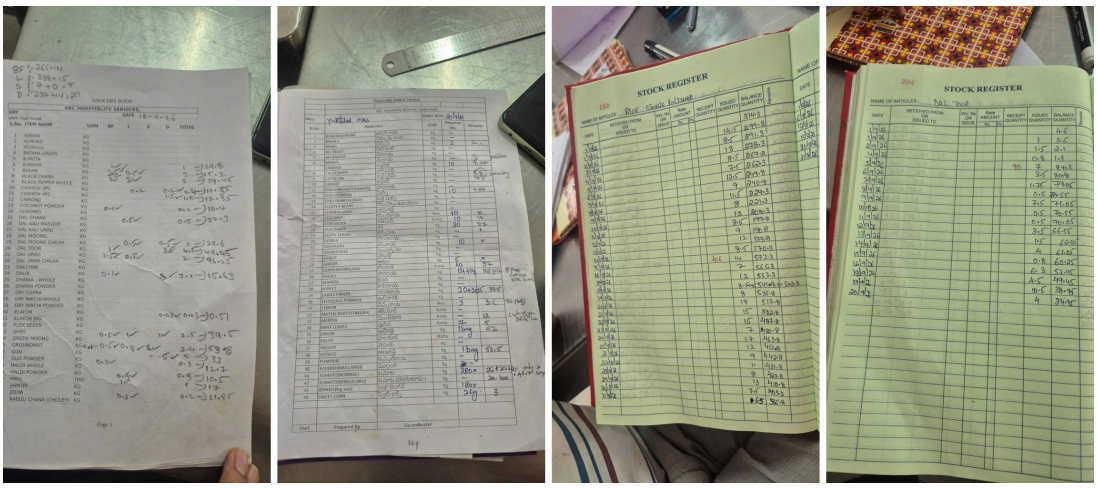

Figure 5.18. The issue and stock books: Grocery Book page 1, the vegetable indent and two stock register pages. Source: field photograph, 26 September 2026.

Daily logs track milk, fruit, gas and fridge temperatures (Figure 5.19). One rule binds the books: what was issued must match what was cooked, with mismatches raised the same day. It covers materials, not diner counts.

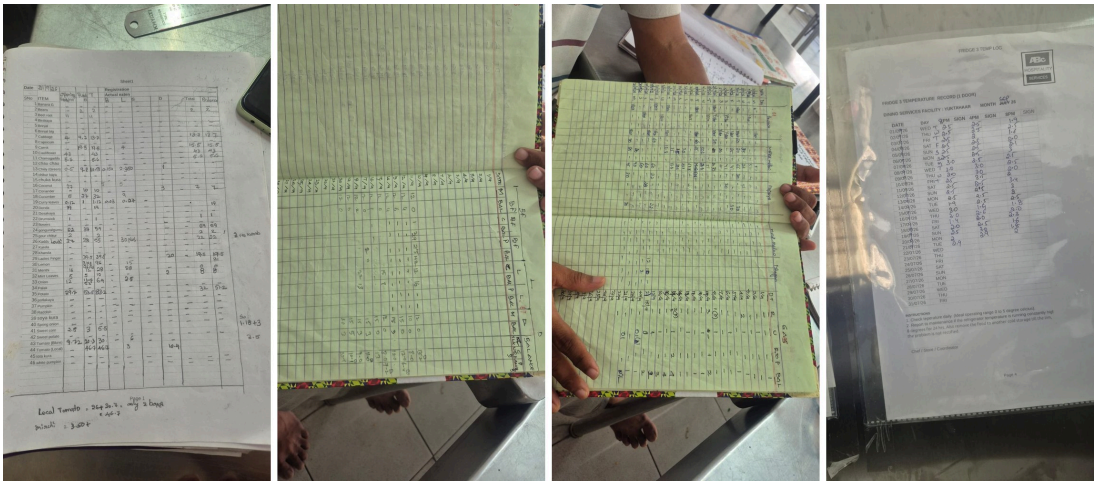

Figure 5.19. The daily logs: vegetable stock, milk and curd, fruit and gas, and the Fridge 3 temperature record. Source: field photograph, 26 September 2026.

The swimlane runs from the daily meeting to the Wastage Book and back (Figure 5.20). Notice the two-week delay before a day's waste resizes the matching day of the menu cycle.

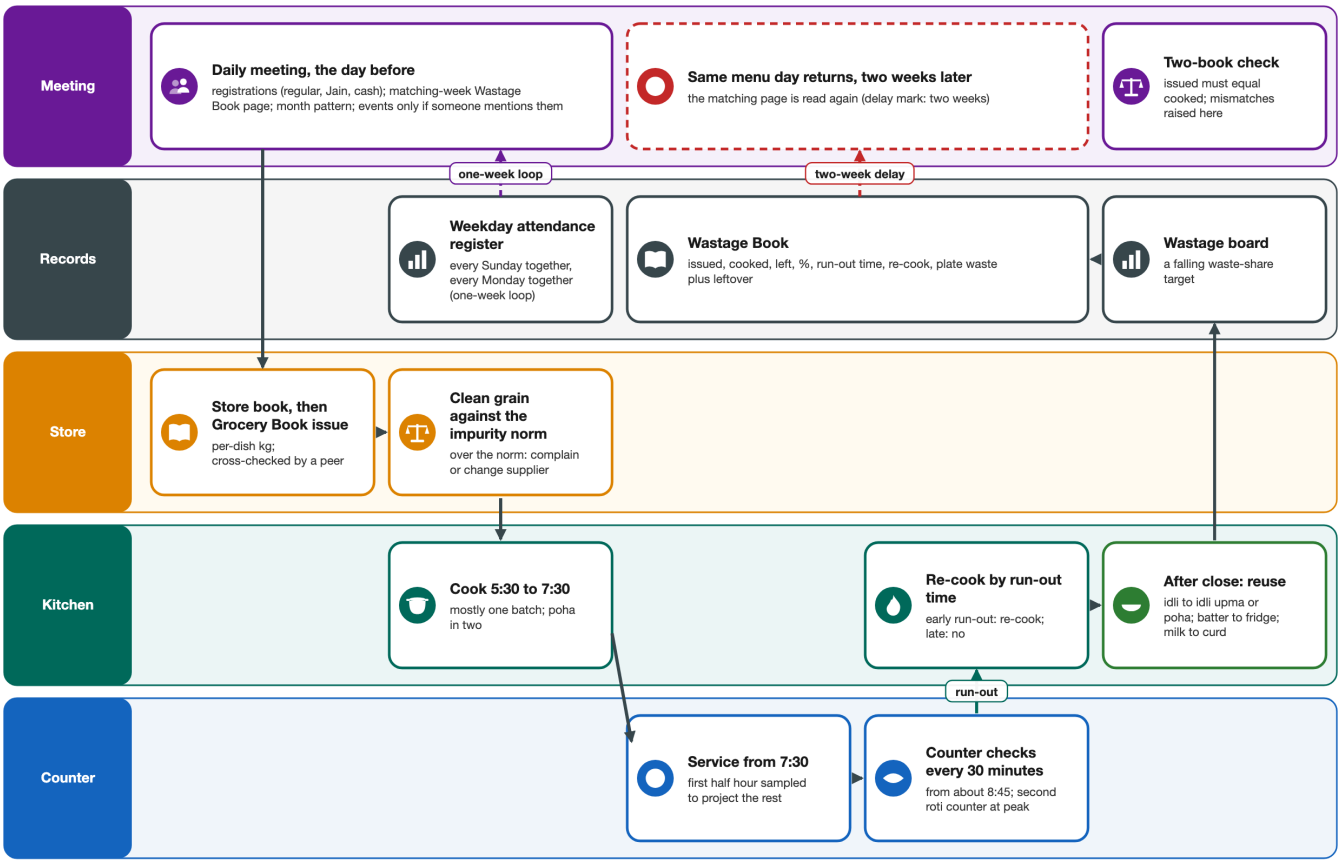

Figure 5.20. Yuktāhār's operator swimlane, from the daily meeting to the Wastage Book and back, with the two-week delay marked. Source: the Yuktāhār site head's walkthrough, checked against the photographed registers.

The breakfast day, operator side

Kadamba's lane ends at the outside diners and the bin, and no record carries the day's outcome into the next first batch (Figure 5.21). About 30 extra portions feed the outsiders, and after 9:30 latecomers enter by the back door (single-sourced).

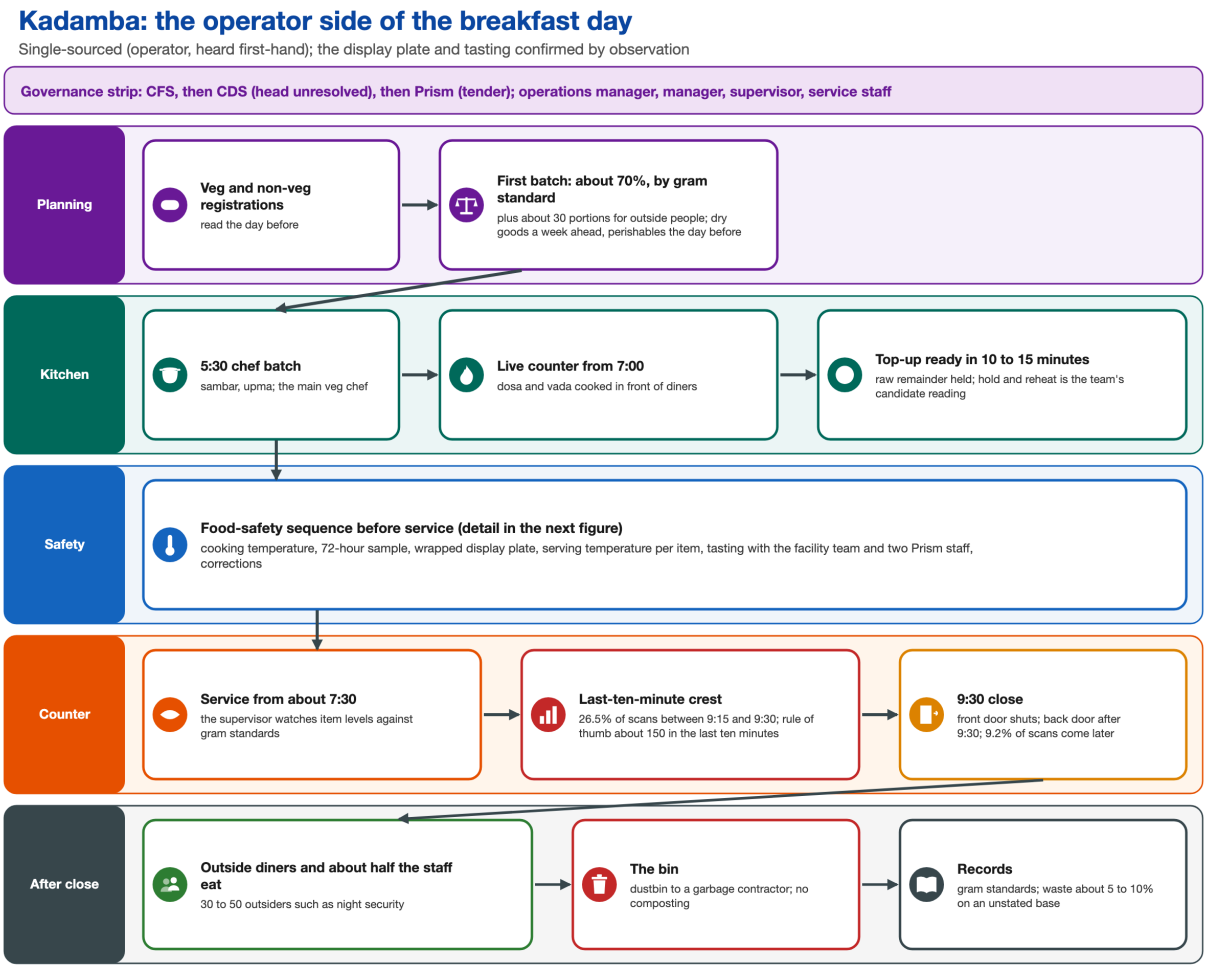

Figure 5.21. Kadamba's operator swimlane: from registrations to the 9:30 close, the outside diners and the bin. Source: the Kadamba manager's account and the team's first-hand account (single-sourced); display plate observed.

Before service a food-safety sequence runs, from cooking temperature and a 72-hour sample to tasting with the facility team (Figure 5.22). Every check concerns safety and taste, and all of them happen before any sign of turnout.

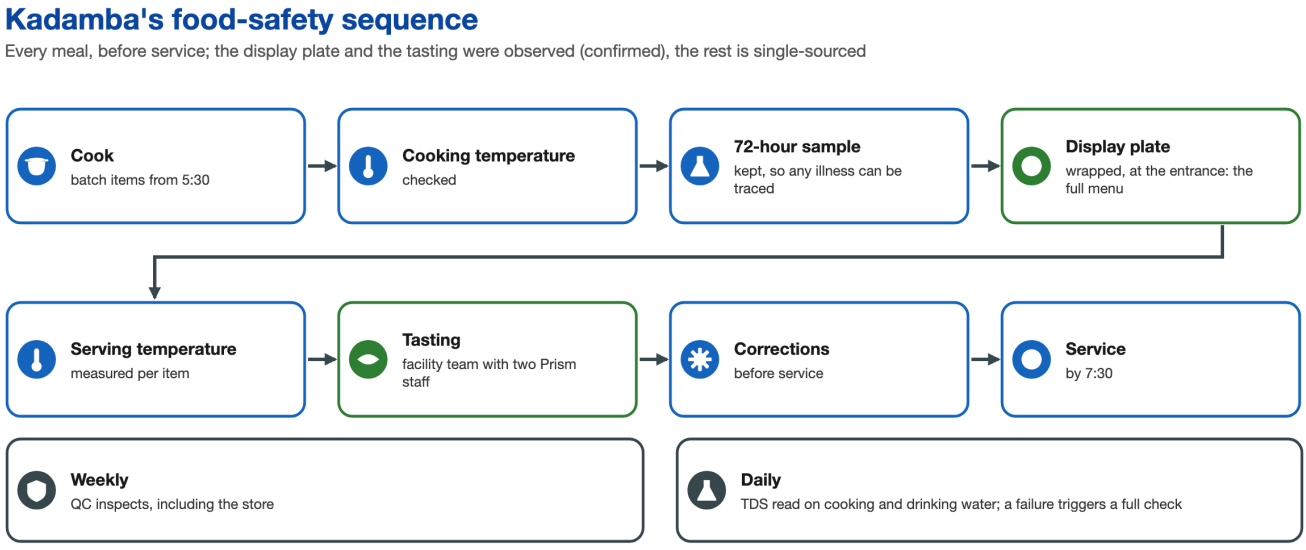

Figure 5.22. The Kadamba food-safety sequence before 7:30. Source: display plate and tasting observed (confirmed); the remaining steps single-sourced.

Rules, formal and informal

Rule	Formal or informal	Effect
Automatic registration; five cancellations per meal type per month, four days ahead	Formal	Registration becomes a standing booking
Skip Meal without refund	Formal	Unused by all six respondents
Each kitchen's ratio, floor or lookup	Informal	Every ratio sits above recorded turnout
Issued must equal cooked (Yuktāhār)	Informal but documented	Mismatches surface the same day
Blame-free waste weighing (Yuktāhār)	Informal norm	Staff resisted in August; waste falling by December

Escalation, exceptions and workarounds

None of the five new student respondents knew of anyone who had tried to change a mess rule. An operational complaint path works for same-incident fixes (single-sourced), which revises the first version's finding that no escalation path was visible, but no path reaches the rule owner.

Exceptions on both sides end in workarounds that absorb the gap before it shows (Figure 5.23). Even Yuktāhār's books disagree: on 3 September the Wastage Book records 77 breakfast diners (28%) and the register, on a partly obscured row, 110 (40%).

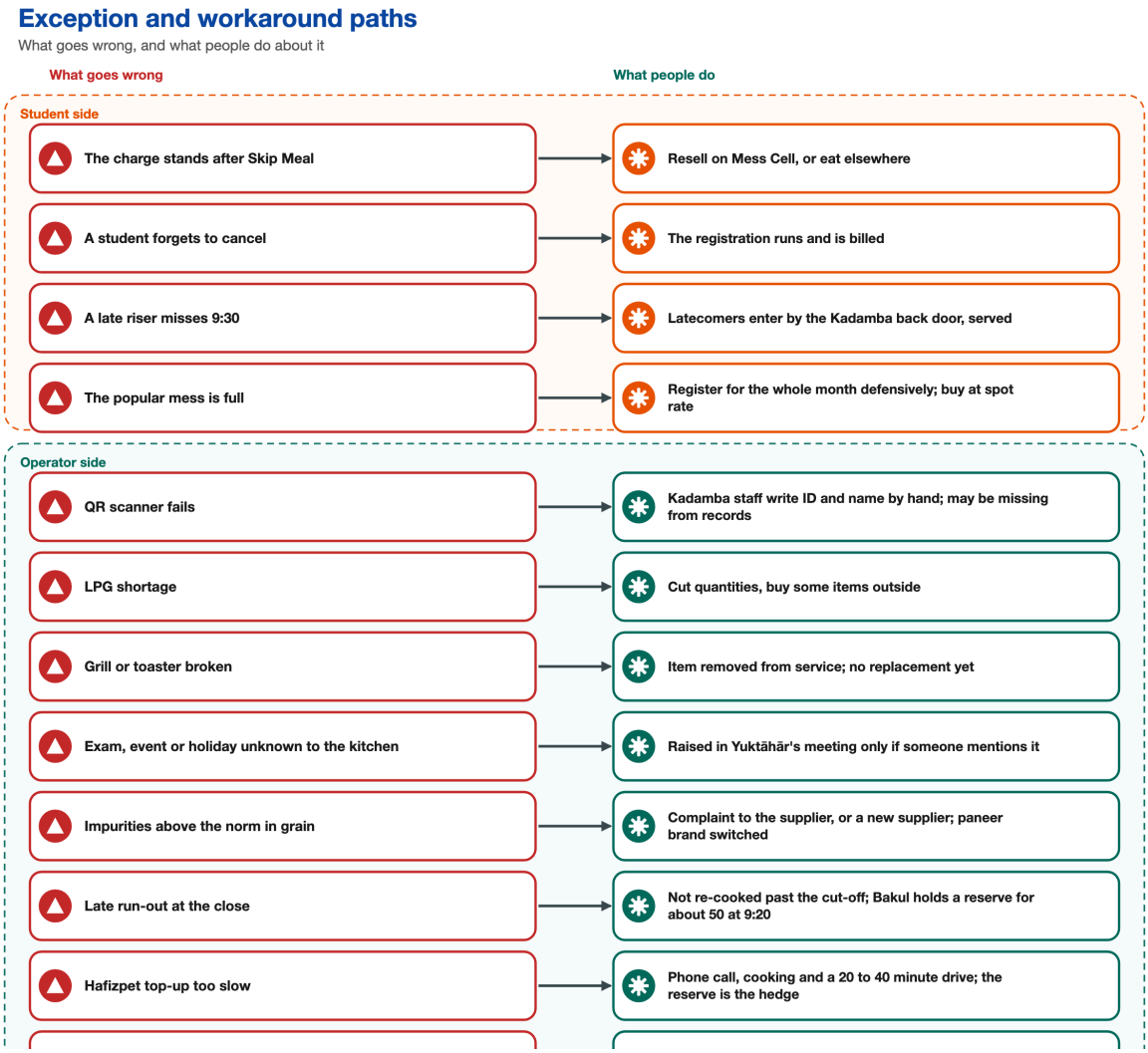

Figure 5.23. Exception and workaround paths, student side and operator side. Source: student interviews, operator accounts and Kadamba scan records.

Decisions and revisions

- We wrote the student half first and marked the kitchen half unobserved until operator evidence existed.
- We traced each caterer as its own system, since merging the three would have hidden three forecasting logics.
- An independent second reading of the interviews brought two corrections: the registration cycle is monthly, and the apparent two peaks are one building crowd.

- Three rewrite passes (two on 27 September, one on 28 September) replaced a four-day procurement lock with a registration cut-off and one passive vendor with three operators. They withdrew composting as an absorber, an unattributed operator chain, and the claim that the best-documented correction acts only on a one-to-two-week delay.
- Scan records stood in for watching one full breakfast, since they show the crowd's shape and timing better; disposal after close stays open.

Activity 5: Analysing System Behaviour over Time

Activity 5 asked what pattern appears when the process repeats over time. Its deliverable is the System Timeline / Behaviour-Over-Time Map, built from measurement wherever records existed.

The records

Kadamba's records hold 32,162 registrations and 11,941 timestamped scans for 1 to 30 April 2026. Yuktāhār's registers for 1 to 20 September 2026 keep breakfast above lunch on one page, a control within the same hall (Figure 5.24).

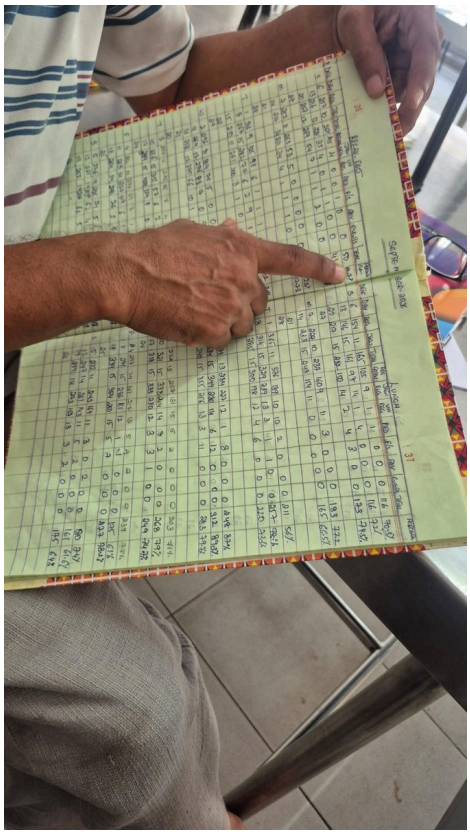

Figure 5.24. Yuktāhār attendance register, September: breakfast above, lunch below. Source: field visit photograph, 26 September 2026.

Timescale	What the records show	Confidence
A morning	One building crowd; 26.5% of Kadamba scans between 9:15 and 9:29	Confirmed
A week	Registrations flat at about 1,070 a day; share eaten from 42.1% (Tuesday) to 27.3% (Sunday)	Confirmed
A fortnight	Yuktāhār's waste records follow the two-week menu cycle	Confirmed
A month	Each seven-day block of Kadamba's April between 35 and 39%	Confirmed at the aggregate
A semester	Respondents diverge; no multi-month record	Unmeasured

A morning

The kitchens start two hours before the first student arrives, and Bakul's labour-heavy items from about 3 AM (Figure 5.25). Notice how much is fixed before the first scan.

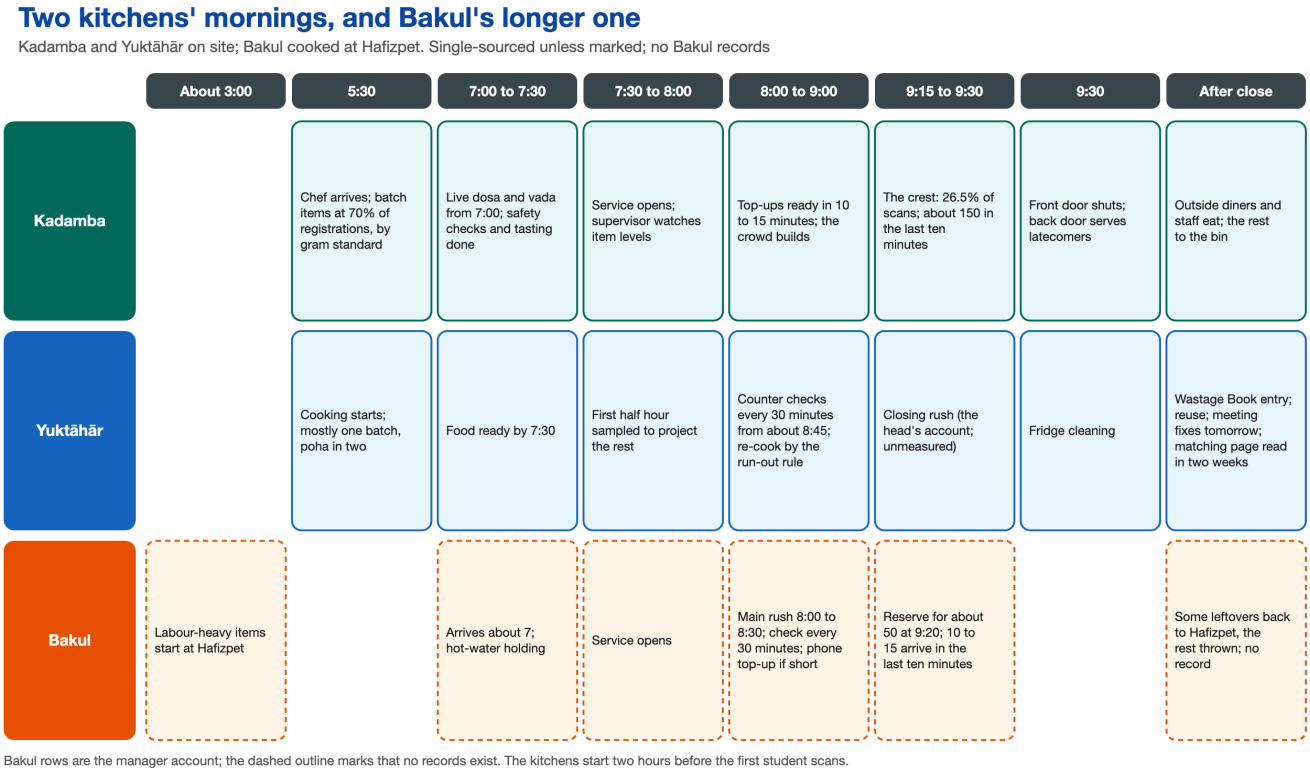

Figure 5.25. The mornings of the three kitchens side by side; Bakul rows are the manager's account, with no records. Source: operator interviews, field account (single-sourced unless marked), Kadamba scans.

Recall gave two competing peaks; the scans show one building crowd that crests just before 9:30 (Figure 5.26). The 9:15 to 9:29 slot carries 26.5% of scans, rising to about nine a minute by 9:29. Kadamba's rule of thumb of 150 diners in the last ten minutes slightly overstates the surge: the records show about 79.

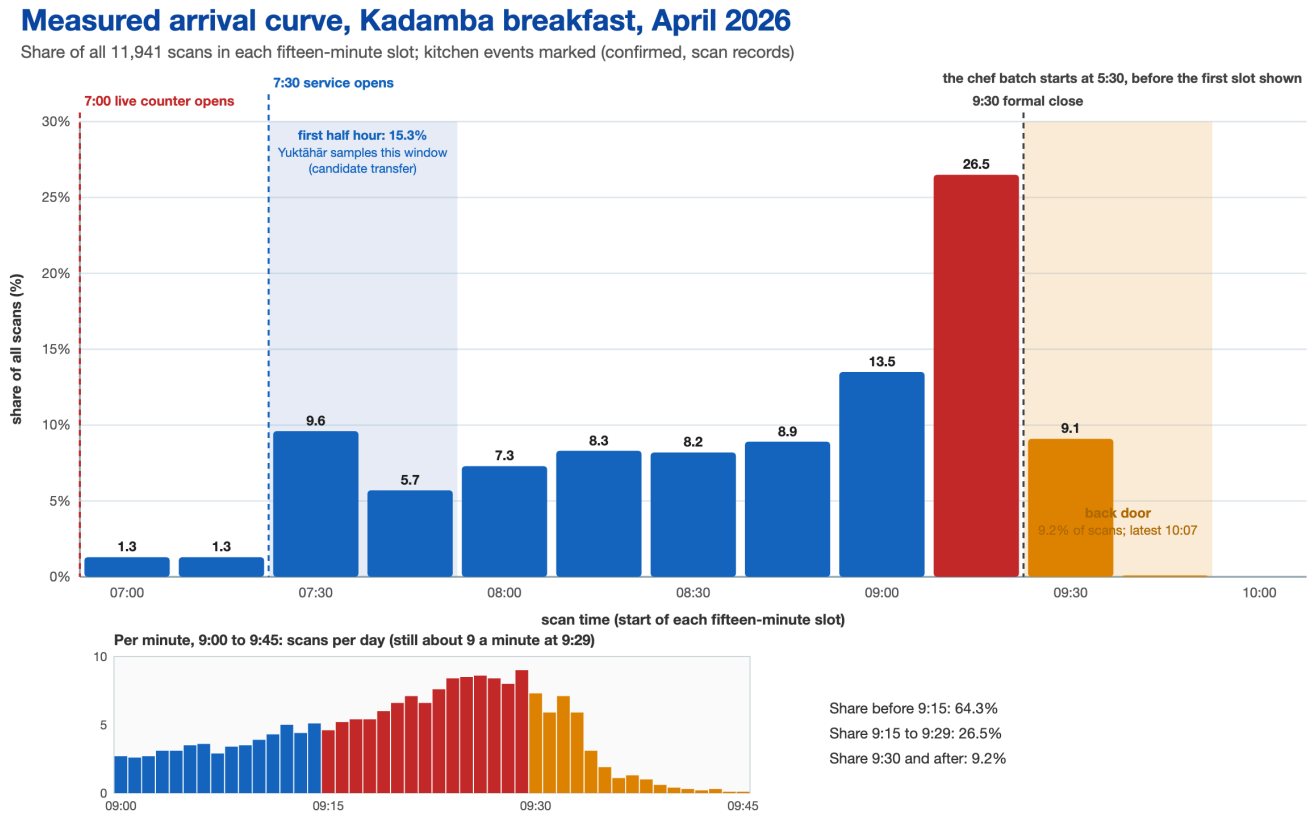

Figure 5.26. Measured arrival curve, Kadamba breakfast, April 2026, with per-minute detail from 9:00 to 9:45. Source: Kadamba scan export, April 2026 (confirmed).

A week and a month

Of 32,162 April registrations, 37.1% were scanned and 62.9% were not (confirmed). From 6 April registrations sit between 1,039 and 1,123 every day, Sunday included, while the eaten line swings (Figure 5.27).

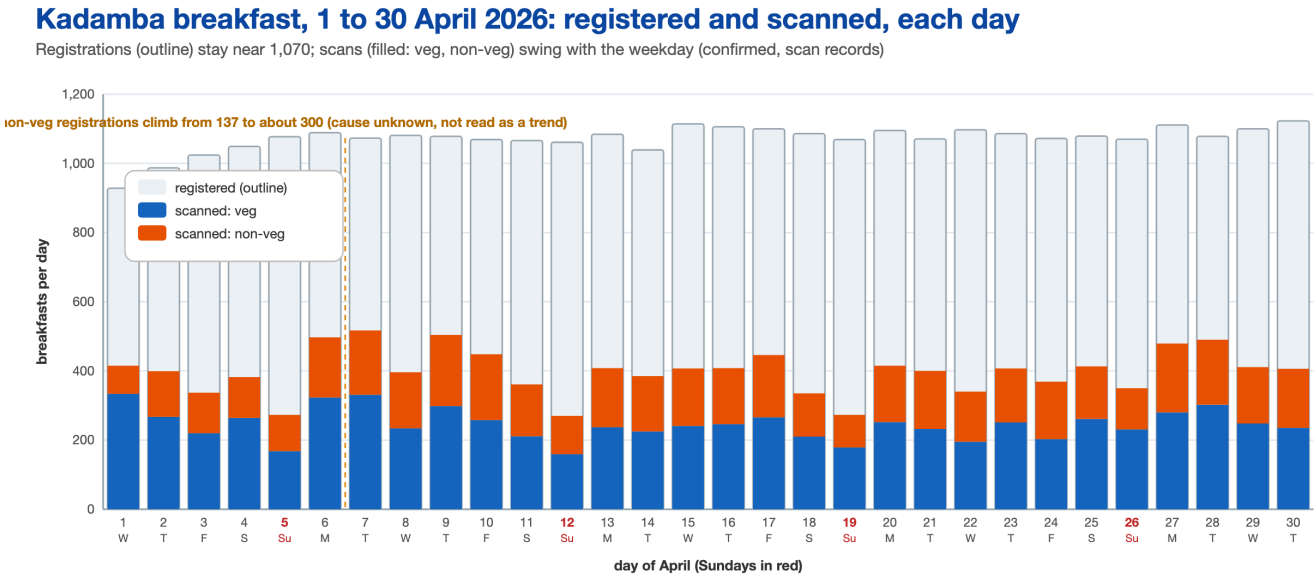

Figure 5.27. Kadamba, April 2026: registered and scanned breakfasts each day. Source: Kadamba scan export, April 2026 (confirmed).

The share eaten runs from 42.1% on Tuesday to 27.3% on Sunday, and the three lowest days of the month were Sundays (Figure 5.28). Weekdays together run at 39.3% and weekends at 31.1.

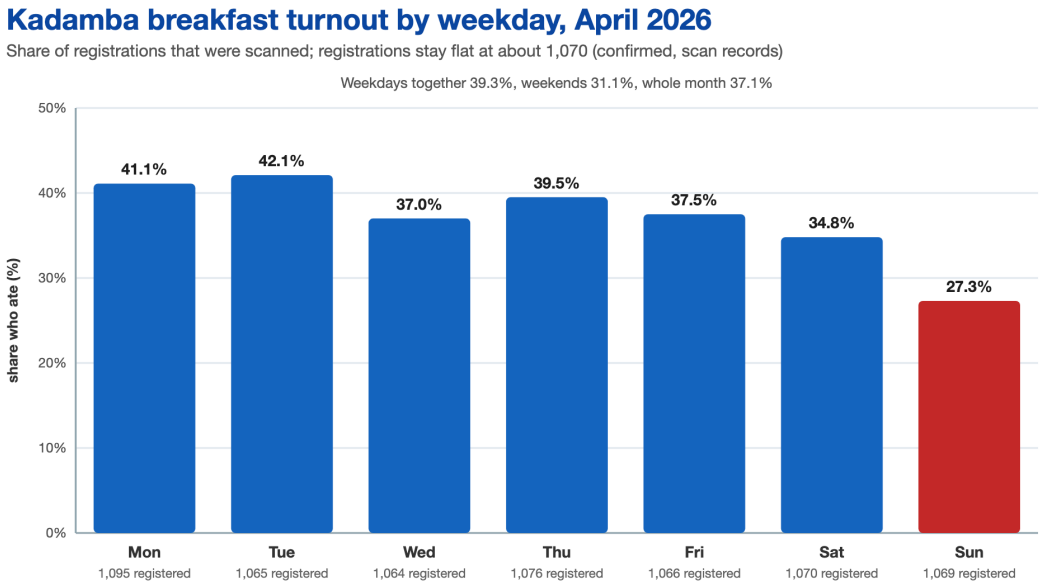

Figure 5.28. Share of Kadamba registrations eaten, by weekday. Source: Kadamba scan export, April 2026 (confirmed).

At Yuktāhār breakfast averages about 28% against lunch at 56 to 89%, and the site head's belief, the dashed line, sits far above the breakfast record (Figure 5.29). Lunch registrations fall on Sundays and breakfast registrations do not, so breakfast may be a standing default in a way lunch is not (candidate, three Sundays).

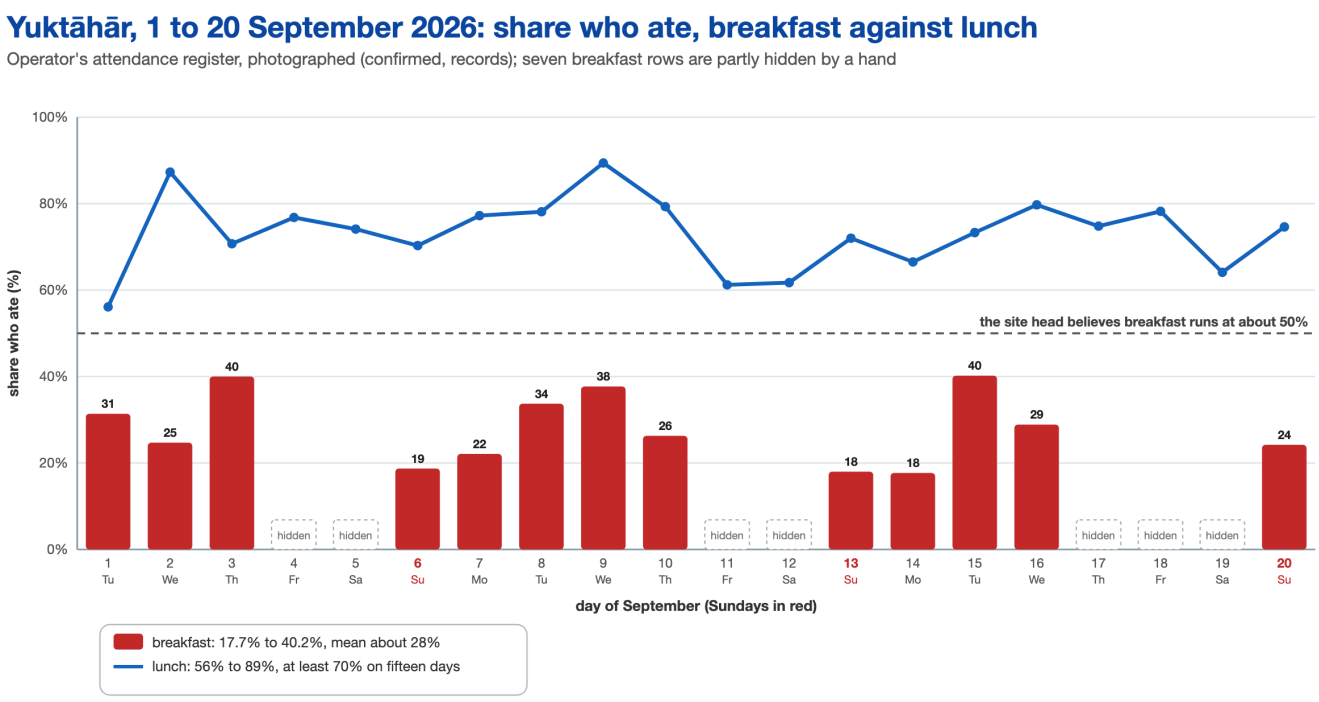

Figure 5.29. Yuktāhār, 1 to 20 September 2026: breakfast against lunch; the dashed line is the site head's belief about breakfast. Source: Yuktāhār attendance registers (confirmed); seven breakfast rows partly hidden.

What is fixed when

Part of the gap comes from timing (Figure 5.30). The matching Wastage Book page sizes a dish about two weeks ahead, cancellations close four days ahead and the first batch is fixed before service. Real-time signals arrive only after the first batch is cooked, and only Yuktāhār turns the first half hour into a forecast.

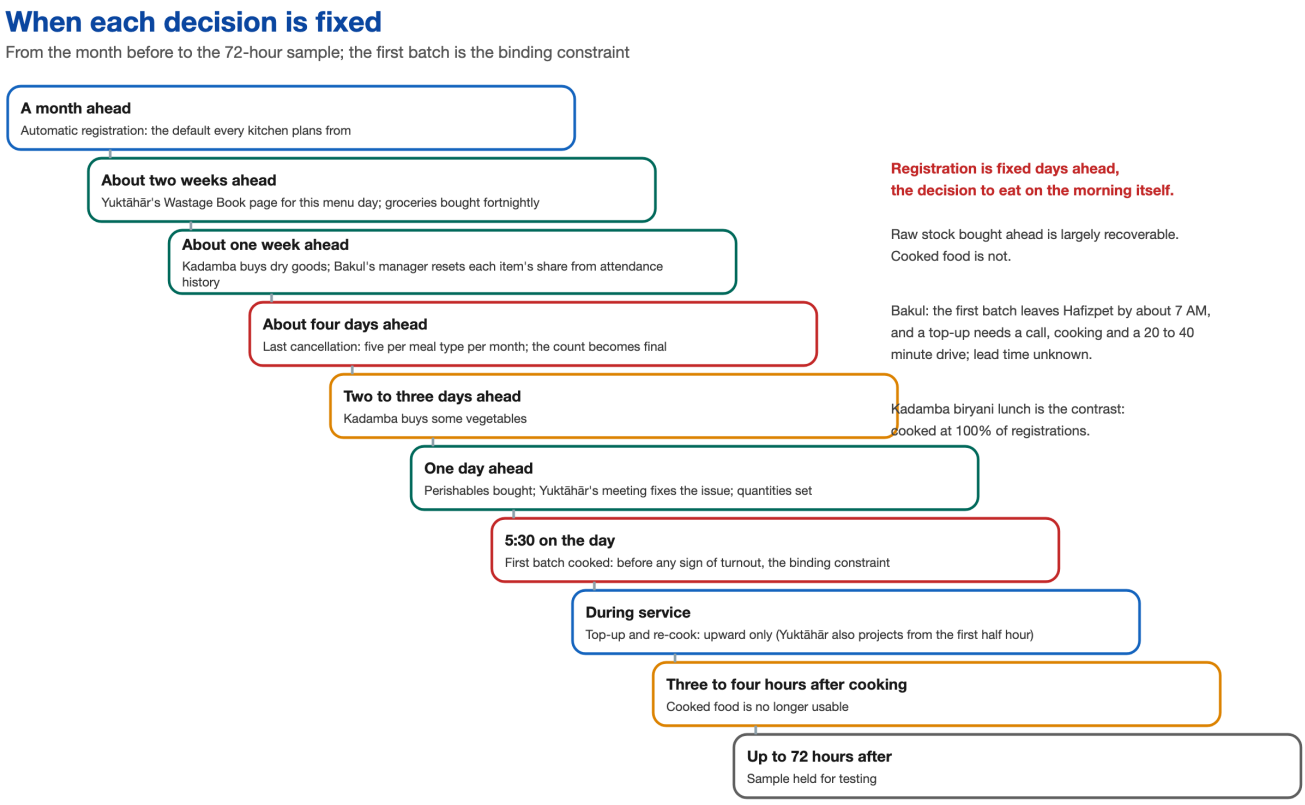

Figure 5.30. When each decision is fixed, from the month before the meal to the 72-hour sample. Source: published rules, operator interviews and Yuktāhār's books.

Significant changes and their triggers

Governance changes cluster in late 2024 and early 2025 and are secondhand, while operator changes run on into 2026 (Figure 5.31). No student mentions a recent rule change.

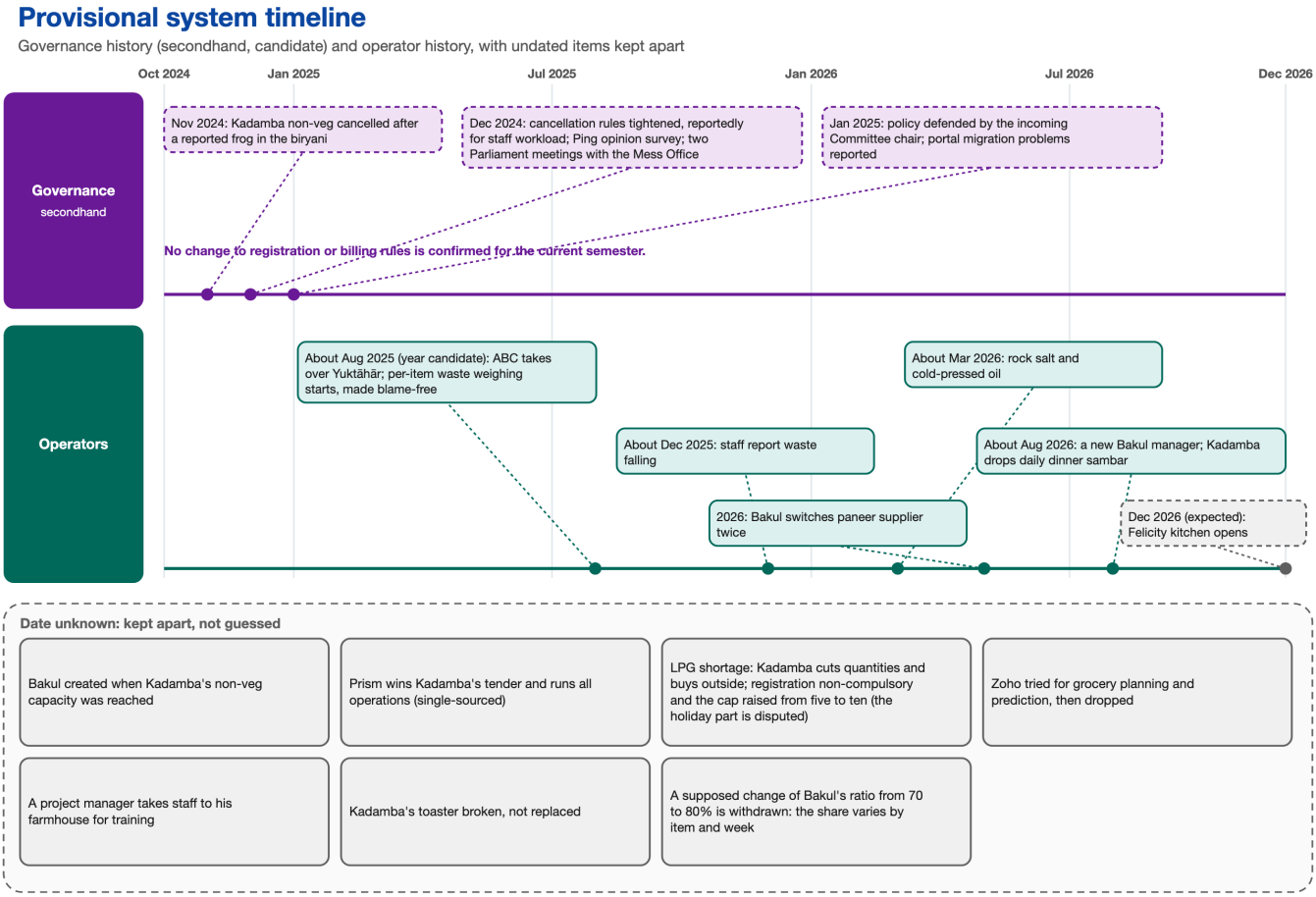

Figure 5.31. Provisional system timeline: governance history above, operator history below. Source: precedent research (secondhand) and operator interviews (single-sourced).

When	Change	Trigger
Date unknown	Bakul created	Kadamba's non-vegetarian capacity ceiling
November 2024	Kadamba non-vegetarian meals reportedly cancelled (candidate)	A frog reportedly found in a biryani
December 2024	Cancellation rules reportedly tightened (candidate)	Mess Office workload, Kadamba renovation
August (year candidate: 2025)	Yuktāhār begins per-item waste weighing (confirmed)	Operator initiative for lower cost and higher profit
Date unknown	Registration non-compulsory, cap raised from five to ten, then reversed (single-sourced)	LPG shortage
September 2026	No confirmed change to registration or billing rules	None identified

Feedback loops and recurring patterns

R1 is a candidate reinforcing loop that sustains the dysfunction without growing it (Figure 5.32). Resale on Mess Cell relieves the student's loss, so pressure to fix billing stays low. The dashed links are inferences no respondent stated.

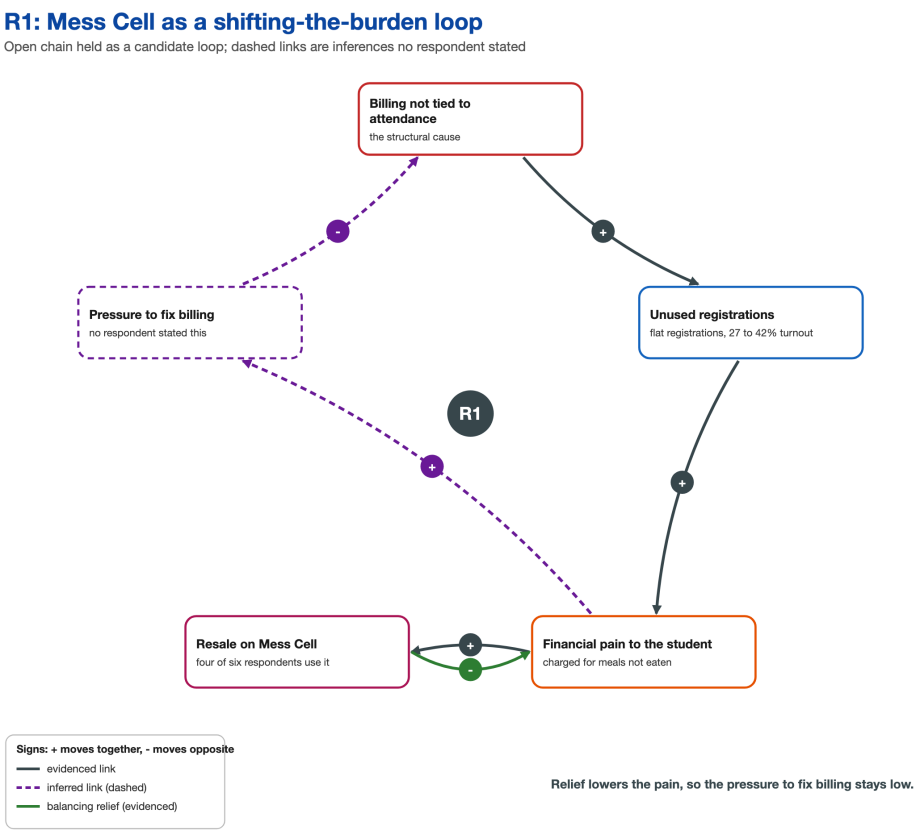

Figure 5.32. R1: Mess Cell as a shifting-the-burden loop; dashed links are unstated inferences. Source: student interviews; authors' analysis.

The kitchen loop, B1, closes, but each hall closes it with a different delay (Figure 5.33). Yuktāhār's site head, seeing 18.7% on Sunday 6 September, said the quantity would be cut "not next month, next week" (confirmed). Kadamba tops up in ten to fifteen minutes and nothing shrinks the next first batch. The dashed amber link is Skip Meal: Kadamba's operator says it cooks on registrations, the institution that declarations reach the kitchen, and the conflict stays open.

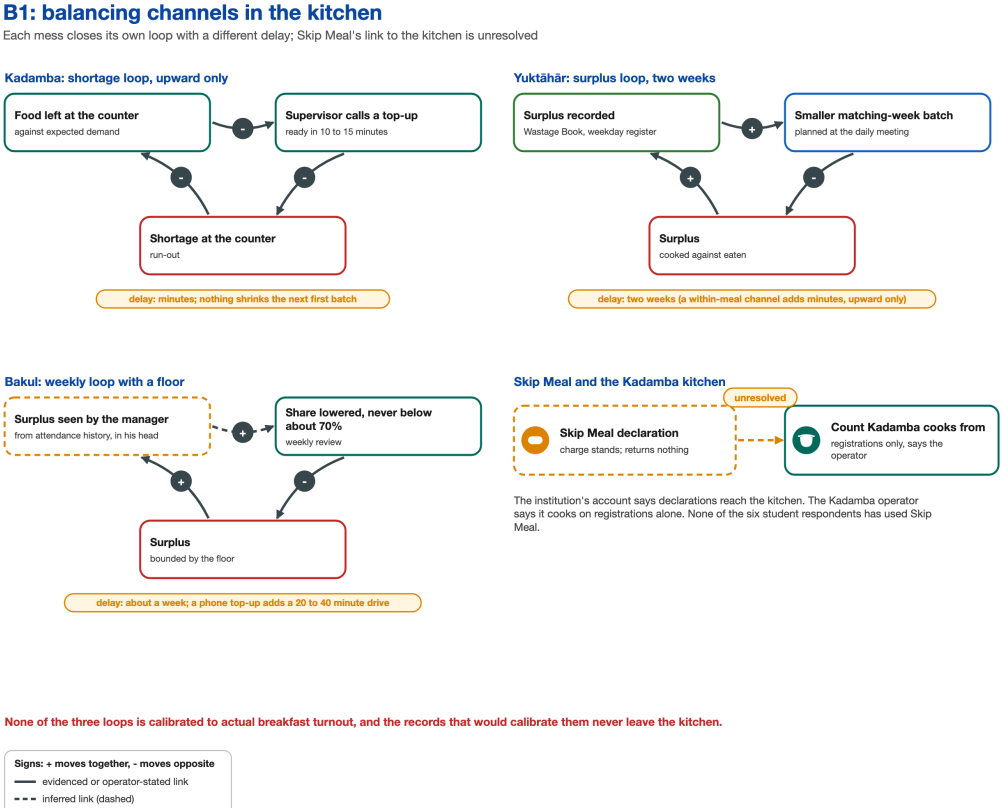

Figure 5.33. B1: balancing channels in the kitchen; the Skip Meal link is unresolved (dashed amber). Source: operator interviews and Yuktāhār's books.

The same meal fails twice. Kadamba's first batch, about 750 portions a day in April, is roughly double the about 398 eaten (candidate), yet food still runs short at the close.

Decisions and revisions

- We froze the evidence at a stated date and listed what is missing, such as Bakul records and other months.
- We trusted records over recall. Two students had said only 30% turn up, a figure the records contradict.
- On 27 September the measured curve replaced the one drawn from recall, and "no rule change" became "no rule change, several operator-side changes".
- We kept a loop as an open chain whenever one link is only inferred (R1, same meal two failures).

Synthesis of Phase 1

The five activities describe one system from five sides, and the records measured what the others could only describe. Nine findings follow.

1. Registration is a default and not a demand signal: automatic booking, a narrow early exit, billing at registration, a Skip Meal that refunds nothing and a walk-in premium all reward staying booked.
2. Every kitchen guesses its own conversion, and every guess is too high.
3. The gap is measured twice, in scan records and operators' books, and neither reaches the owner of the registration and billing rules.
4. The binding constraint is the cooked first batch, not procurement.
5. Shortage and waste happen on the same morning, because a late crest meets a fixed first batch.
6. Academic administration is the clearest sleeping stakeholder: it holds the class start time and has no link to mess governance.
7. Oversight and feedback change kitchens, not rules.
8. Surplus is absorbed out of sight, partly by real people, so any cut to the first batch must plan for them.
9. The fix respondents requested most was a longer close, with pay-as-you-go registration proposed as a structural option.

Two gaps stay open: disposal after close is unobserved, and semester-level trends are unmeasured. Phase 2 turns these findings into a model of the system.

Phase 2: Modelling the System

Phase 2 turned the evidence into a model of the system. Its central finding is that the breakfast gap is a stable equilibrium: measured every day, absorbed before anyone with authority sees it, and never routed to the people who set the rules.

Activity 6: Developing the System Map and Causal Loop Analysis

Activity 6 asked for one picture of the system's actors, processes, structures, bottlenecks and feedback loops, and produced the System Map and the Causal Loop and Feedback Analysis. The map puts the binding constraint at the cooked first batch. The loop test shows that every working loop corrects the kitchen and none touches the booking.

The system map

The map runs from who decides, through how a breakfast happens and what it runs on, to what holds it in place (Figure 6.1). The bottom band carries the finding: five constraints, from the four-day exit to records that never leave the kitchen. Operators appear as three, not one box, and Kadamba also feeds 30 to 50 outsiders and about half its staff (single-sourced).

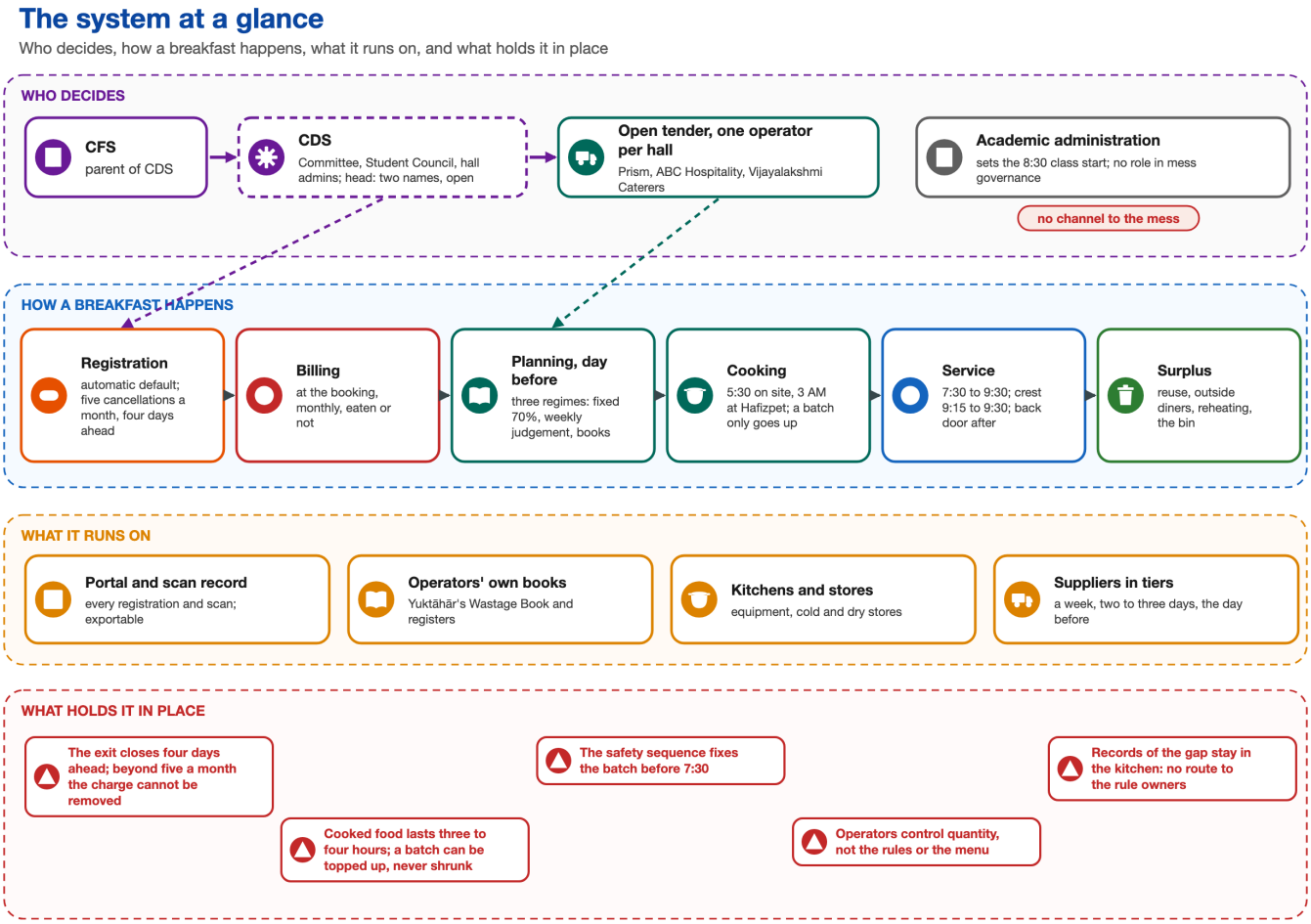

Figure 6.1. The system at a glance: who decides, how a breakfast happens, what it runs on, and what holds it in place. Source: authors' analysis of interviews, records and posters.

Every first batch sits above turnout, and only the solid bars are records (Figure 6.2). Kadamba's 70% batch is about 1.9 times the 37.1% who eat, and 2.6 times on a Sunday (candidate arithmetic). The four-day lock applies to registration, not to buying, so what binds waste is the cooked first batch, which can be topped up but never shrunk.

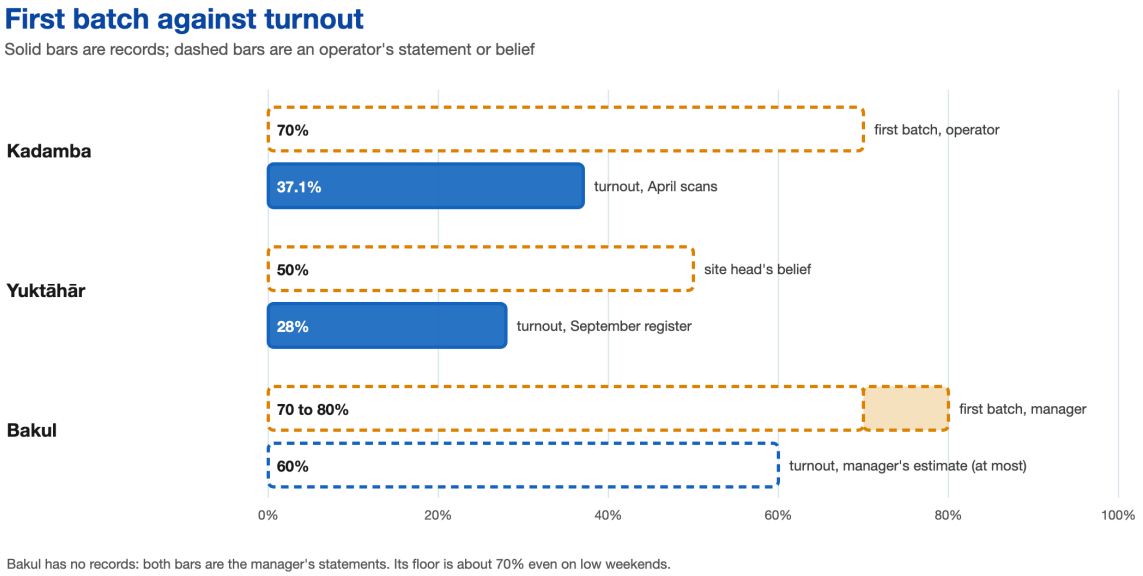

Figure 6.2. First batch against turnout at each hall; solid bars are records. Source: Kadamba scan records, April 2026; Yuktāhār registers, September 2026; operator statements (single-sourced).

The halls hold similar data and differ in learning. Only Yuktāhār feeds the outcome back into the next decision, through paper books read at a twice-daily planning meeting. That chain stops at the kitchen door: nothing reaches CDS, the Mess Office or the menu owner (Figure 6.3). The books reconcile materials, not diners: for 3 September the Wastage Book records 77 breakfasts and the attendance register 110 (confirmed).

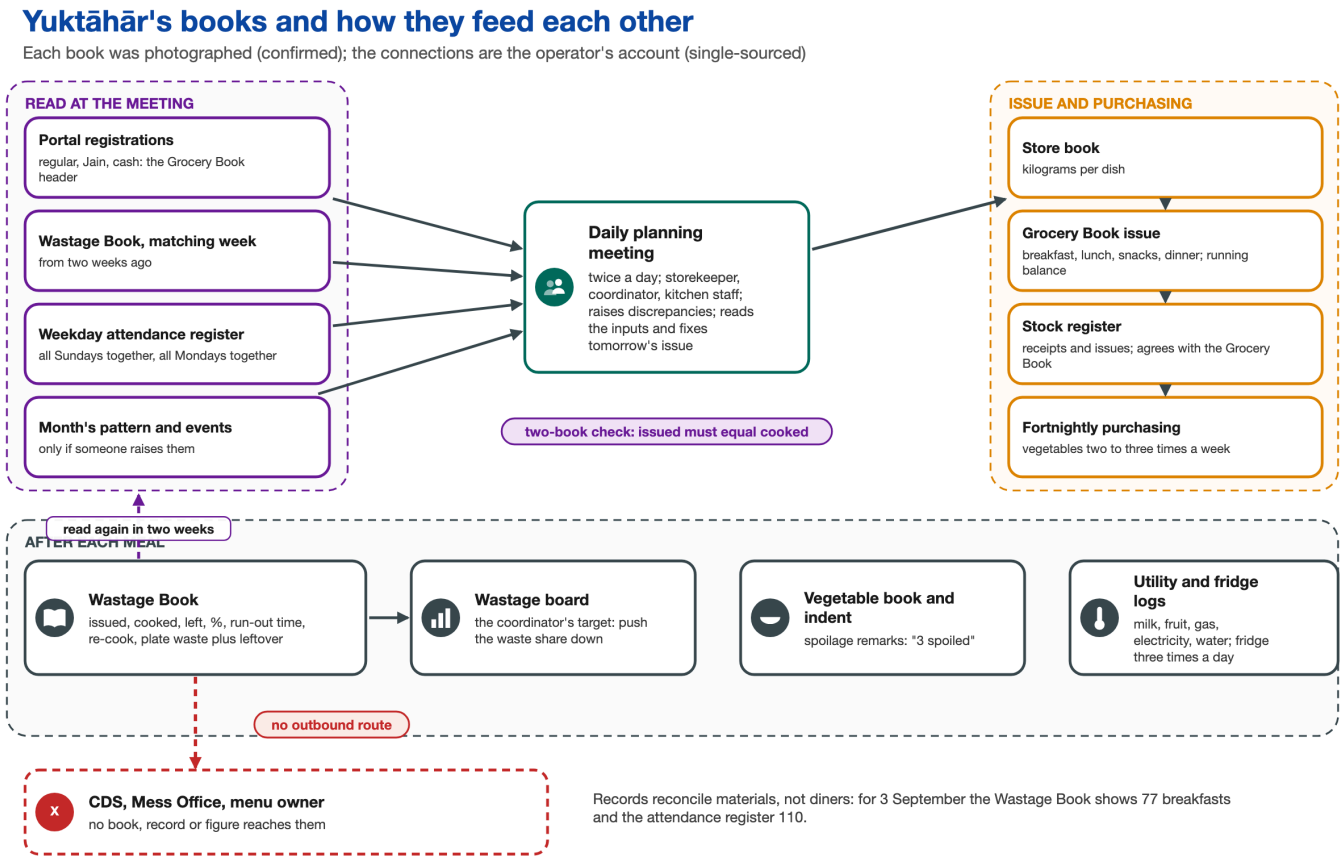

Figure 6.3. How Yuktāhār's books feed each other, and where the chain stops. Source: photographed books (confirmed); connections from the operator's account (single-sourced).

Surplus, service and governance

Surplus is absorbed before the rule-setters see it: reuse, outside diners and staff, reheating (candidate), latecomers, and finally a contractor's bin. Each absorber does real good, so no fix should remove one. Nobody has measured the cost of the 20,221 Kadamba breakfasts registered and never eaten in April.

The service blueprint splits what the student sees from what the kitchen does (Figure 6.4). The kitchen reacts within a meal, and only upward. One artefact crosses the line of visibility, the display plate, and it shows the menu, never quantity. The Kadamba operator's own picture overstates the final rush: the records put 19.8% of scans in the last ten minutes, against about 43% in his example (candidate).

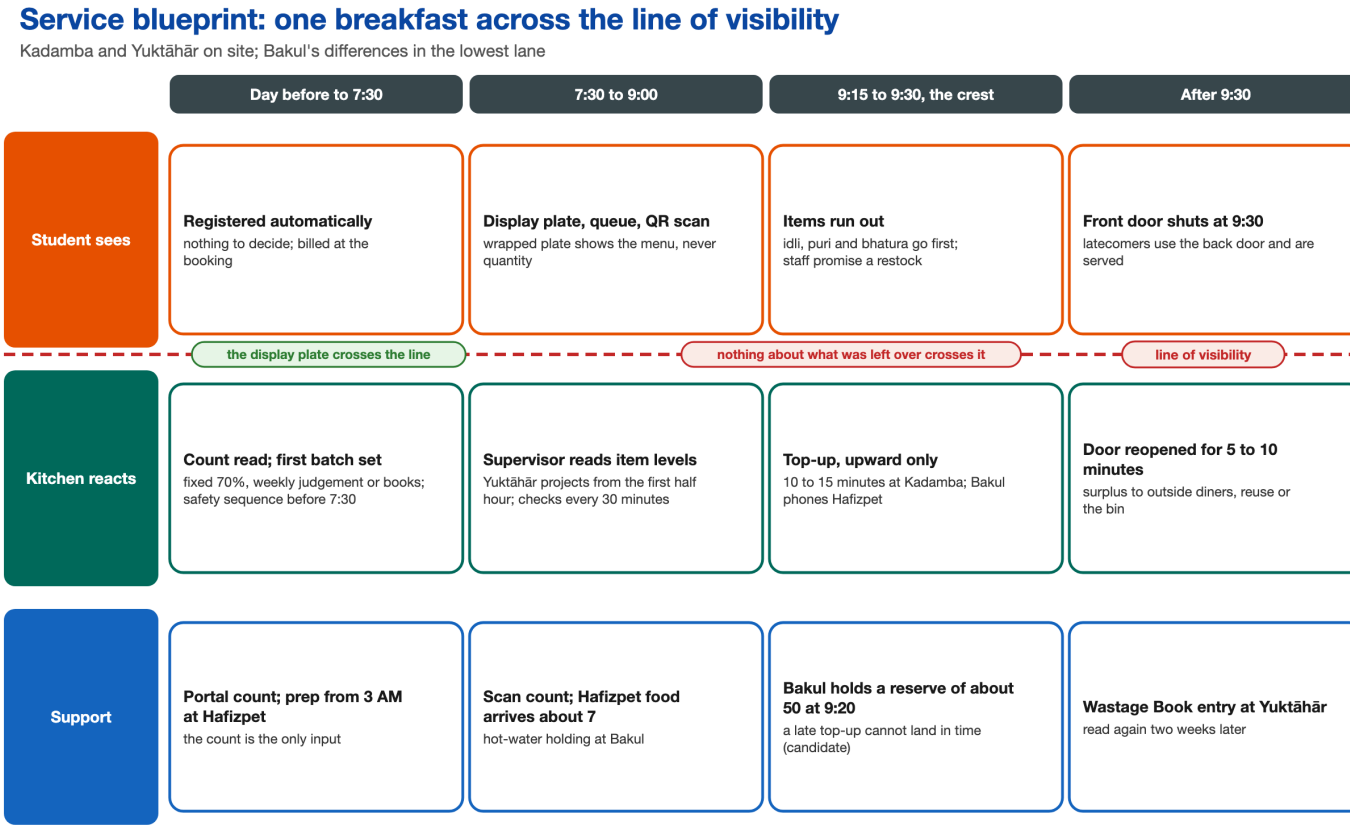

Figure 6.4. Service blueprint: what the student sees, what the kitchen does, and the one signal that crosses the line. Source: authors' analysis; Bakul differences from the manager's account (single-sourced).

Governance offers three routes (Figure 6.5): a complaint route that fixes incidents on the day, an escalation route that produced the menu rotation, and no route for changing rules. Who heads CDS is unresolved, with two names on record. Operators see which items waste (uttapam, set dosa and upma at Kadamba) and have no route to the menu owner. Academic administration has no link to mess governance, a confirmed absence.

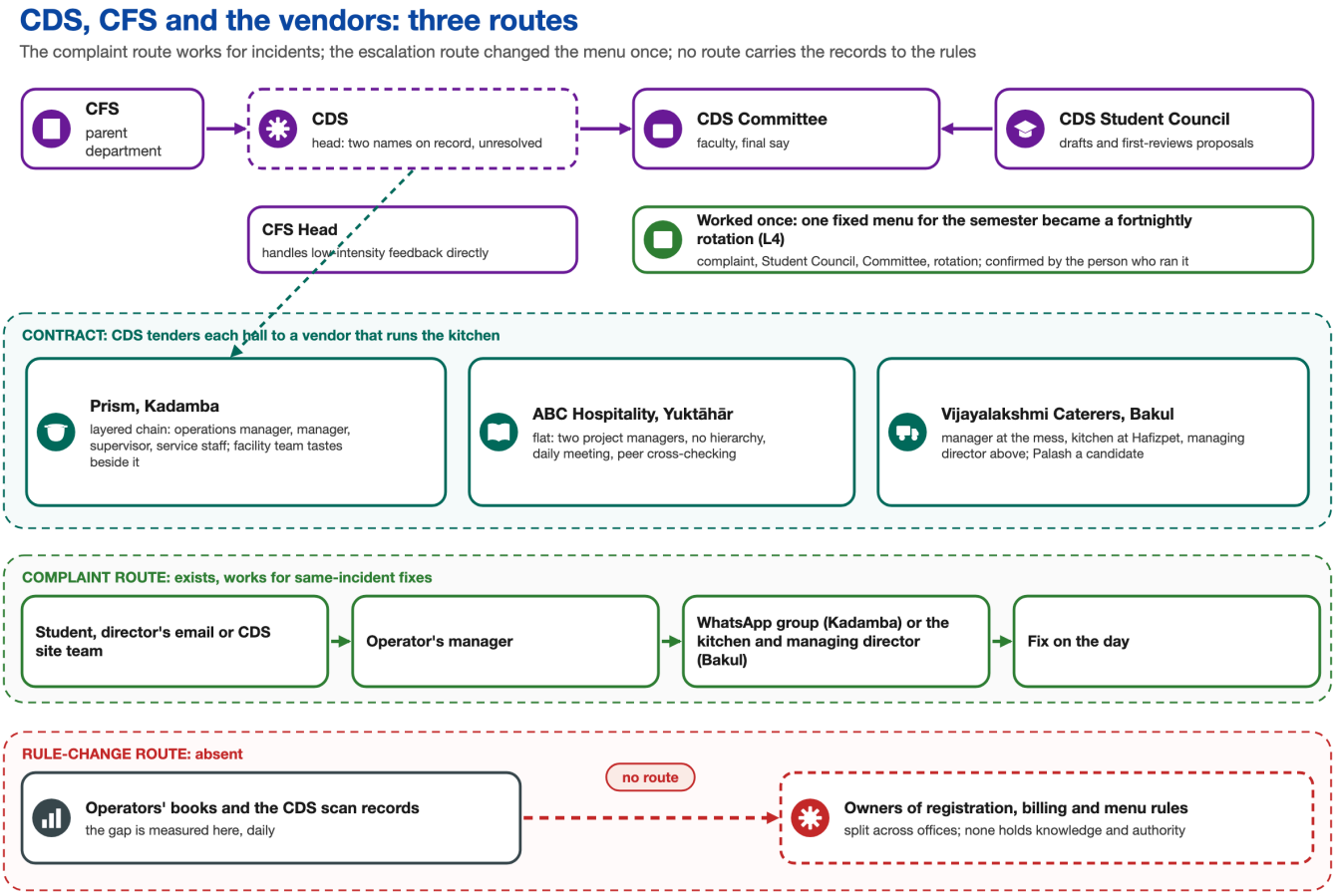

Figure 6.5. CDS, CFS and the vendors: a complaint route that works for incidents, an escalation route that produced the menu rotation, and no rule-change route. Source: authors' analysis of interviews and posters.

We had treated the 8:30 class start as an obstacle, which is probably unfair: the timing may suit harder lectures or let each programme fit five days on shared halls.

The system already solved this once

During a cooking-gas shortage the mess cut to one item at the same price and the cancellation limit rose from five to ten a month. In holidays and long weekends such as the campus festival, registration stopped being required: students walked in and scanned, and nobody who neither registered nor came was charged (single-sourced). The holiday part is disputed, since our own account is that students stay registered through holidays. If it holds, "nobody has worked this out" becomes "this was worked out and switched off".

Bottlenecks

The bottleneck is the cooked first batch: a ratio of a standing booking, set against a crowd that at Kadamba crests in the last fifteen minutes (Figure 6.6). Each link has a lock beneath it, from the four-day exit to safety checks that commit the batch before 7:30. The chain ends in shortage and surplus on the same morning, and the record of the gap stays in the kitchen.

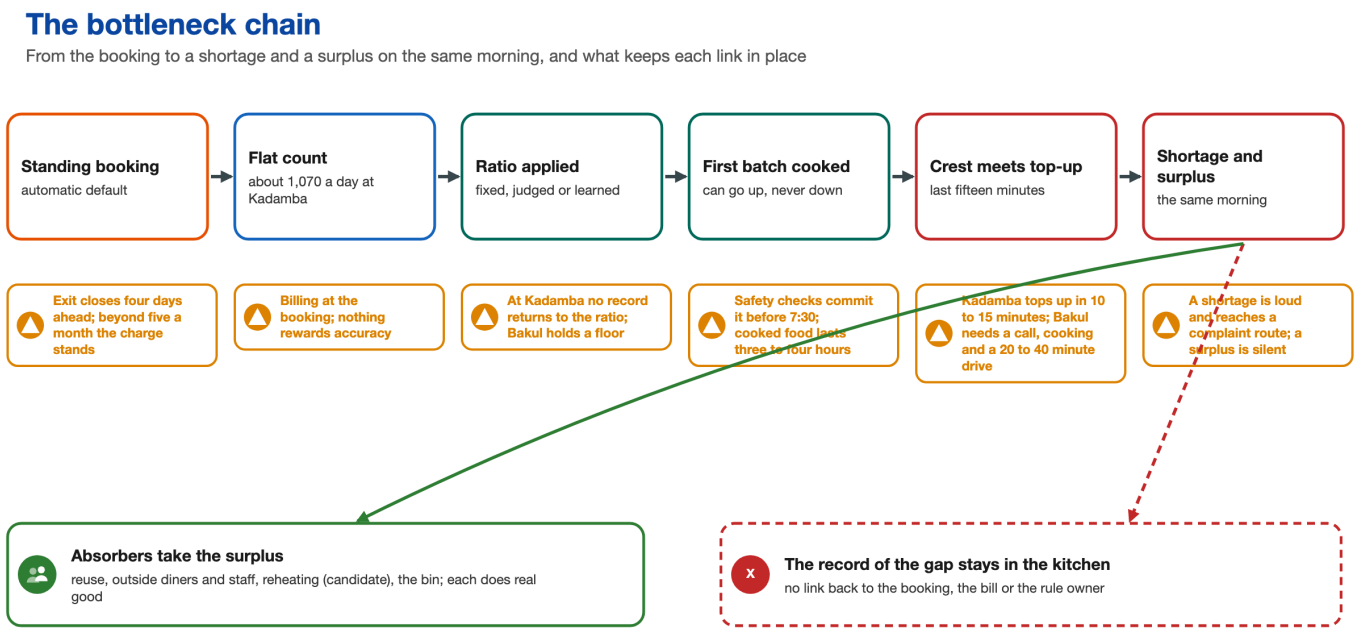

Figure 6.6. The bottleneck chain from the booking to the same-morning shortage and surplus, with what keeps each link in place. Source: authors' analysis.

The closure test and the loop picture

A causal loop exists only if the last variable in a chain causes the first; an odd number of negative links balances and an even number reinforces. We asked of each mechanism whether every link is evidenced, whether the last really causes the first, and what the delay is. We tested eighteen: L1 to L17 plus L3b (Figure 6.7).

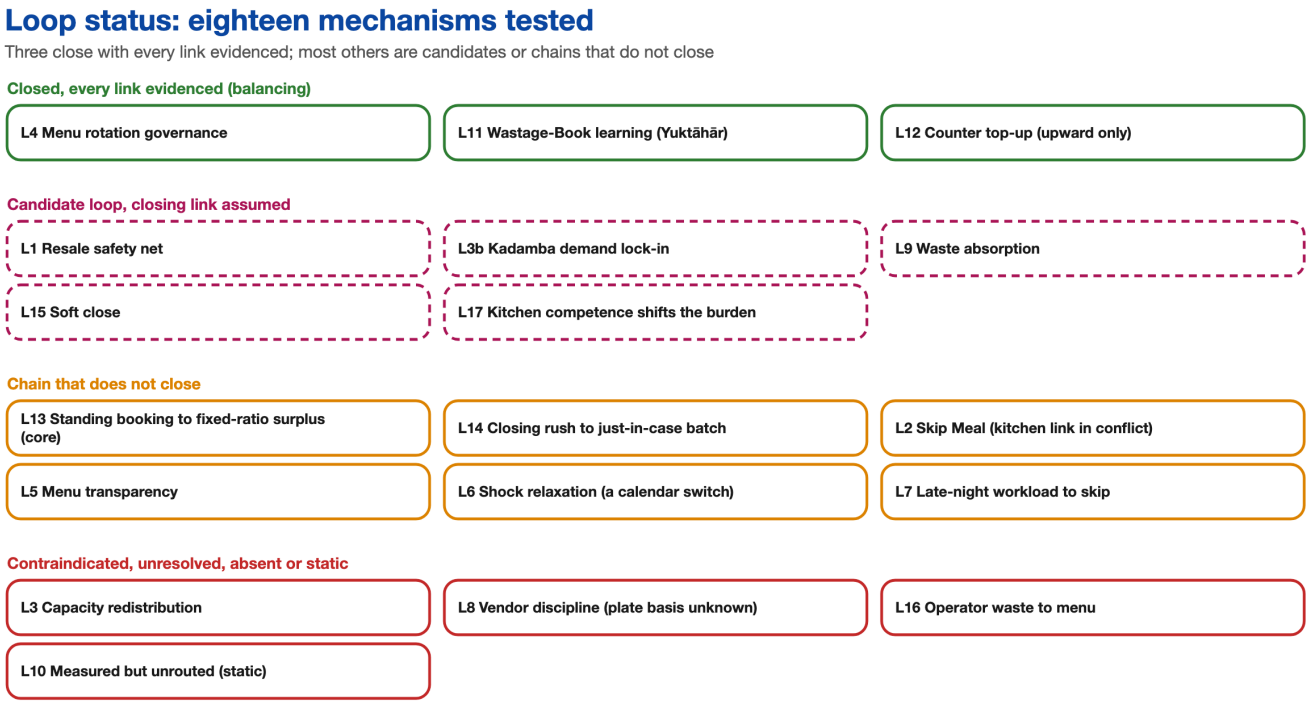

Figure 6.7. The eighteen mechanisms by status. Source: authors' analysis; closure test applied to each mechanism.

ID	What it is	Status
L4	Menu rotation governance	Closes: balancing, confirmed
L11	Waste-learning through the Wastage Book	Closes: balancing; confirmed at Yuktāhār, absent at Kadamba, candidate at Bakul
L12	Counter top-up within a meal	Closes: balancing, confirmed; upward only
L1	Speculative registration and resale	Candidate (reinforcing)

ID	What it is	Status
L3b	Kadamba demand lock-in	Candidate (reinforcing)
L9	Waste absorption	Candidate (reinforcing)
L15	Soft close	Candidate (reinforcing)
L17	Kitchen competence shifts the burden	Candidate
L13	Standing booking to fixed-ratio surplus, the core	Open chain; quality branch open
L14	Closing rush to just-in-case batch	Open chain; candidate ratchet
L2	Skip Meal	Open chain; kitchen link in conflict
L5	Menu transparency	Open chain
L6	Shock-adaptive relaxation	Open chain; a calendar switch
L7	Late-night workload to skipping	Open chain; first link confirmed
L3	Capacity redistribution	Disconfirmed: strongly contraindicated
L8	Vendor quality discipline	Unresolved; depends on the plate basis
L16	Operator waste to menu choice	Absent
L10	Measured-but-unrouted gap	Static condition

Three mechanisms close with every link evidenced, all balancing: L4, L11 and L12. Every working loop corrects the kitchen's conversion of bookings into food; none touches the booking.

Loops that close

L11 is Yuktāhār's Wastage Book loop: a larger leftover lowers the next matching day's issue (Figure 6.8). It exists because weighing was made blame-free after staff feared blame ("they are catching us"), and by December staff reported lower waste (self-report). It adjusts quantity, never registration.

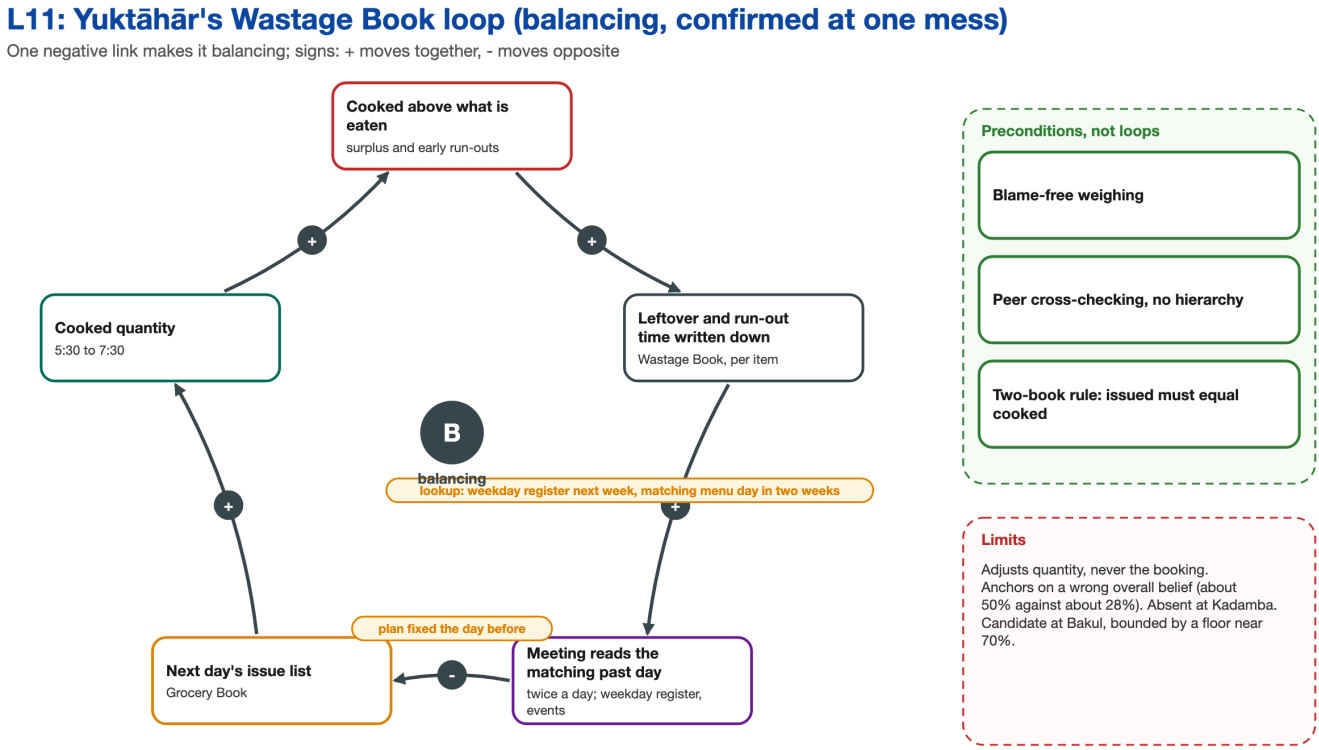

Figure 6.8. L11: Yuktāhār's Wastage Book loop, its two delays, and the preconditions that keep the record honest. Source: Yuktāhār registers (confirmed); operator account.

L12, the top-up, runs at all three halls, and what differs is the delay (Figure 6.9). At Kadamba a top-up takes 10 to 15 minutes (single-sourced), so one asked for at the final surge lands at or after the close. Off site none can land late, so Bakul's manager holds food for about 50 at 9:20: as a candidate reading, an insurance premium paid in surplus.

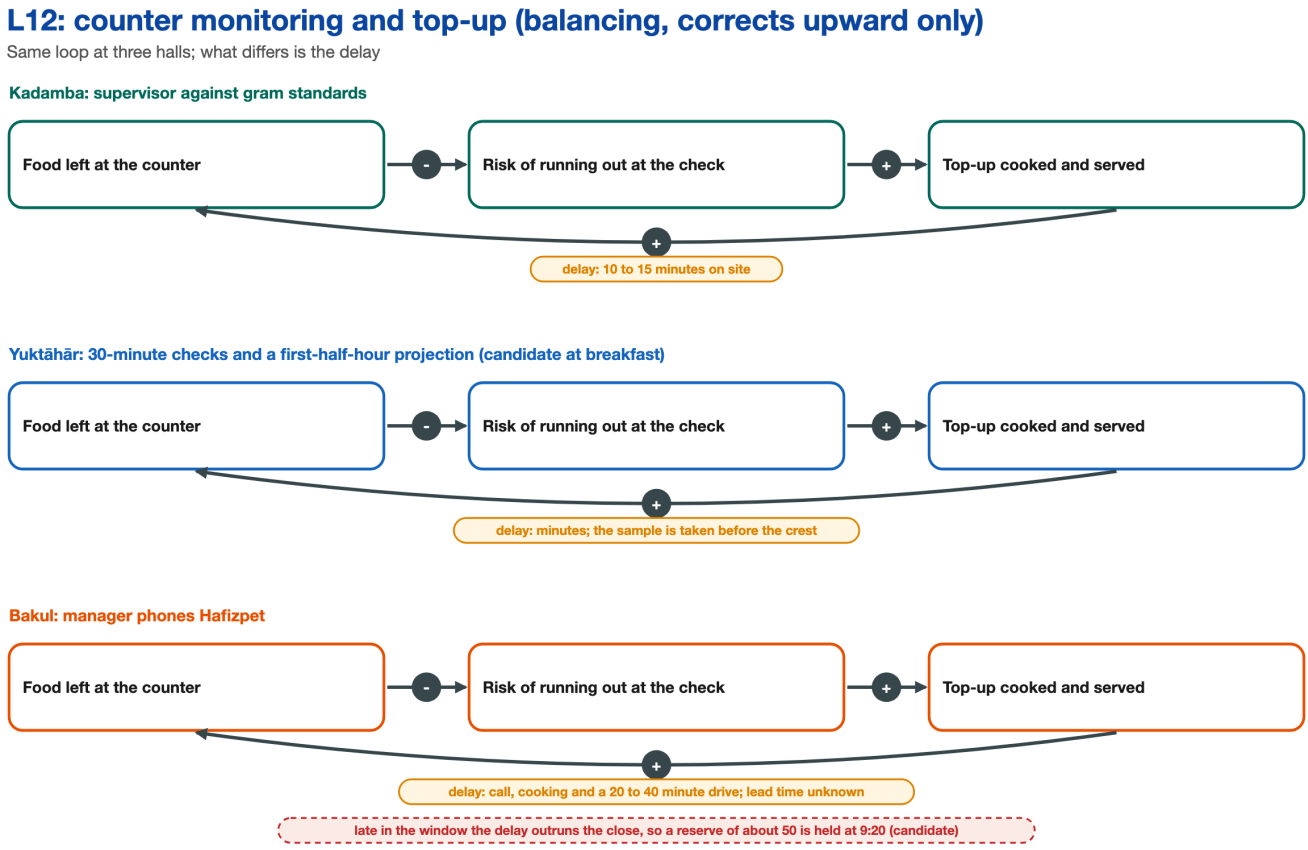

Figure 6.9. L12: top-up at three halls, and the delay that decides how well each works. Source: operator accounts (single-sourced); Yuktāhār registers (confirmed).

L4 carried menu fatigue through the Student Council and Student Parliament to a biweekly rotation approved by the CDS Committee, and operators' records confirm the cycle (Figure 6.10). A loud complaint with a route can close a loop end to end; that route has never carried the booking.

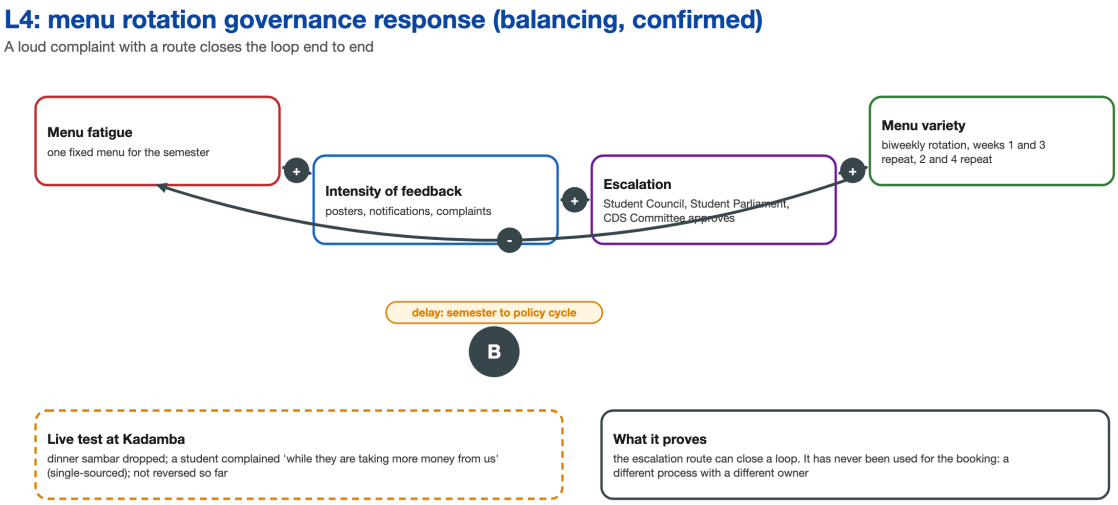

Figure 6.10. L4: the menu-rotation loop, closed end to end by a loud complaint with a route. Source: facilities head's account and operators' records.

Candidate loops and open chains

Five candidate loops would close if their last link held: resale (L1), waste absorption (L9), Kadamba demand lock-in (L3b), a soft close (L15) and kitchen competence shifting the burden (L17). At Kadamba a first batch of about 750 against about 398 eaten a day leaves roughly 350 portions before any top-up (candidate arithmetic), and 9.2% of April scans came after 9:30 (confirmed).

In L17 the dashed link is the assumed one: relief each morning plausibly lowers pressure for the fundamental fix, a booking that tracks the daily decision to eat (Figure 6.11). The shift is partly benign, since outsiders and staff eat the surplus.

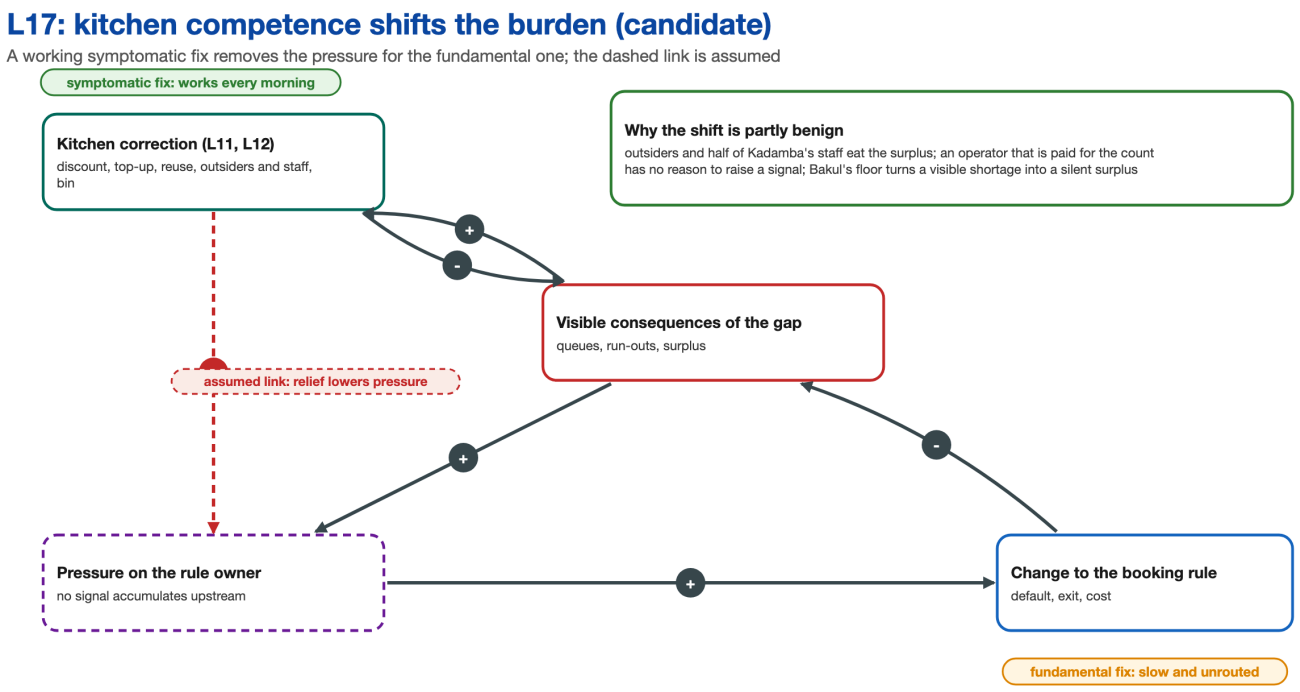

Figure 6.11. L17: kitchen competence relieves the symptom each morning, so the fundamental fix never gets pressure. Source: authors' analysis; the dashed link is assumed.

L13 is the core chain (Figure 6.12). Registration is automatic, with five cancellations per meal type per month, each at least four days ahead (confirmed), so the count stays flat at 928 to 1,123 a day at Kadamba in April while turnout swings. Nothing runs back from the surplus to the booking. The quality branch is dashed because more skipping after reheated food is not evidenced.

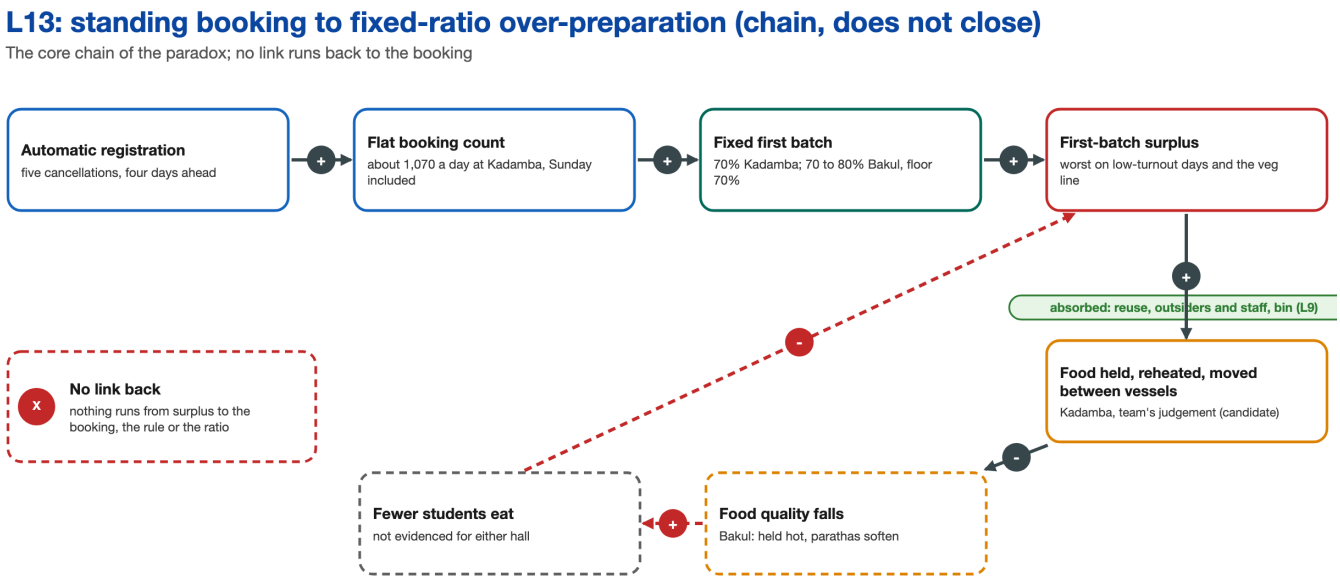

Figure 6.12. L13: the core chain. A flat booking meets a fixed ratio; the quality branch is dashed because its closing link is not evidenced. Source: authors' analysis of Kadamba scan records and operator accounts.

L14 runs from the closing rush, about nine scans a minute at 9:29 (confirmed), to a just-in-case batch (Figure 6.13). Shortage is loud and surplus silent, so an operator that raises its ratio after run-outs and never lowers it would ratchet upward. No change of ratio is on record, so this is an asymmetry of intent, not an observed drift.

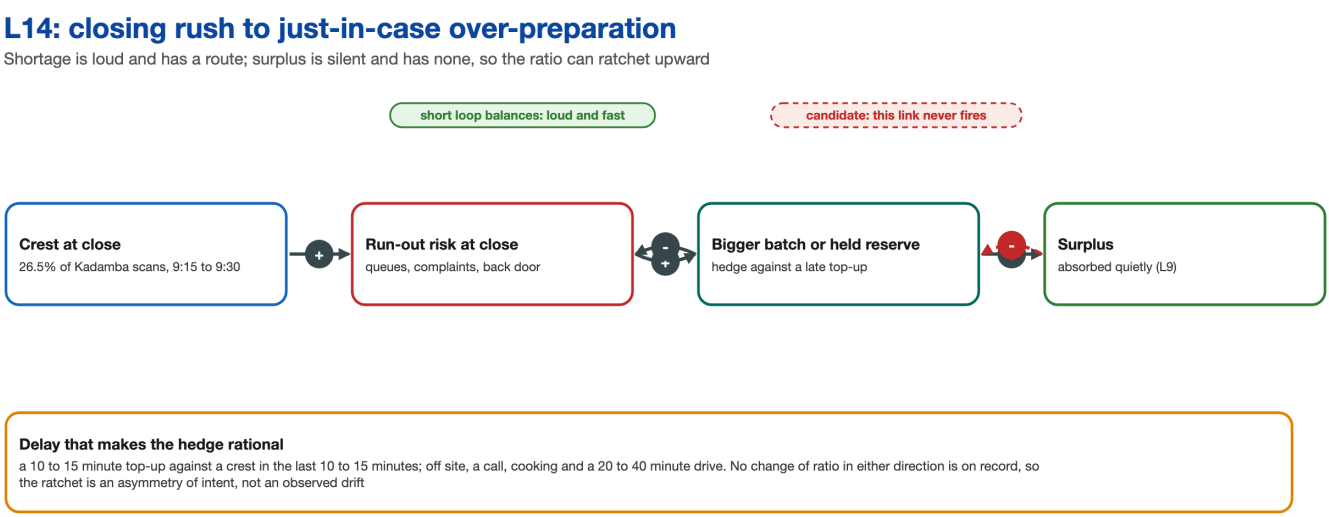

Figure 6.13. L14: shortage is loud and surplus is silent, so the ratio can ratchet upward; the dashed link is a candidate. Source: authors' analysis; arrivals from Kadamba scan records.

L2, Skip Meal, is near zero at the student end: four of six respondents who addressed it had never used it, one because "it doesn't return the money." Its kitchen link is in open conflict (Figure 6.14): the CFS side says declarations reach the kitchen, and Kadamba's operator says it cooks on registrations alone (single-sourced).

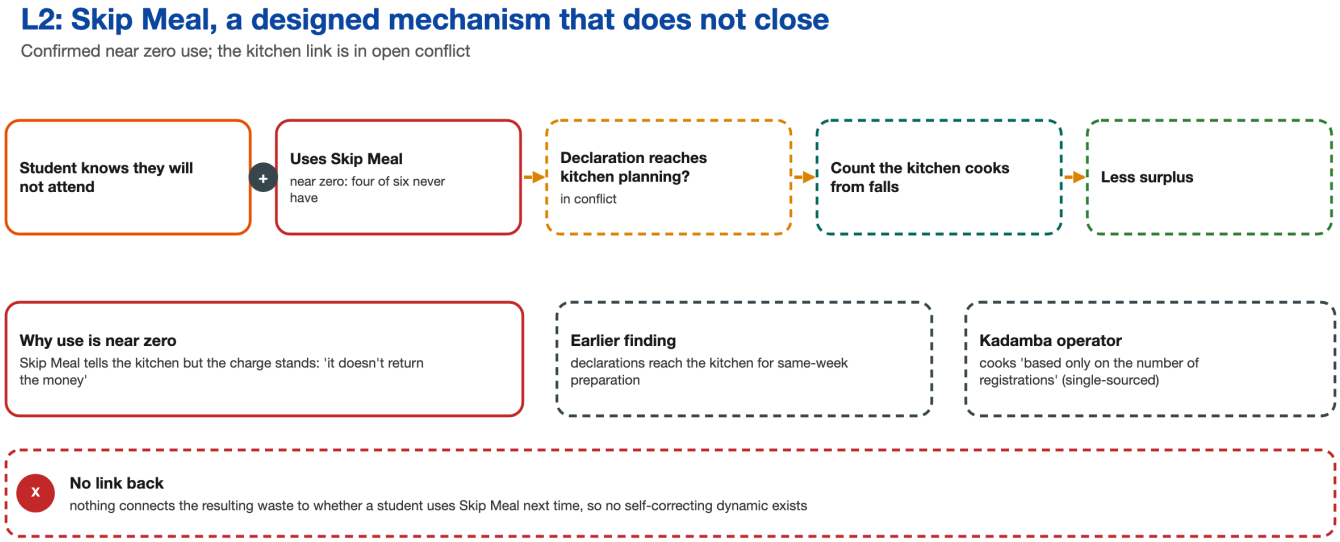

Figure 6.14. L2: Skip Meal. The student link is confirmed and near zero; the kitchen link is in open conflict. Source: student interviews; Kadamba operator's account (single-sourced).

Of the rest, L3 is strongly contraindicated, L8 is unresolved because no contract has been seen, and L16 is absent: operators hold the waste data and the committee holds the menu. L10 is static. The gap is measured in at least three places, including 62.9% no-shows in Kadamba's scan export (confirmed), and the failure is a missing route.

Decisions and revisions

- We added a service blueprint because it showed what crosses the line of visibility, and set aside a rich picture as notation without a finding.
- We drew a loop only where its closing link had direct evidence. Three earlier loops became open chains and one a calendar switch, and the L1 to L17 scheme replaced the first map's R1 and B1 to B4.
- We replaced the four-day procurement lock with the cooked first batch as the binding constraint, since operators buy on their own schedules.
- We removed composting as unverified, since the facilities head's account conflicts with the operator's first-hand one, and dropped the claim that the gap is invisible.
- We rebuilt the early drafts, grouped by users, technology, business and social factors, as a feedback system after the facilities head's interview and the Kadamba walkthrough.

Activity 7: Identifying Systemic Patterns and Root Causes

Activity 7 moved from symptoms to the structures and beliefs beneath them, and produced the Systemic Problem Analysis (Deliverable 3 of Phase 2). It found three structural causes, two conditions that keep them in place, and one actionable mechanism.

Seven chains through the iceberg

We traced seven areas from a dated event down to the belief that keeps it standing, each layer sourced and labelled (Figure 6.15). Two observations recur: billing that followed attendance was run and switched off, and the gap is measured twice and reaches nobody who can change the rules.

Figure 6.15. The seven chains at a glance: a visible event in each traced down to the belief that holds it in place. Source: authors' analysis of interviews, records and field visits.

Chain	Event	Structure or belief beneath it
1. Registered, not showing up	Sunday 6 September: 267 registered at Yuktāhār, 50 ate	Billing counts an automatic booking; registrations stand in for demand
2. Food runs out before the close	Idli, puri and bhatura gone at Kadamba around 9:20	First batch fixed before any turnout signal; top-up only upward
3. Complaints ignored until forced	Cancellation cap tightened in December 2024 without recorded committee sign-off	No confirmed rule requires committee approval; consultation seen as optional
4. A working fix switched off	Gas shortage: one item, cap doubled from five to ten	An automatic count valued over matching demand (candidate)
5. A real fix and its shadow	A fortnightly rotation replaces one semester-long menu	A complaint path that worked; a known menu also enables skipping
6. Not a problem at all	A student fasts on purpose	The default registers everyone, so a fast counts as a no-show
7. Cooking to a guess	Shown his register, the Yuktāhār site head still says "about 50%"	Three operator stances under one booking rule

Chain 4 is the counterexample to Chain 1 (Figure 6.16). By the CFS Chair's account the relaxed model eased the no-show problem every time and was switched off when the trigger passed, with no review; no attendance record has been seen.

Why it was dropped is unchecked: per-plate payment may count registrations, a default is less work, or the model reverts with the calendar.

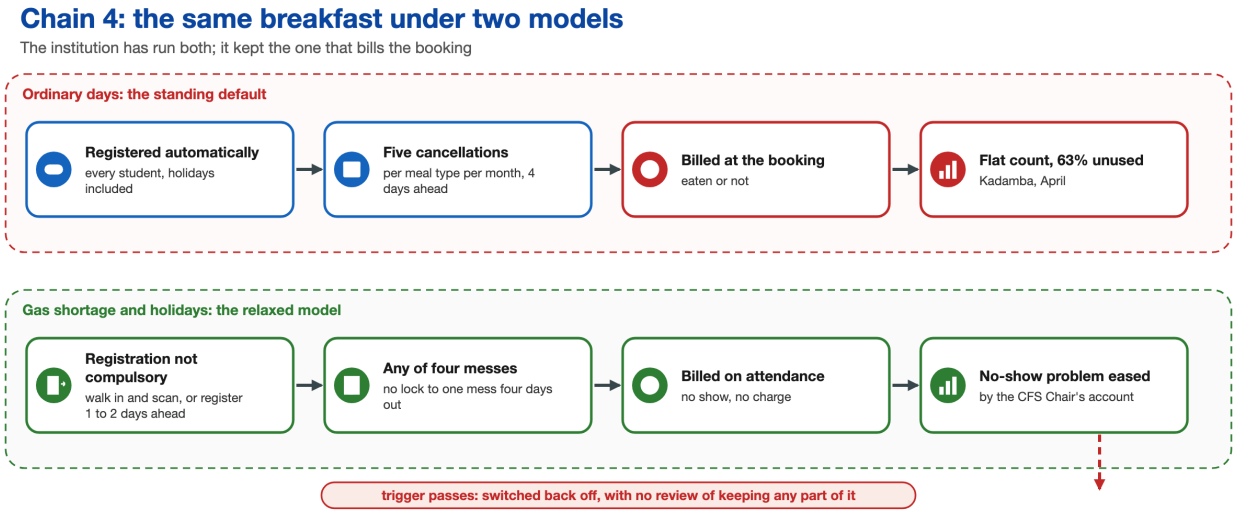

Figure 6.16. The same breakfast under the standing default and under the relaxed shock model. Source: authors' analysis; the relaxed model is single-sourced.

Chain 6 is kept on purpose: late sleep and fasting are real, and too small to explain the gap. Chain 7 finds the gap about as large under opposite kitchen philosophies, so calibration cuts waste and run-outs and leaves no-shows untouched.

Three structural causes

Three structural causes recur and split the problem in two: students not showing up, and kitchens cooking more than is eaten (Figure 6.17).

- Registration is an automatic standing default with a narrow, early exit, so it cannot follow the morning decision. Lunch runs under the same booking and mostly works.
- Billing fires at the booking, not at consumption. This explains the broad booking, the unused Skip Meal and Mess Cell, but not why kitchens cook above turnout.
- The gap is measured but unrouted: scans and Yuktāhār's books reach no rule owner, so a fix that worked twice was never evaluated.

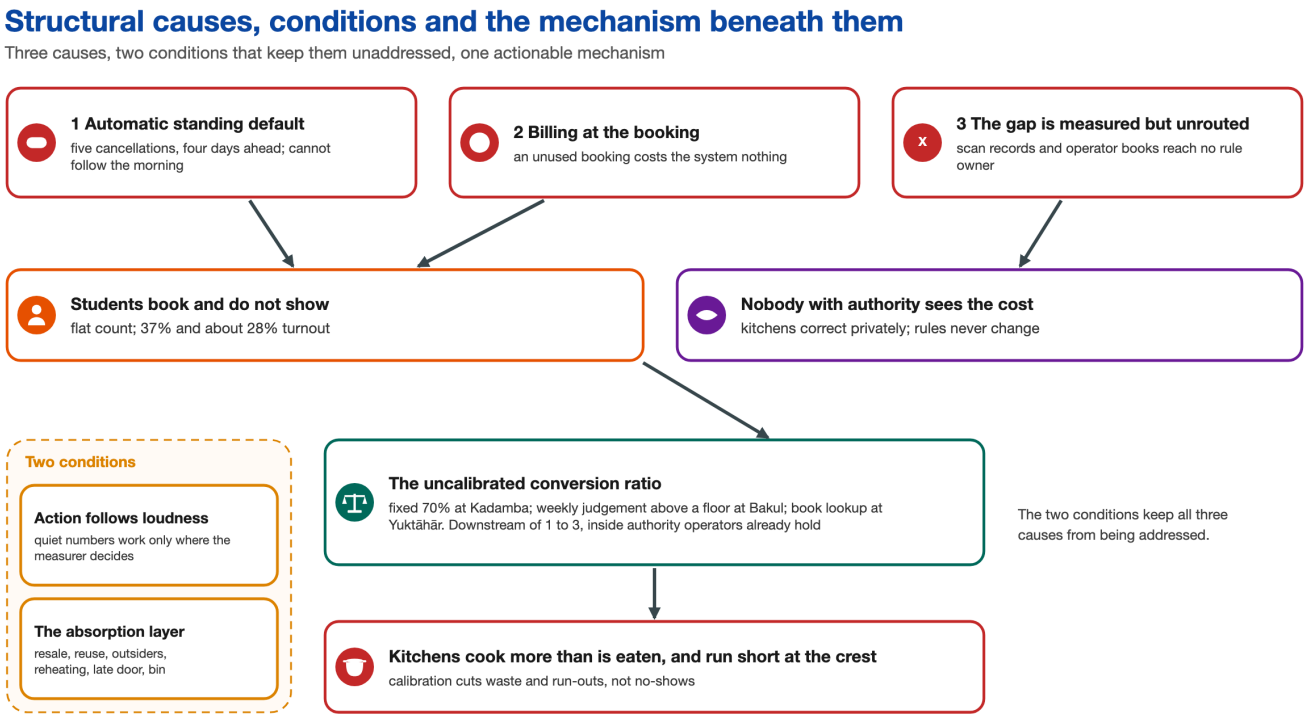

Figure 6.17. Three structural causes, the two conditions that keep them unaddressed, and the conversion ratio beneath them. Source: authors' analysis.

Two conditions keep all three in place (the dashed box in Figure 6.17). Action follows loudness: menu fatigue was loud and got fixed, while the larger registration gap is quiet. An absorption layer takes up each consequence before anyone with authority sees it.

Beneath the kitchen half sits the uncalibrated conversion ratio, in three regimes: a fixed 70% at Kadamba, 70 to 80% reset weekly above a floor at Bakul, and a book lookup at Yuktāhār (Figure 6.18). It is downstream of the booking, so not a root cause, but it is the most actionable structure because it sits inside authority operators already hold. What a plate counts for in payment is contested: the Bakul manager says plates served (single-sourced), Kadamba's operator says registrations (candidate). No contract has been seen.

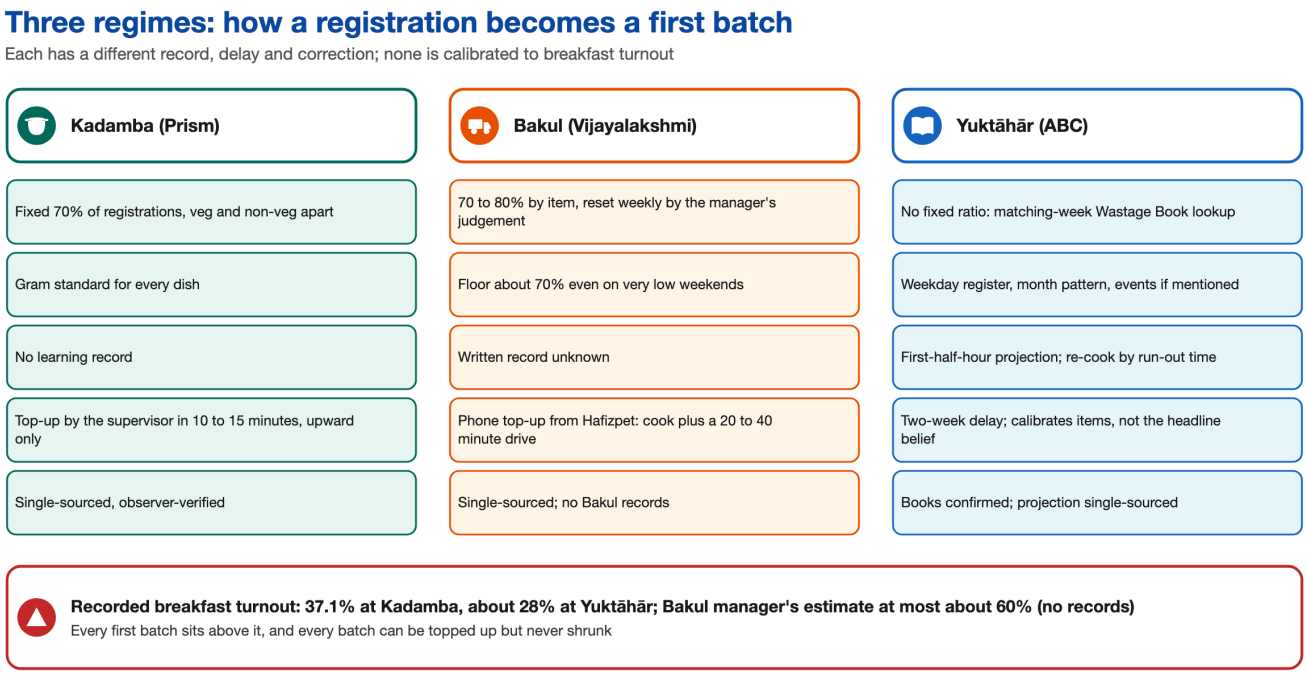

Figure 6.18. The three regimes that turn a registration into a first batch. Source: operator interviews and Yuktāhār's books; Bakul figures are the manager's statements.

Archetypes

Of eleven archetypes checked, five hold up. Shifting the burden appears twice: Mess Cell resale stands in for fixing billing, and reuse, outsiders, held food and the bin stand in for calibrating the batch (Figure 6.19). Growth hitting a ceiling fits Kadamba's popularity, with Hafizpet's kitchen a candidate ceiling for Bakul and Palash. Success to the successful fits Kadamba's scarcity feeding defensive registration while Bakul stays underfilled. Fixes that fail stays a candidate, on second-hand accounts of the December 2024 tightening.

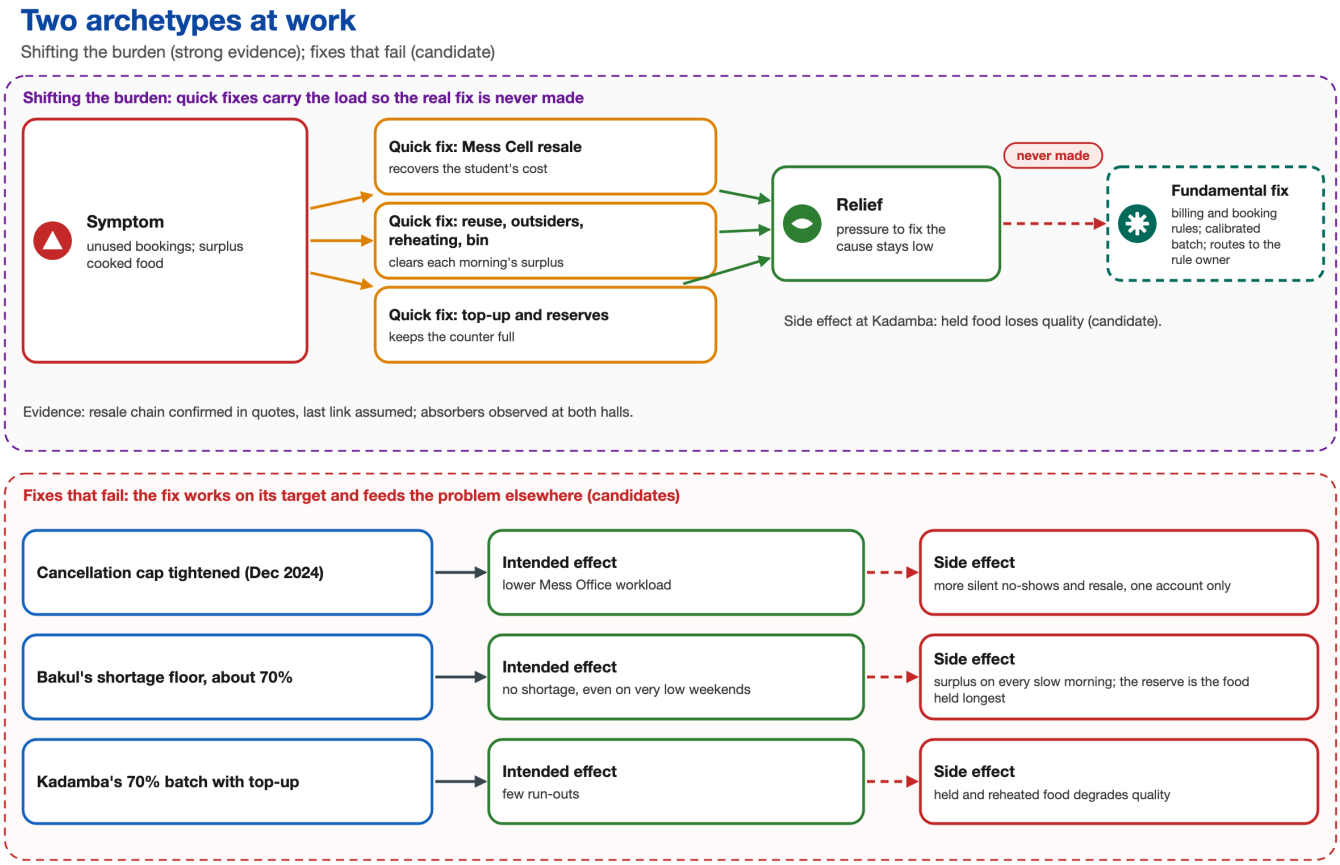

Figure 6.19. Two archetypes at work: shifting the burden, and fixes that fail. Source: authors' analysis; fixes that fail rests on second-hand accounts (candidate).

Tensions, power and workarounds

Eleven tensions pull both ways at once (Figure 6.20). Registering broadly and reselling is individually smart and, summed, is the waste; at Bakul a shortage has a route to the managing director and a surplus has none. Power is lopsided: students carry nearly all the visible cost and control almost none of the rules, and operators are data-rich and authority-poor.

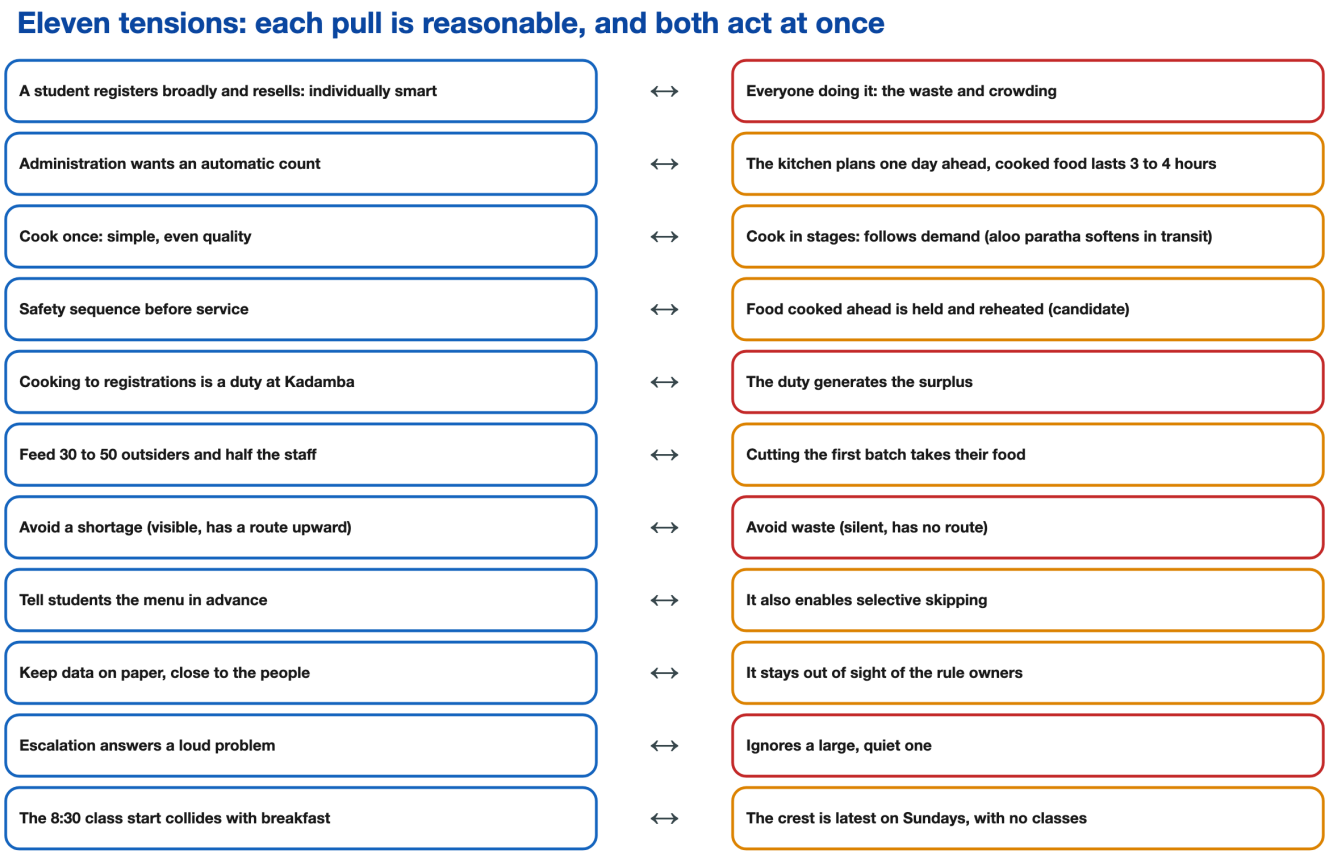

Figure 6.20. Eleven tensions: each pull is reasonable, and both operate at once. Source: authors' analysis.

Every workaround absorbs a cost that would otherwise force a fix (Figure 6.21). Mess Cell absorbs the pain of over-registering, so pressure on billing never builds, and a resold meal probably still shows as eaten by the original registrant.

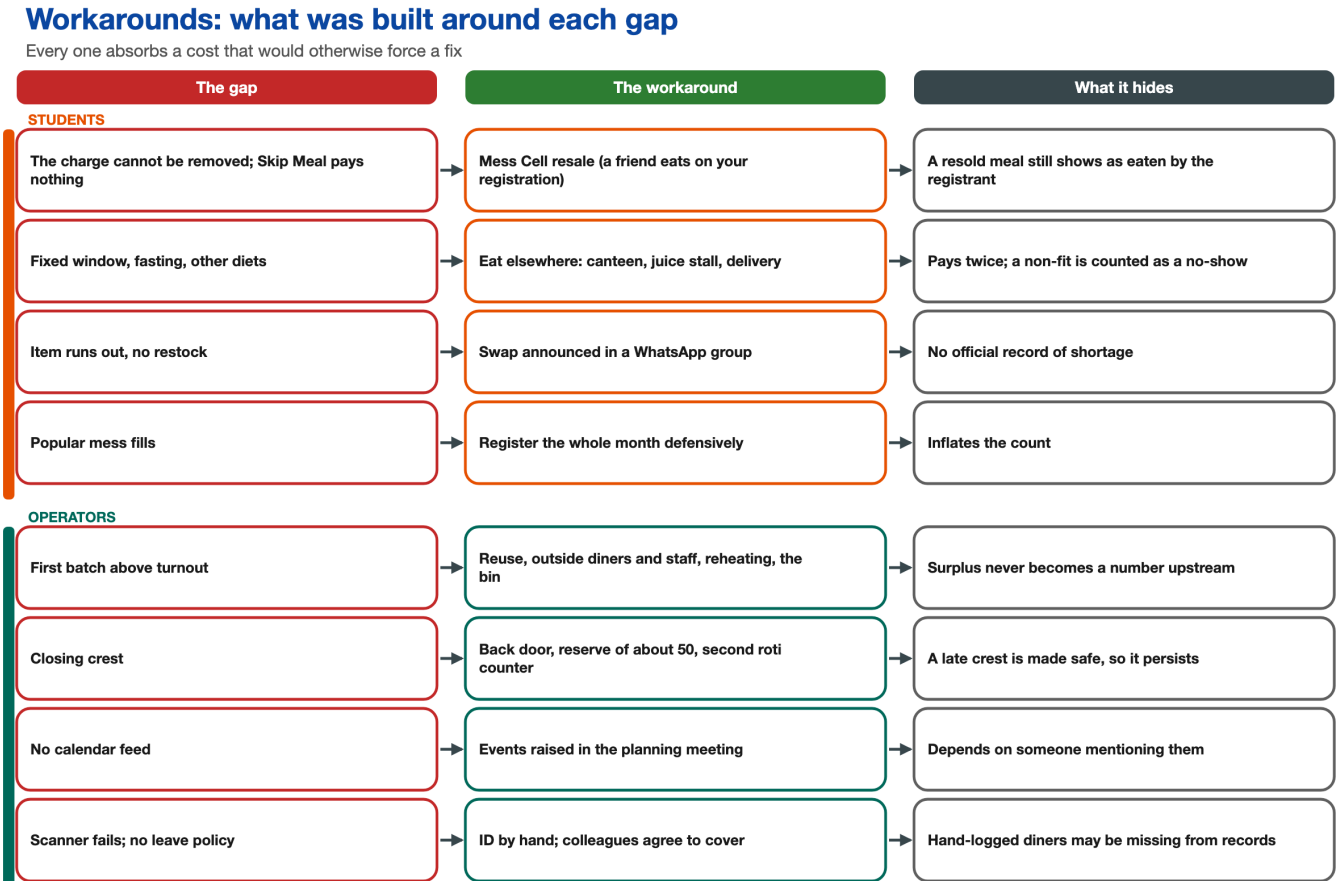

Figure 6.21. Where the system leaves a gap and what people have built to absorb it. Source: student interviews and operator interviews and visits.

Decisions and revisions

- We rebuilt the first analysis on 13 September so that every layer is sourced and labelled, and relabelled the iceberg so that an event is one concrete happening and standing rules sit at structure, because a mislabelled layer sends every lower one wrong.
- We kept Chain 6 so that legitimate non-attendance is not counted as dysfunction, and turned no single respondent into a population claim.
- We counted only structures that pass a root-cause test as causes, demoting the four-day procurement lock and coarse information. The plate basis, Skip Meal and the head of CDS stay open.

Activity 8: Reframing the Problem

Activity 8 converted the operational problem into a design opportunity. It produced the Reframed Systemic Problem Statement (Deliverable 4 of Phase 2) and the Design Opportunity Statement (Deliverable 5). Our first version, written on 14 September, read as a consulting memo naming a fix and an owner, so we rebuilt it.

The reframed problem statement

> The Breakfast Paradox is a stable equilibrium. The system holds the gap between registration and attendance in place because every actor's behaviour inside it is individually reasonable, and every consequence it produces is absorbed before it reaches anyone who sets the rules.
>
> Three structures hold it. Breakfast is booked by default: every student is registered automatically, holidays included, and can leave only through five cancellations per meal type per month, each made at least four days ahead, so the default cannot follow the decision a student makes on the morning. Billing fires at the booking, not at consumption, so the inaccuracy costs the system nothing and, if operators are paid on booked plates, the operator nothing either. And the gap is measured twice, in the CDS scan record and in the operators' own books, but neither measurement is routed to the people who set registration, billing and menu rules, whose decision rights are split so that none holds both knowledge and authority. Lunch runs under the same rules and mostly works, which makes this a breakfast problem.
>
> What follows is quietly absorbed. Resale takes the student's loss. The operator's discount on the booking, live top-ups, leftover reuse, outside diners and staff who eat the same food, and late-door service take most of the physical surplus. Holding and reheating turn some of the rest into lower quality, and the remainder leaves in the bin. Outside venues take the dissatisfaction and informal workarounds take the cultural mismatch. Most of this is useful and should not be removed. Together it keeps the gap from producing a signal loud enough to reach a system that responds to complaint intensity.
>
> No single actor feels this as their whole problem. The student feels a charge. One operator measures the gap every morning and has nobody to tell, another cooks to the count as the obligation it is paid for, a third adjusts weekly but never below a floor held against shortage. The office holds a statistic. The paradox is seen in aggregate only by the actors with the least authority over the rules.

Bakul keeps no records, so every Bakul figure behind the statement is its operator's own account.

Challenging the initial assumptions

Interviews, Yuktāhār's books and 11,941 Kadamba scans tested five starting assumptions.

Assumption	What the evidence shows
Students not showing up is the problem	A symptom, and partly no problem: one respondent fasts by choice
Students are careless or irrational	Most never choose the booking, and the exit is narrow and early (confirmed)
Nobody has solved this	Attendance-tracked billing ran twice, in a gas shortage and in holidays such as Felicity, and was switched off
Nobody knows how big the gap is	Measured in at least three places: 37.1% turnout at Kadamba, about 28% at Yuktāhār, lunch at 56 to 89%
This is one problem	Three structural causes plus one thing that is no problem

This revises an earlier gap of about 30%, which was a lunch figure. Beliefs still sit above the records: Yuktāhār's site head says up to about 50%.

Stakeholder perspectives and tensions

Asked for their problem, no one describes the paradox, and each feels a smaller piece (Figure 6.22). Students feel a charge and never the aggregate cost. The Yuktāhār operator measures the gap and has nobody upstream to tell. The facility team

checks safety and taste, never quantity.

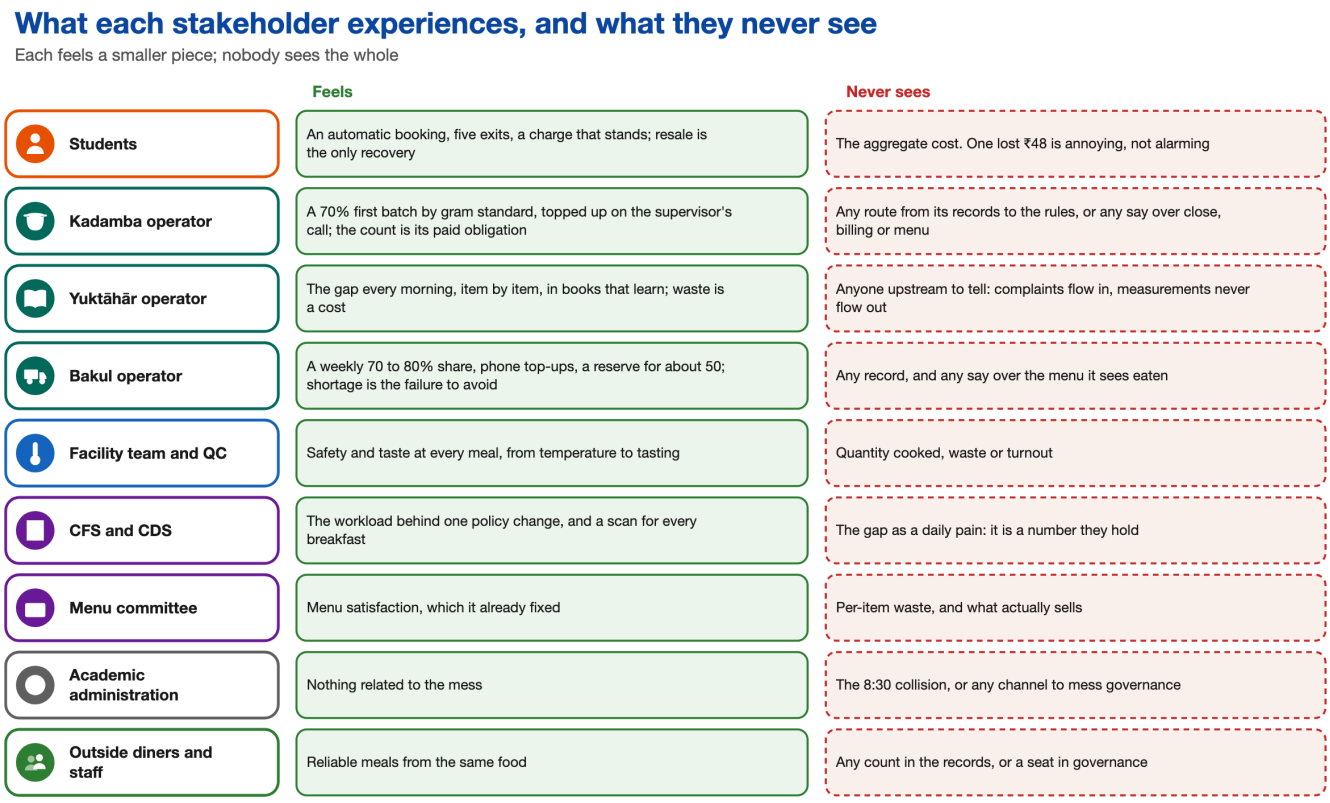

Figure 6.22. What each stakeholder experiences and what stays invisible to them. Source: authors' analysis of interviews and visits; operator entries are single-sourced.

Ten further tensions are rational on both sides (Figure 6.23). The institution chose billing certainty over demand accuracy each time the trigger passed. A first batch of 70% of registrations meets 37.1% turnout: on April's average day about 530 vegetarian portions against about 245 eaten (candidate arithmetic), yet late students still met run-outs.

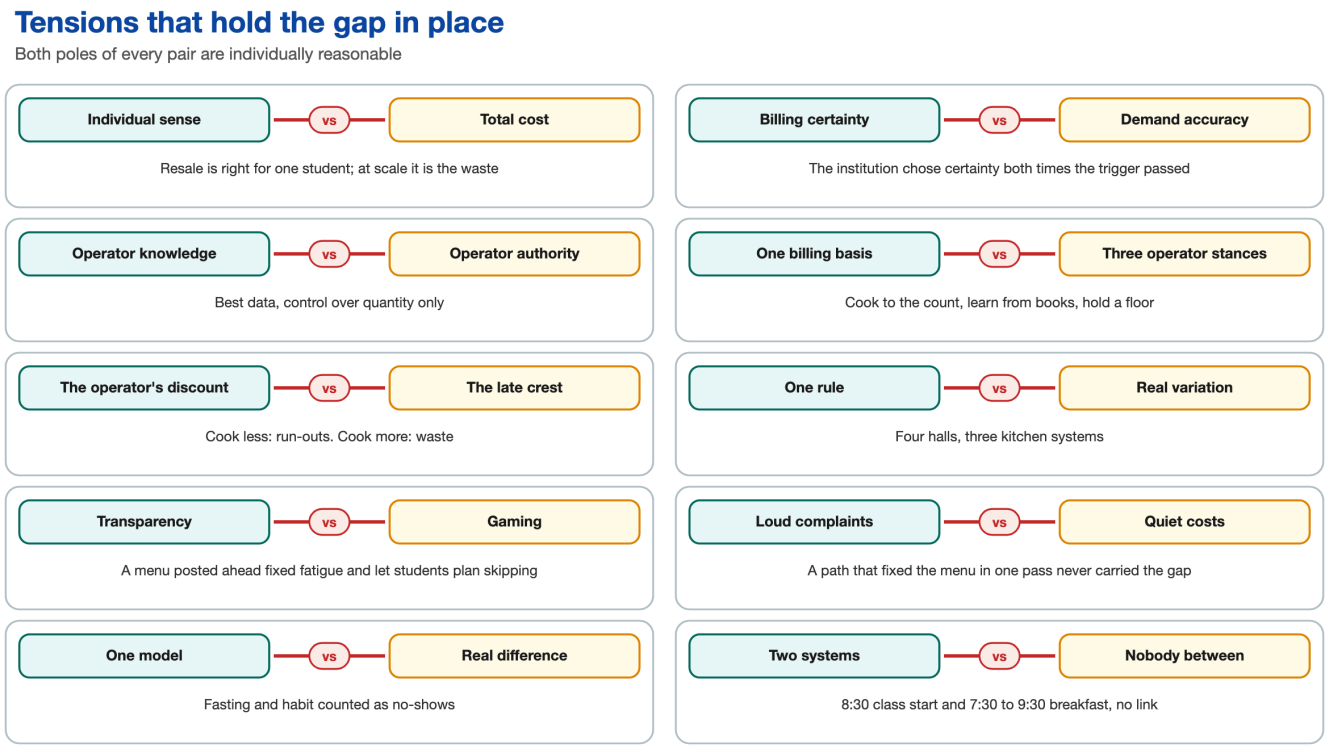

Figure 6.23. Ten tensions in which each pair is rational on both sides. Source: authors' analysis.

Why the system holds steady

The original framing stayed in the top two layers of the iceberg; the causes sit below, down to beliefs such as "registration is demand" and "measurement means blame" (Figure 6.24).

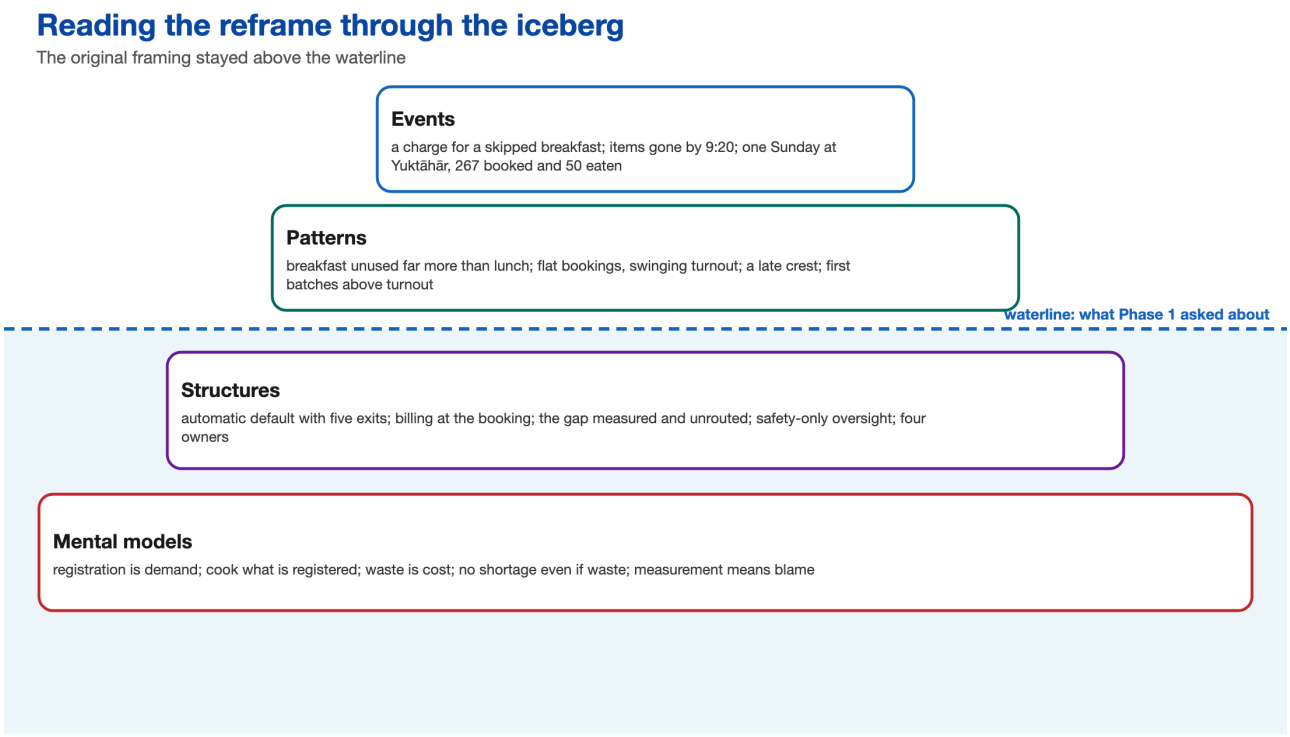

Figure 6.24. The paradox read at four depths. Source: authors' analysis.

The gap persists neither because nobody solved it nor because somebody benefits. Three forces create it, and eight absorbers take up its cost before it reaches the rule setters (Figure 6.25). Most absorbers help people. Three protect no one (the over-cooked first batch, repeated reheating, the bin), and nothing the kitchens learn crosses upward.

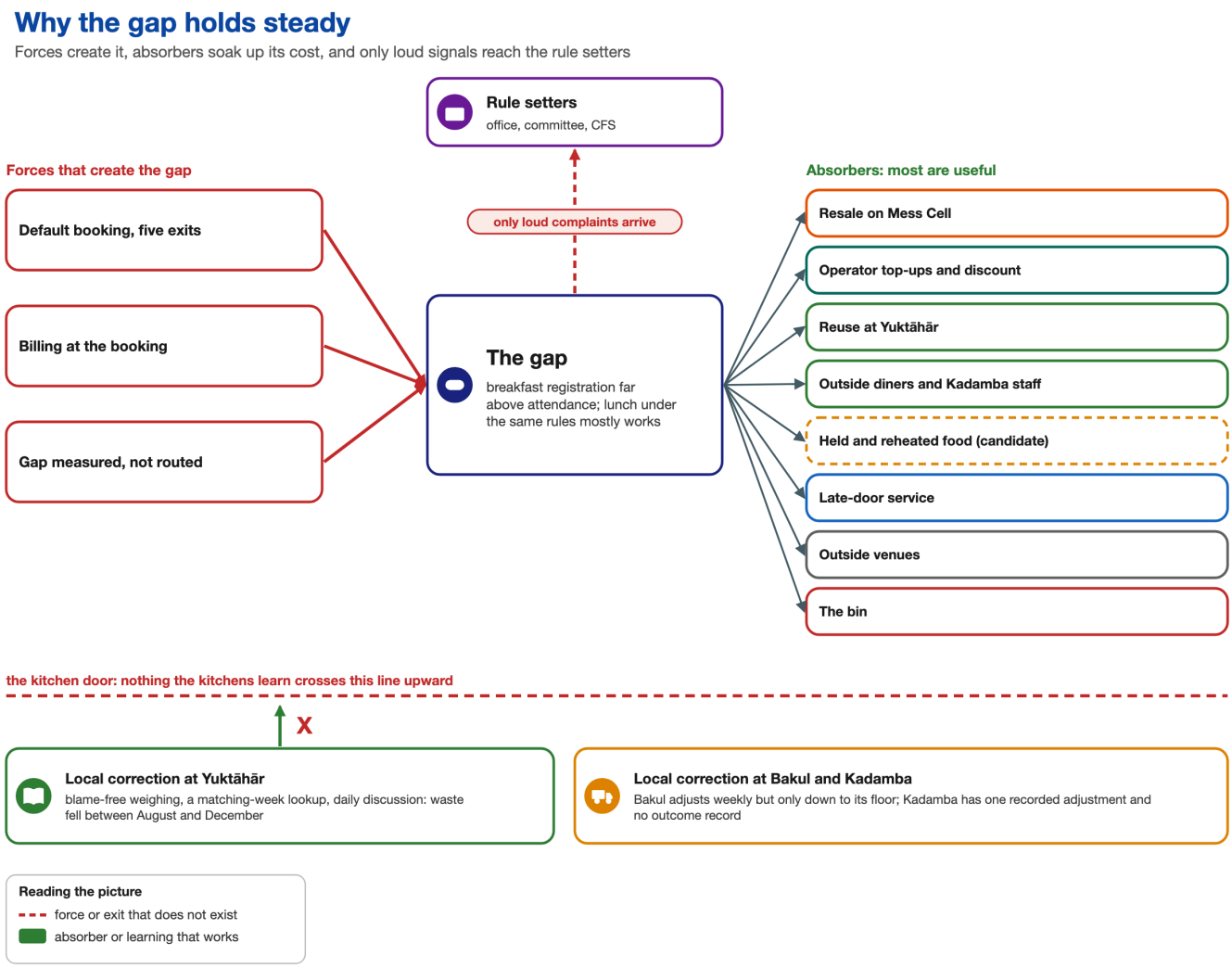

Figure 6.25. Why the gap holds steady: three forces create it, eight absorbers take up its cost, and only a loud signal reaches the rule setters. Source: authors' analysis; the held and reheated food absorber is a candidate.

Alternative framings and validation

We scored six alternative framings on four tests: explaining breakfast against lunch, explaining why the gap persists unseen, reaching the booking, and fitting the evidence. Only the reframing passes all four (Figure 6.26). "How do we help kitchens forecast better?" fails because operators already forecast: the best-instrumented believes up to 50% where its register says 28.

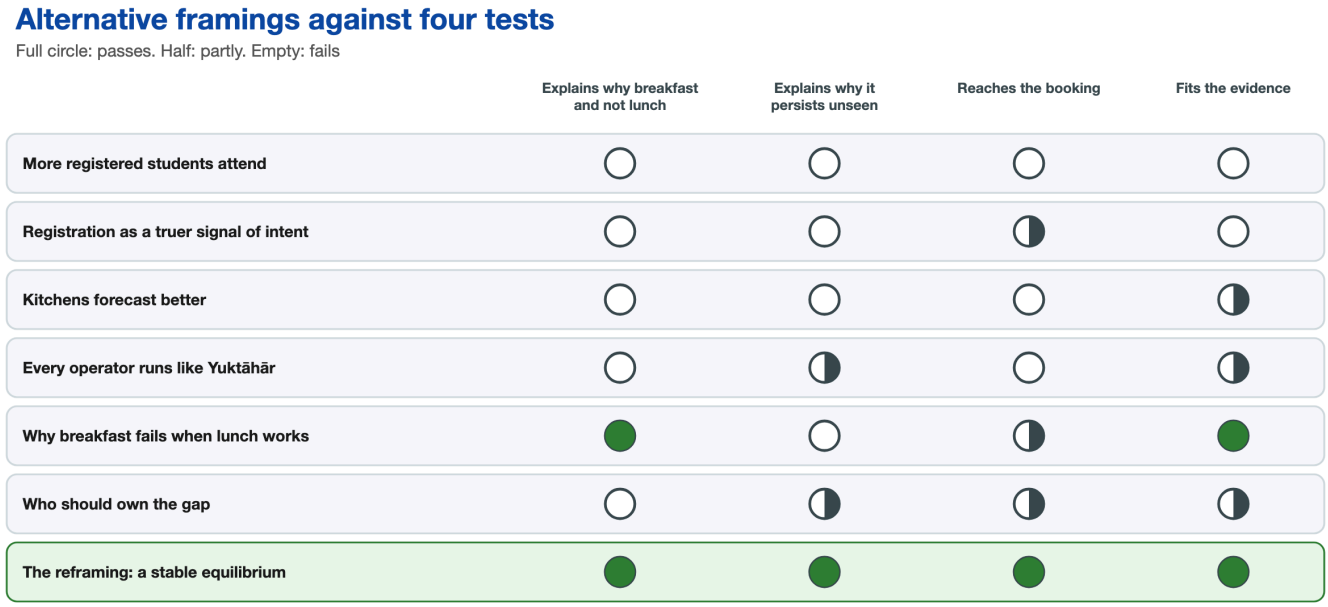

Figure 6.26. Six alternative framings against four tests; only the reframing passes all four. Source: authors' analysis.

Most starting assumptions were revised or overturned (Figure 6.27), and whether Skip Meal reaches the kitchen stays conflicting. One question limits the framing: what a plate counts for in each operator's payment, which decides who bears a no-show. Eight claims from the Bakul manager's account are carried as hypotheses (H1 to H8), never findings.

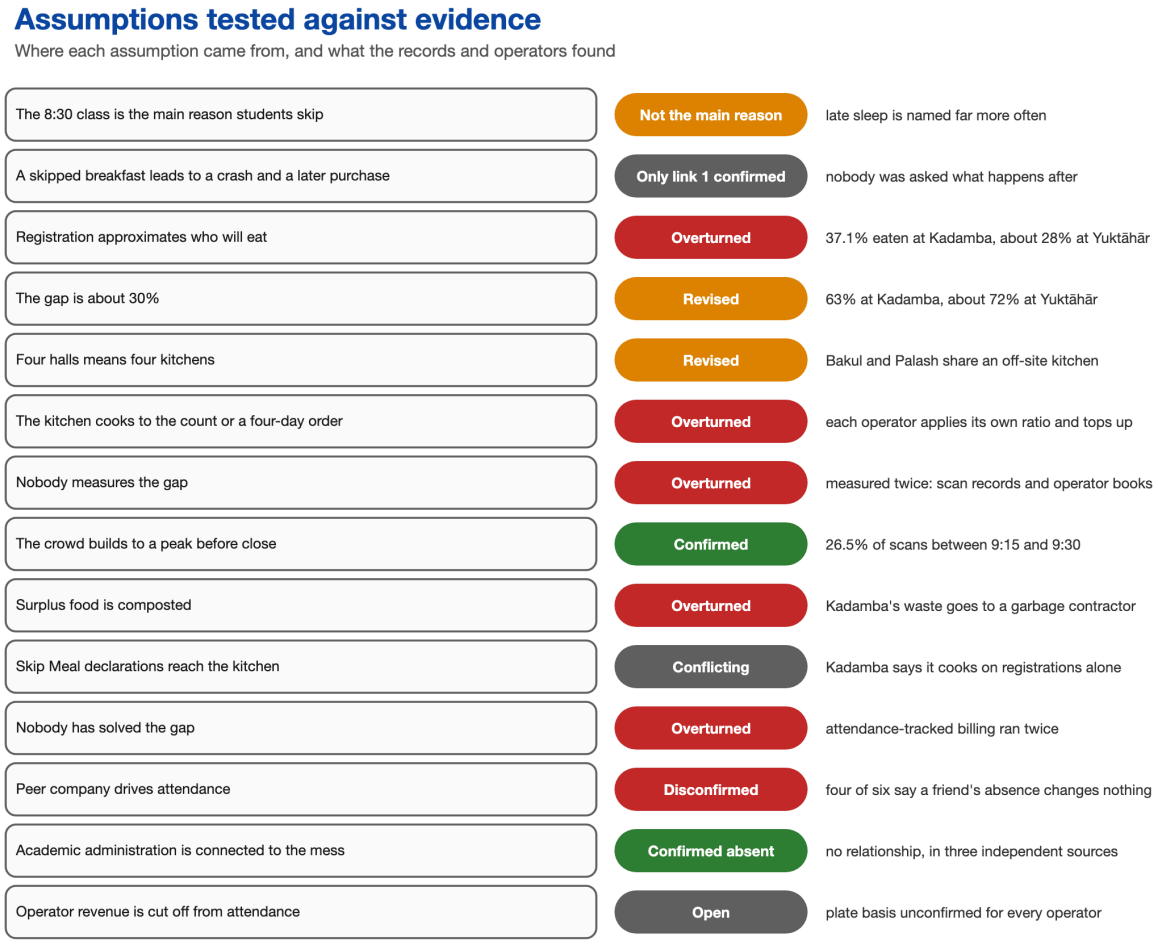

Figure 6.27. Assumptions tested against evidence: what held, what changed and what is open. Source: authors' analysis of records, interviews and field accounts.

The design opportunity statement

Read against the iceberg, intervening pays most at information flows, because the records already exist (Figure 6.28). The cancellation cap has moved from five to ten and back (confirmed) and left the default alone, and lowering the batch ratio by decree moves cost into run-outs.

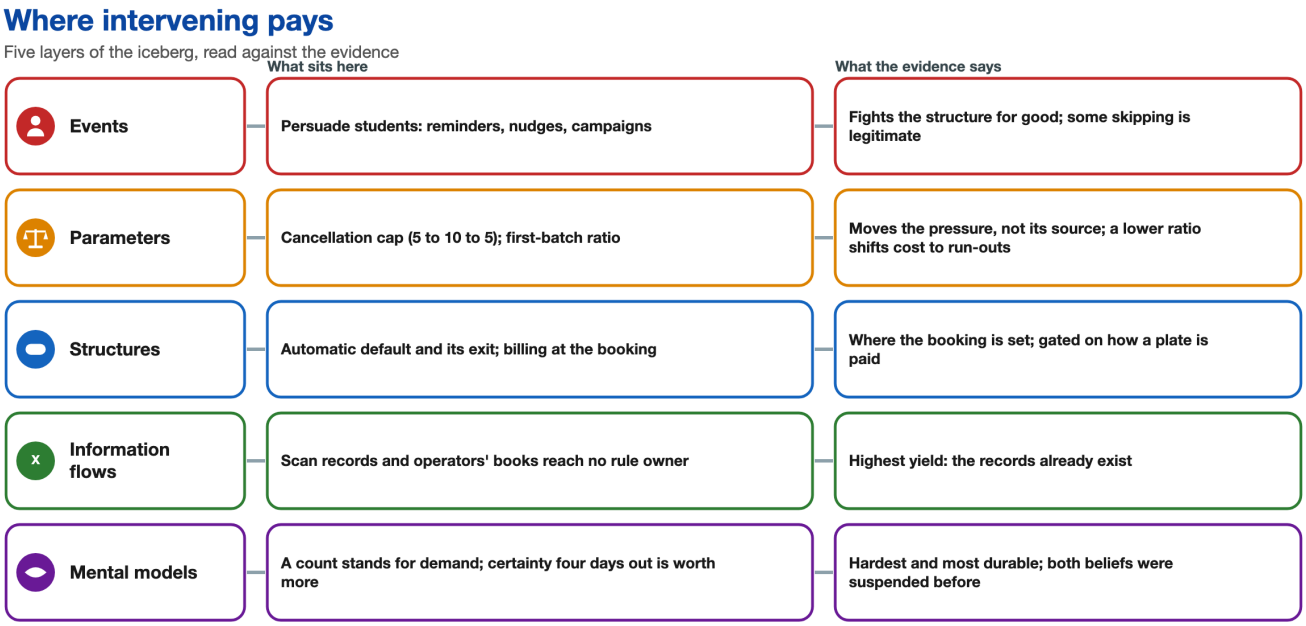

Figure 6.28. Where intervening pays: five layers of the iceberg read against what the evidence shows. Source: authors' analysis.

The opportunity is to give this system a way to notice its own cost.

The Breakfast Paradox persists because the gap goes unmeasured, although it is measured daily, and because the fix is unknown, although it has been run twice. The measurements, in scan records and operators' books, reach no one with authority to change the rules that produce the gap. Every consequence is absorbed before it becomes visible, and the institution's attention triggers on complaint intensity, and this problem is quiet.

The opportunity is to build the missing routes: from the records that exist to the kitchen decision that fixes how much is cooked, and from the same records upward to a decision about registration and billing. Four intervention points follow. One sits in the kitchen, where a registration becomes a cooked quantity. Three sit in governance, in decreasing depth: the registration and billing rules, the ownership gap and the attention mechanism. The last is the cheapest and most overlooked, and would have surfaced the CFS Chair's own figures years ago.

The four intervention points

The four points sit on the chain from booking to bin (Figure 6.29). Food quality as eaten, demand by item and the menu's pairings rank below them as secondary areas.

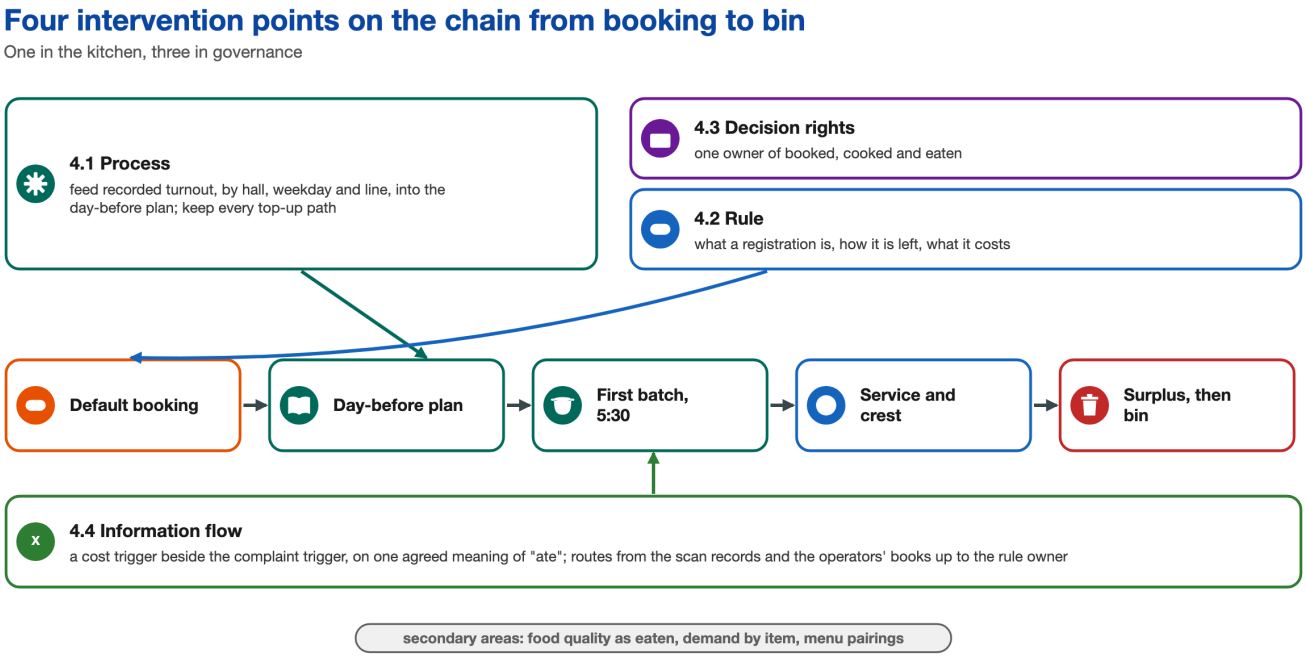

Figure 6.29. The four intervention points on the chain from booking to bin. Source: authors' analysis.

- 4.1, process: feed recorded turnout by hall, weekday and line into the first-batch decision, keeping every top-up path.
- 4.2, the rule: a standing, partial version of the attendance-tracked model, using scans already collected.
- 4.3, decision rights: one named owner of the booked, cooked and eaten outcome.
- 4.4, information flow: an escalation trigger on measured cost beside the complaint trigger, the cheapest by far.

What any intervention must respect

Eight constraints follow from the evidence (Figure 6.30). Keep the absorbers that feed or pay people, and target only the three that protect no one. Act before the first batch is cooked, never raise shortage risk, and measure cooked against eaten with run-outs and quality. Leave the 8:30 class start alone, aim the fix at neither students nor operators, and keep measurement blame-free, starting from the paper books.

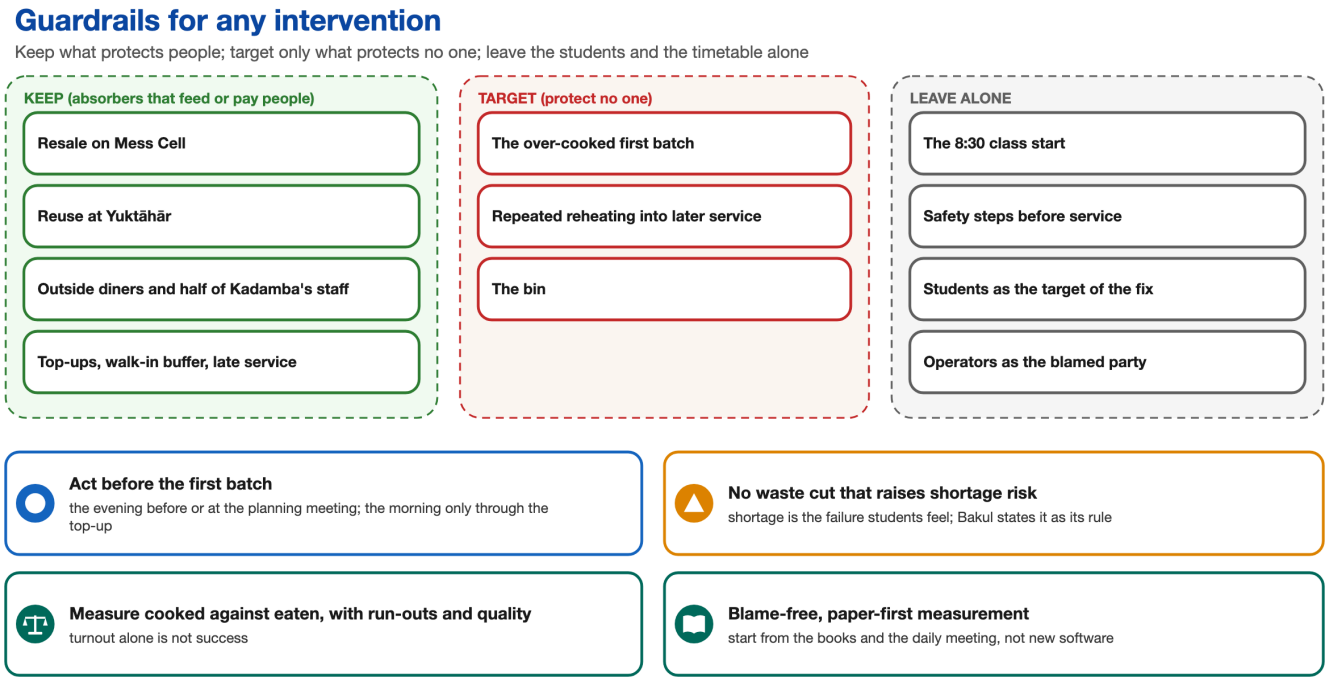

Figure 6.30. Guardrails: what to keep, what to target and what to leave alone. Source: authors' analysis.

Sequence and the gating questions

The order runs from cheap to deep (Figure 6.31). Information flow needs no new data, process shows a physical effect fastest, ownership turns a number into a decision, and the rule comes last.

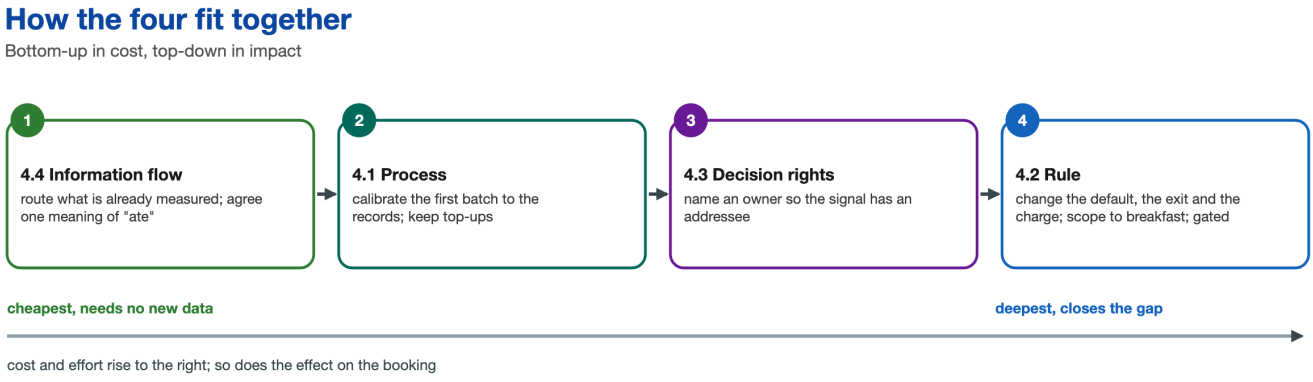

Figure 6.31. How the four intervention points fit together: cheap and enabling first, deep and gated last. Source: authors' analysis.

Two open questions gate the deepest point (Figure 6.32). Does a plate for payment mean served, as Bakul's manager says, or registered, as Kadamba's operator's remark suggests? And who set the four-day lead and the five-cancellation cap, and does anything operational depend on them? Either way the other three points stand.

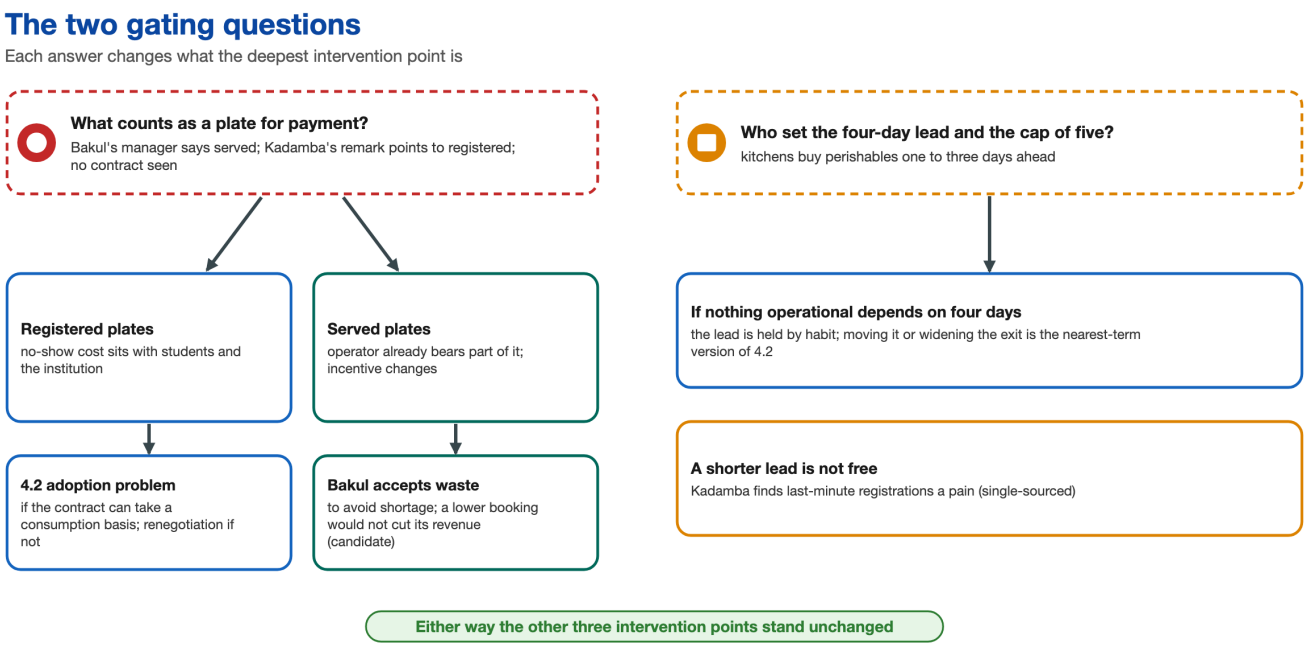

Figure 6.32. The two gating questions and what each answer would change. Source: authors' analysis; both questions are open.

Decisions and revisions

- We rejected our first reframe, which named a fix and an owner, and asked instead what would make anyone fix the gap.
- We kept only the reframing of six scored framings, leaving the rest on record so their rejection can be reviewed.
- We stopped before designing and ranked the four points by depth and cost, so Phase 3 starts from a shortlist filtered against the evidence.
- We held the plate basis open as the gating question and kept conflicting accounts side by side: Skip Meal, the head of CDS, and the registration cadence (operators say weekly blocks, students monthly).

Phase 3: Identifying Leverage Points and Design Principles

Phase 3 asks where the equilibrium found in Phase 2 can change.

Activity 9: Defining Design Principles and Intervention Opportunities

Activity 9 asked where and how the system could be changed. It produced the Leverage-Point and System Intervention Map and seven design principles. The deepest workable levers sit on routes, goals and beliefs that nobody holds today, and the kitchen levers are the quick fix.

Twelve leverage points

We placed twelve levers on Meadows' (1999) twelve places to intervene, counting from place 1, the deepest, and writing "place n of 12" on each. Most sit at places 2 to 6, from mindset to information flows (Figure 7.1). None claims place 9 (delays), and places 11 and 12 appear only as settings inside deeper levers: cutting Kadamba's 70% ratio by decree would only move cost from surplus to run-outs.

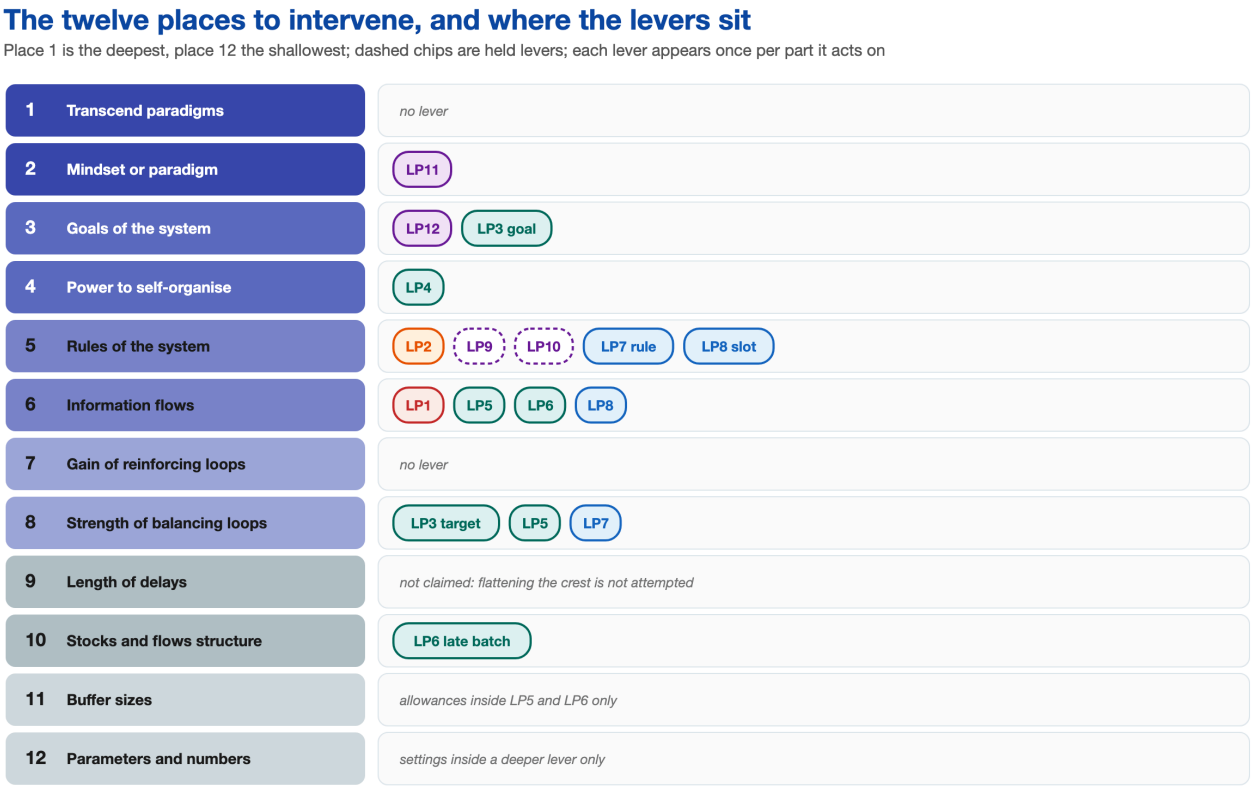

Figure 7.1. The twelve places to intervene, and where each lever sits. Held levers are ringed with a dashed line. Source: authors' analysis, after Meadows (1999).

LP	Lever	Place (of 12)	Can a student team act on it
LP1	Build the missing routes	6	Yes: high to prototype, medium to deliver
LP2	Change the breakfast default and its exit	5	Propose only; the rule owner (Mess Office or CDS, unconfirmed) decides
LP3	Operator goal: surplus is a cost	3; target at 8	Low to medium; Prism management, Bakul manager
LP4	Operator forum that runs its own trials	4	Medium; operators, with CDS convening
LP5	Record-calibrated first batch, daily record	6, 8	Operator decides; high at Yuktāhār, medium at Kadamba, unsizeable at Bakul
LP6	Crest-aware top-up, staged late batch	6, 10	Operator decides; high at Kadamba, low at Bakul
LP7	Cost trigger on the oversight line	8; rule at 5	Medium, after LP1
LP8	Operator voice in the menu	6; slot at 5	Medium; menu committee (unconfirmed)
LP9	Named owner of booked, cooked, eaten	5	Held; low

LP	Lever	Place (of 12)	Can a student team act on it
LP10	What a booking costs	5	Held; low
LP11	Test governing mental models	2	Medium; each belief's holder
LP12	Change the goal actually sought	3	Low to medium; CFS and CDS

The levers rest on records. Of Kadamba's 32,162 April registrations, 37.1% were eaten (confirmed), yet no route runs from any record to any rule owner (LP1). Registrations hold near 1,070 every weekday while turnout runs from 42.1% on Tuesdays to 27.3% on Sundays (LP2). The 70% batch is about 2.2 times what the veg line ate (candidate arithmetic; LP5), and 26.5% of scans land between 9:15 and 9:30 (confirmed; LP6).

LP11 sets beliefs against records, such as the Yuktāhār site head's "about 50%" against about 28 in his own register. LP12 turns a goal the CDS poster already states, minimising food wastage through advance meal planning (confirmed), into two working indicators: booked against eaten and cooked against eaten.

Two levers are held. Decision rights are split at least five ways (confirmed), and whoever sets registration and billing bears none of the cost. LP9 would name an owner of booked, cooked and eaten. LP10 rests on a precedent: attendance-tracked billing ran during a gas shortage and some holiday and fest periods, and worked (confirmed). Both wait until the head of CDS and the payment basis are known.

The archetypes check

Of six archetypes checked, three read as present: shifting the burden and fixes that fail (both candidate), and a balancing process with delay. Shifting the burden orders the set (Figure 7.2). Kitchen competence and resale relieve the symptom within minutes to weeks, while each route to the fundamental fix has withered: resale replaced Skip Meal, and attendance billing reverted (confirmed). The dashed side effect (L17) is the trap, since a better-run kitchen eases pressure on the people who could fix the cause.

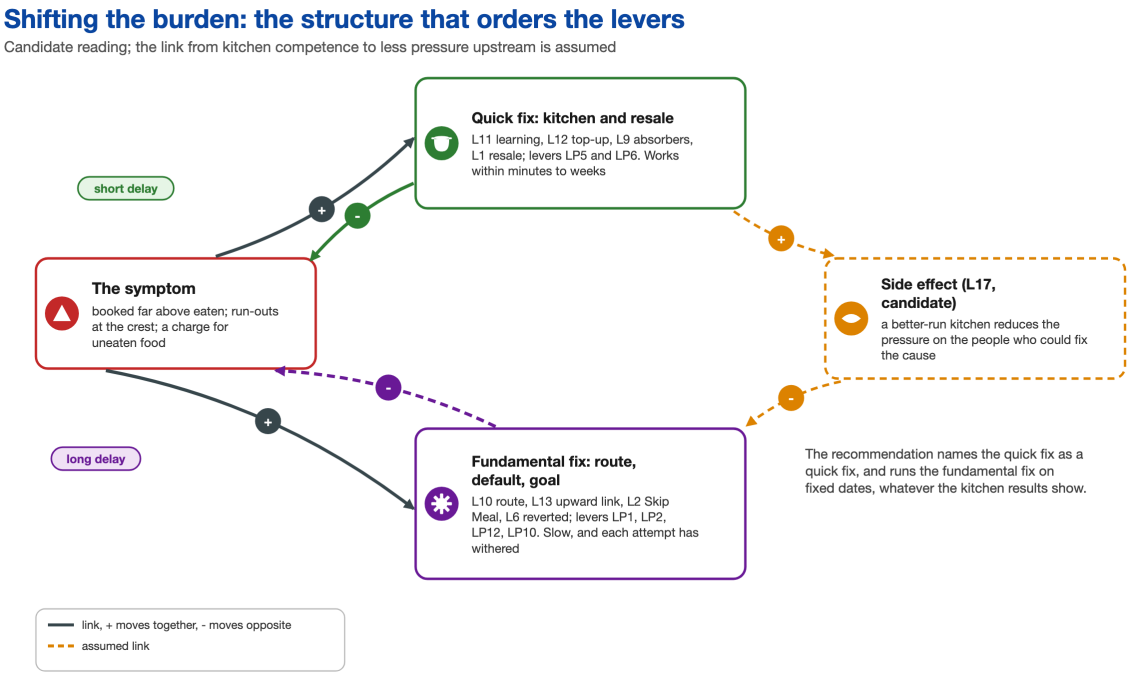

Figure 7.2. Shifting the burden: the structure that orders the levers. Source: authors' analysis; the link from kitchen competence to less pressure upstream is assumed.

LP5 and LP6 are therefore the quick fix, worth doing because students eat the food saved and they buy time. LP1, LP2, LP11 and LP12, with LP10 once released, are the fundamental fix.

Ranking and changes by type

LP1 ranks first though three levers are deeper, because every other lever needs its input. LP11, the deepest, is second because it needs the ledger first. LP2 is third as the deepest rule lever not gated on the plate basis, and LP12 fourth because nobody able to carry it has engaged. By change type, the levers propose:

- Rules: a shorter lead, an evening skip that lowers the count, Sunday opt-in, consumption billing as already run (LP2, LP10, LP6).
- Roles: a quantity line at the tasting, a CDS forum, a named owner, an operator menu slot (LP7, LP4, LP9, LP8, LP12).
- Incentives: a later exit that removes the charge, surplus shown as cost, blame-free recording (LP2, LP3, LP5, LP6).

- Information flows: the ledger, a weekly calendar note, item waste into the menu cycle (LP1, LP8).
- Processes: a calibrated batch with a daily record, an arrival card, a late batch after 9:00 (LP5, LP6).

The widest rule change costs the kitchen least, because the four-day lead protects only dry goods.

The natural comparison between the halls

Kadamba and Yuktāhār share registration, billing and, as far as is known, tendered plate payment. Yet Yuktāhār plans from records and weighs blame-free on paper, while Kadamba cooks a fixed ratio. The halls part in the planning-rule row (Figure 7.3), so a learning practice is possible under current rules. Five other differences separate them, so the comparison cannot prove cause and stays candidate.

The natural comparison: three halls, three planning rules

Same registration and billing rules; the halls differ in five other ways, so the comparison cannot prove the cause

	Kadamba Prism	Yuktāhār ABC Hospitality	Bakul Vijayalakshmi Caterers
Scale	about 1,070 registrations a day	about 230 to 400 a day	no records
Menu lines	veg and non-veg	veg only, with Jain service	veg and non-veg; Palash shares the kitchen (candidate)
Firm and layers	four-level chain	flat team, two project managers	one manager at the mess, kitchen 8 km away
Records, month	CDS scan export, April	handwritten registers, September	none
Planning rule	fixed 70% of registrations, top-up by supervisor	matching-week lookup, first-half-hour check, run-out rule	70 to 80% by item, floor near 70%, phoned top-up
Gap measured	63% unused (confirmed)	about 72% unused (confirmed)	unknown; manager estimates at most about 60% turn up

Figure 7.3. The three halls side by side, from scale to measured gap. Source: authors' analysis of both halls' records and the operators' accounts.

The design objective and the case for the status quo

The objective is a breakfast system in which registration matches each student's intent, the kitchen cooks to expected eaters, and the cost of any gap reaches whoever sets the rules. It keeps registration and resale, and accepts some surplus as the price of never running out. The status quo persists because it pays each actor: students make no decision, Prism has few run-outs, CDS keeps a low workload. So LP1, LP11 and LP12 must reach CDS and the Mess Office before any rule proposal.

Design principles

Every Phase 4 concept must pass all seven principles. Each card in Figure 7.4 names what its principle rules out.

1. P1 Route before you measure, and route on paper.
2. P2 Make measurement blame-free, and judge it over months.
3. P3 Change the default, the goal and the belief, not the people.
4. P4 Act on the first batch the day before, and in the morning only through the top-up.
5. P5 Feed the absorbers by plan, name them as symptomatic outlets, and target only the over-cooked batch, repeated reheating and the bin.
6. P6 Measure success as cooked against eaten, run-outs and food quality, not turnout.
7. P7 Measures are diagnostic and plural, never a single target.

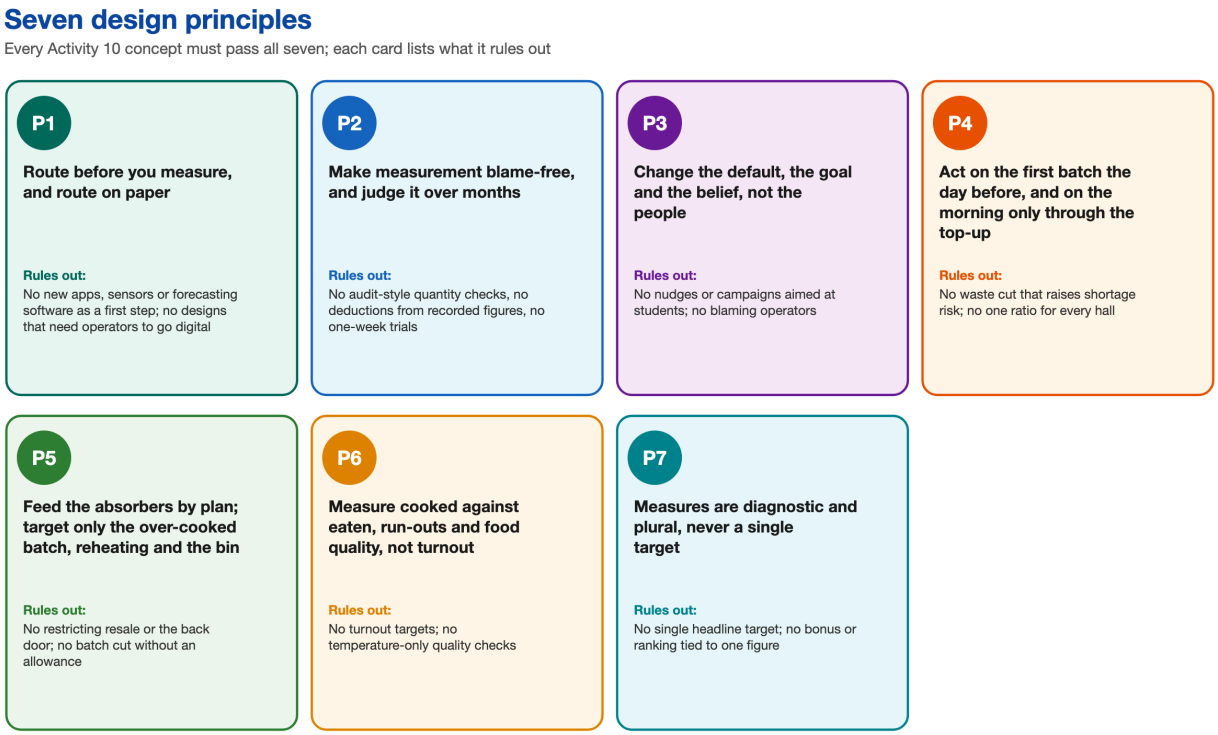

Figure 7.4. The seven principles, with what each rules out. Source: authors' analysis.

Intervention points

Fourteen intervention points follow one breakfast from booking to bin, then into governance (Figure 7.5): default, exit and kitchen count (IP1 to IP3), day-before plan (IP4), first batch and tasting (IP5, IP6), projection, top-up and crest (IP7 to IP9), surplus, record and disposal (IP10 to IP12), records sent upward (IP13) and the menu (IP14). Five levers crowd IP13, where the scan export is reviewed by no one. Only LP5 and LP6 touch an absorber that protects someone, and both keep it.

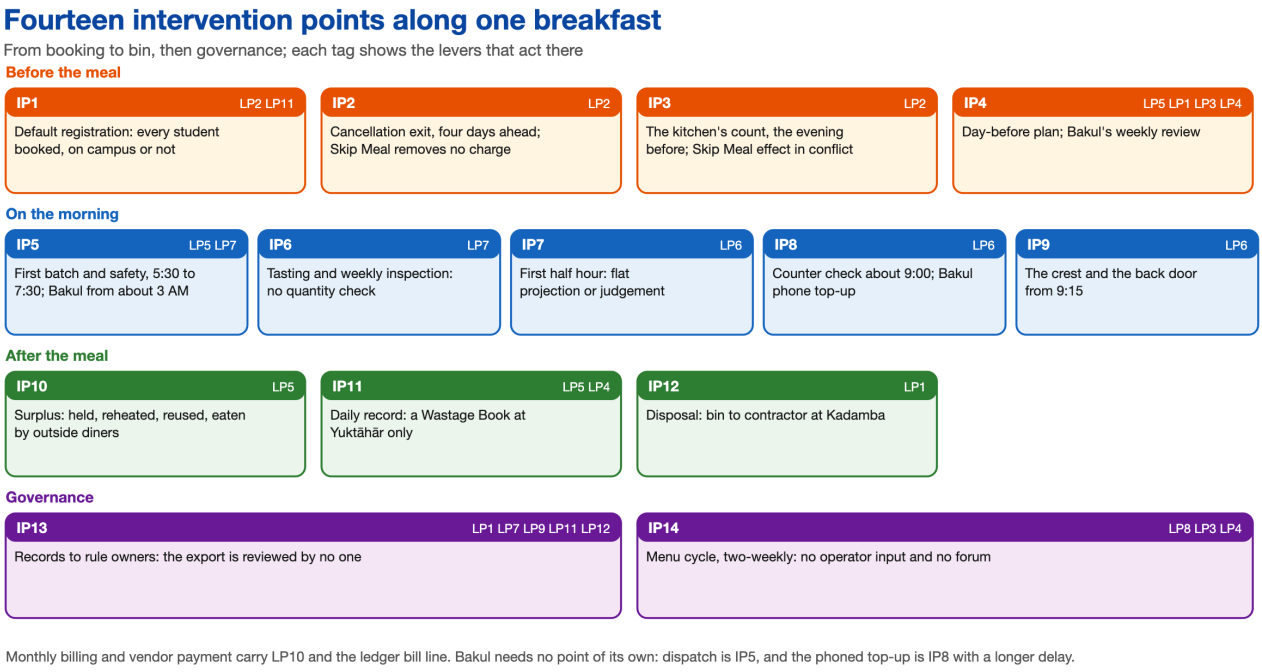

Figure 7.5. Fourteen intervention points along one breakfast, with the levers that act at each. Source: authors' analysis.

Sunday breakfast at Kadamba

Sunday is where three levers meet (Figure 7.6). It has Kadamba's lowest turnout: a 70% veg batch is about 3.1 times what Sunday diners ate, against about 1.9 times on Tuesdays (candidate arithmetic). LP2 (IP1 to IP3), LP5 (IP4) and LP6 (IP8, IP9) all bite hardest there, so Kadamba's Sunday veg line became the first test slice for Phase 4.

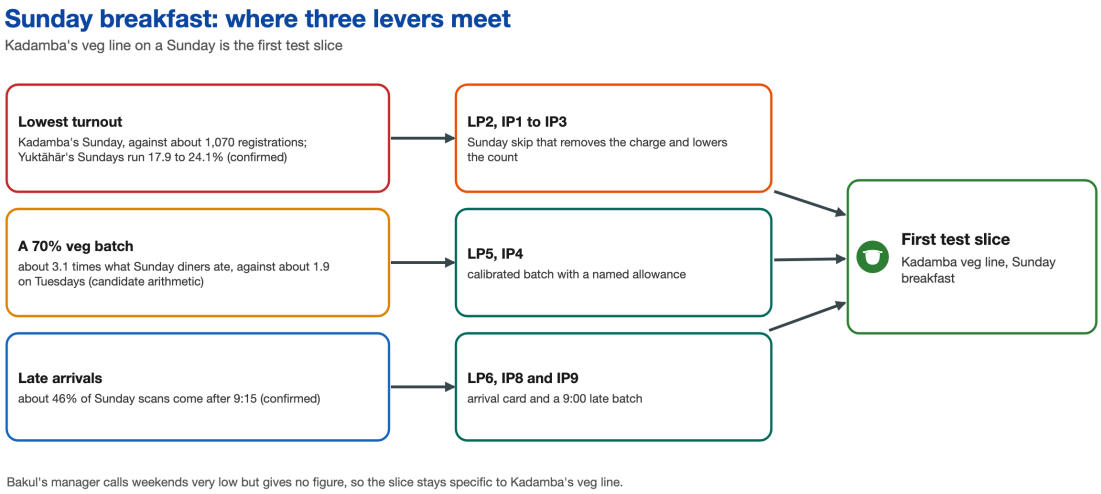

Figure 7.6. Where the three levers meet on a Sunday. Source: Kadamba scan export, April 2026 (confirmed); levers from authors' analysis.

The Leverage-Point and System Intervention Map

The map draws breakfast as one chain from the automatic default to the bin, with records and money above it and governance on top (Figure 7.7). Every solid flow runs along the chain or up into a record, and stops there. Dashed red lines are routes that do not exist today. The dashed green learning link, confirmed at Yuktāhār only, is the sole working feedback into the day-before plan. Kadamba's operator loses most from LP2 yet is most needed for LP5, so at Kadamba LP3 and LP4 come first.

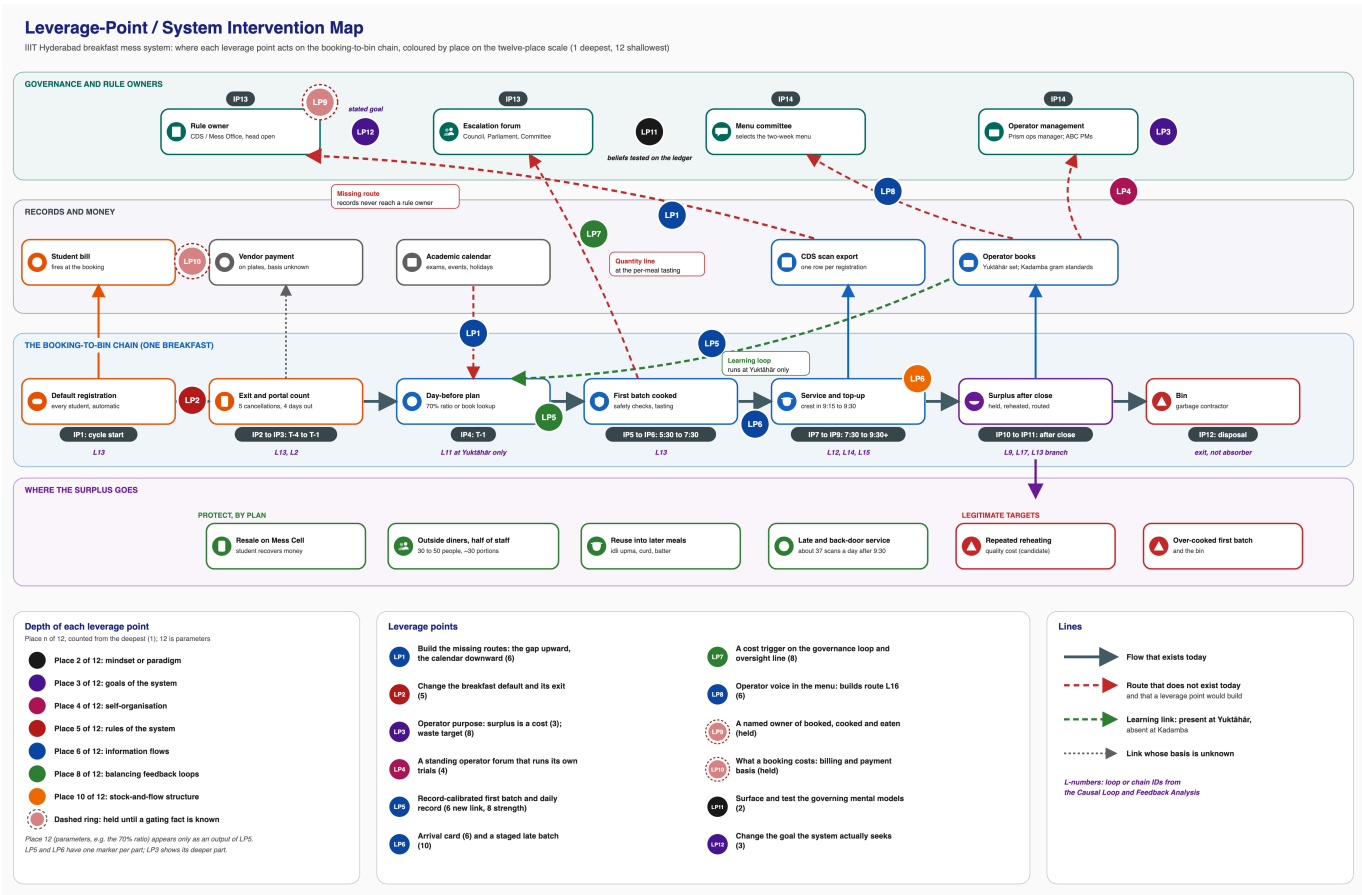

Figure 7.7. Leverage-point and intervention map for the breakfast mess system. Source: authors' analysis, built from the loop analysis and the booking-to-bin chain.

Decisions and revisions

- We wrote "place n of 12" on every lever, because two opposite numbering conventions had mixed in the first draft.
- We ruled out any lever that removes an absorber protecting someone, such as resale or meals for outside diners and staff, because that raises the cost of someone not at fault.

- We held LP9 and LP10 unscored, because a billing change made without the payment terms could reverse the operator incentive.
- We labelled LP5 and LP6 the quick fix in every report, because a kitchen presented as the answer is the shifting-the-burden trap in miniature.
- We made measurement blame-free (P2) and added P7 in revision, because a figure used against its recorder stops being recorded honestly.

Phase 4: Designing, Testing and Evaluating Interventions

Activity 10: Designing, Testing and Evaluating Interventions

Activity 10 asked us to develop and assess interventions. It produced intervention concepts and prototypes, a scenario model with an impact evaluation, and a final recommendation: route and read the gap together, open the Sunday exit on a fixed date, and calibrate the kitchen to buy time. No pilot was possible, so model outputs are projected, stakeholder reactions simulated, and co-design tests planned.

Eight intervention concepts

Seven concepts aim at a lever, and G is a deliberate foil. “B with C” is the calibrated first batch plus the 9:00 late batch.

Concept	What it is	Levers	Where it stands
A The Breakfast Ledger	Weekly per-hall sheet of scans and books, with the charge for uneaten breakfasts	LP1, LP7	Fundamental; top single score
B The Kitchen Learning Kit	Plan card, named allowance, daily sheet, operator forum	LP3, LP4, LP5	Quick fix; in the bundle with C
C The Crest Plan	Weekday arrival card; late batch at 9:00	LP6	Quick fix; in the bundle with B
D The Sunday Exit	Sunday skip that removes the charge and lowers the count	LP2	Fundamental; proposed at week 6
E The Operator Menu Slot	Operator slot at each two-week menu cycle	LP8	Fundamental; passes, acts on L16 only
F Billed at Consumption	Charge on the scan	LP10	Held behind the plate basis
G Ratio Cut to 50 Percent	First batch cut from 70 to 50% by instruction	None (place 12)	Foil; fails G1 and P4
H The Shared Reading	Joint sessions on each belief and the working goal	LP11, LP12	Fundamental; in the bundle

The quick fixes cluster on LP3 to LP6, the fundamental concepts spread across routes, default, goal and belief, and G pulls on nothing (Figure 8.1).

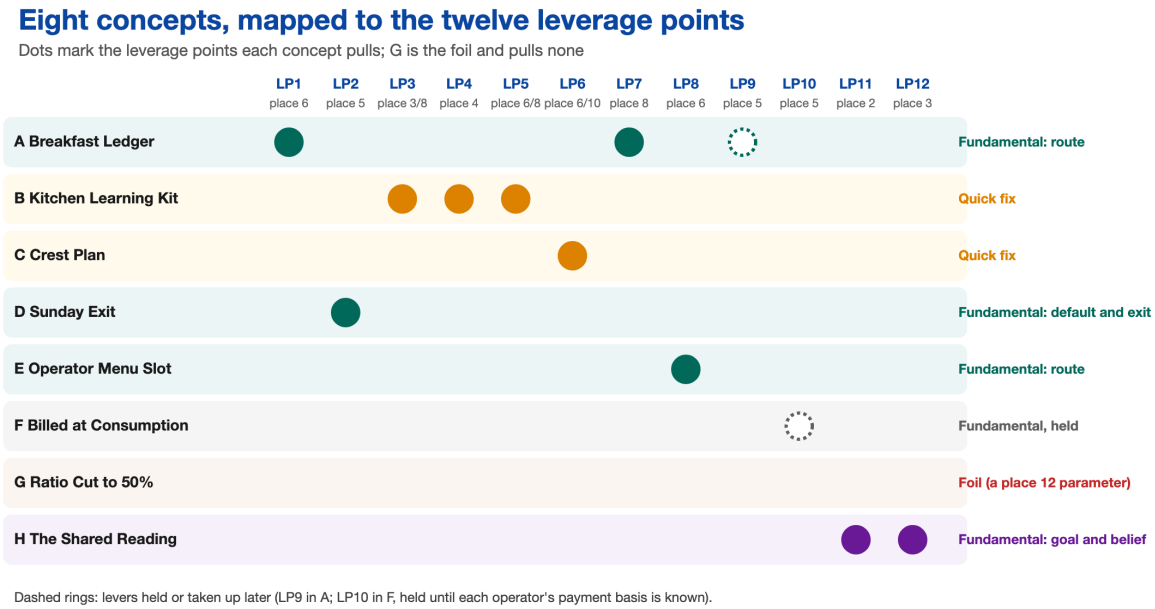

Figure 8.1. The eight concepts mapped to the twelve leverage points. G, the foil, pulls on none. Source: authors’ analysis.

Prototypes

The prototype is the re-sequenced bundle across one breakfast (Figure 8.2). The kitchen lane is the quick fix; the student, records and governance lanes carry the fundamental fix on fixed dates. Each piece is on paper:

- The ledger sheet (A) is one A4 page per hall per week. For Kadamba's veg line on Sunday 19 April 2026 it shows 817 registered, 179 eaten and 638 charged and not eaten at Rs 48 (Rs 30,624).
- The plan card and daily sheet (B) set each dish at the evening meeting from the gram standard, and log batches, top-ups, run-out time and leftovers.
- The 9:00 rule (C): divide diners so far by the day's usual share by then, add 10%, subtract the first batch and cook the difference.
- The Sunday exit (D) is one Skip Meal control, open until 8 pm on Saturday, as an opt-in or a skip that removes the charge.
- The menu note (E) lists items most left over and earliest run out; in the shared reading (H) each participant writes their own figure first.

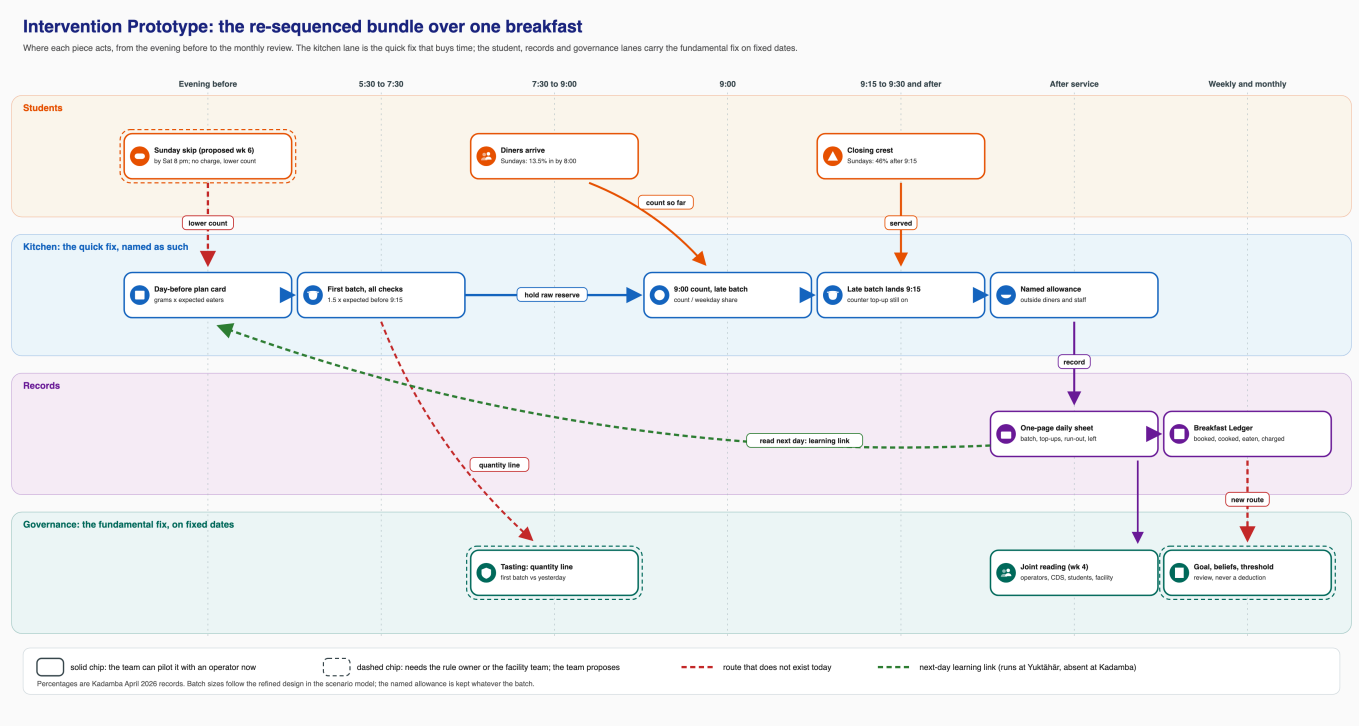

Figure 8.2. The re-sequenced bundle across one breakfast: quick fix and fundamental fix on separate lanes. Source: authors' design, built from the concepts and the booking-to-bin chain.

Applied to 19 April, the plan card and the 9:00 rule cook close to the 179 who ate, far below the baseline batch (Figure 8.3). Six co-design tests are planned for Stage 1, because no model can catch a design nobody would use; none has run. They range from the skip wording with 8 to 12 students to the 9:00 batch with the Kadamba supervisor and cook.

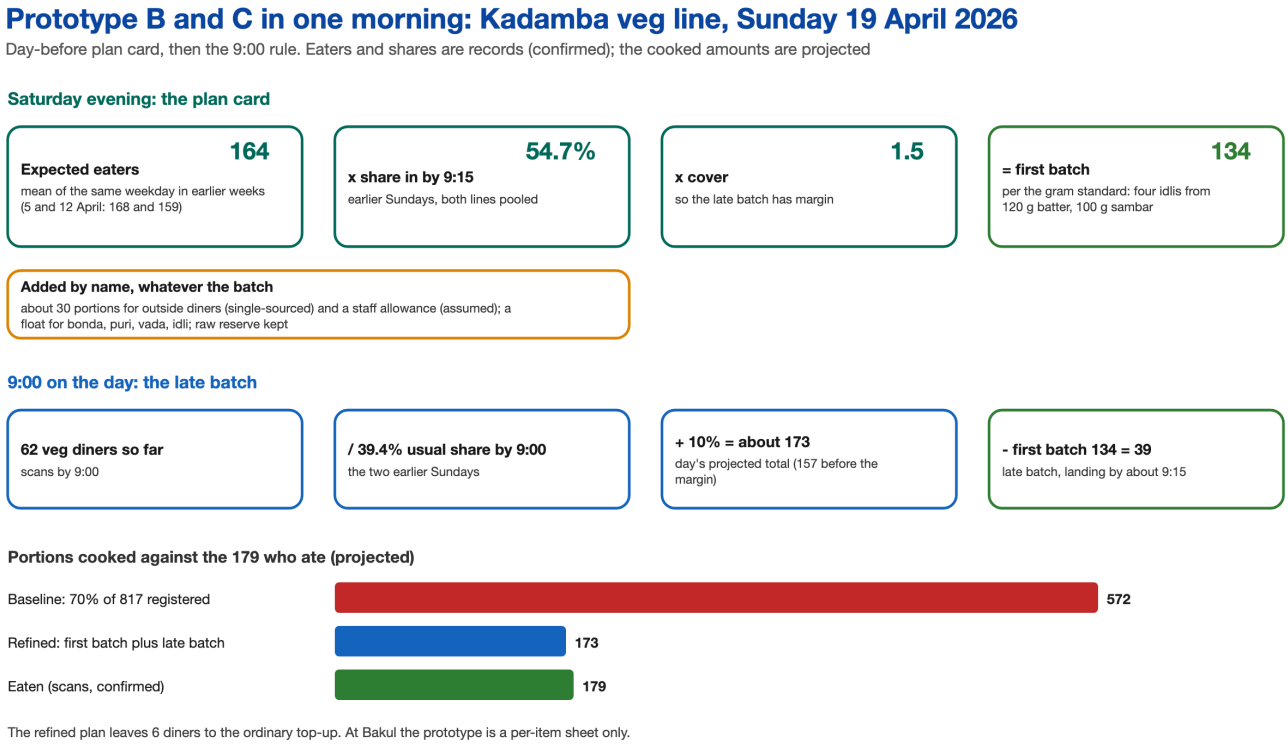

Figure 8.3. Prototypes B and C on one morning: plan card, 9:00 rule, and portions cooked against portions eaten. Source: Kadamba scan export, April 2026 (confirmed); batches are model output (projected).

The scenario model

The model explores how the system might respond and makes no predictions. It replays Kadamba's April 2026 records (32,162 rows with scan times, confirmed) arrival by arrival, over 23 out-of-sample days (8 to 30 April). Inputs include a 70% first batch per line (tested 60 to 80, single-sourced), rates of Rs 48 veg and Rs 66 non-veg, and skip uptake of 25 or 50% of no-shows (assumed). Nothing is sized for Bakul, which has no records.

No single kitchen rule is enough (Figure 8.4). The ratio cut leaves 150 surplus portions, 138 of them veg, and runs non-veg short on 7 of 23 days, because one flat ratio cannot serve two lines 16 points apart. A calibrated single batch cuts surplus to 60 but moves the cost to the crest: on 26 April the veg line would have run out at 9:24, 45 short. Only B with C keeps surplus and shortage both low.

Rule (projected)	Cooked a day	Surplus a day	Short a day	Line-days short (of 46)
Baseline: 70%, one batch	759	362	0	0
G: ratio cut to 50%	542	150	4.5	7
B alone: forecast plus 10%, one batch	440	60	16.8	16
B with C: calibrated plus 9:00 late batch	485	91	2.6	7

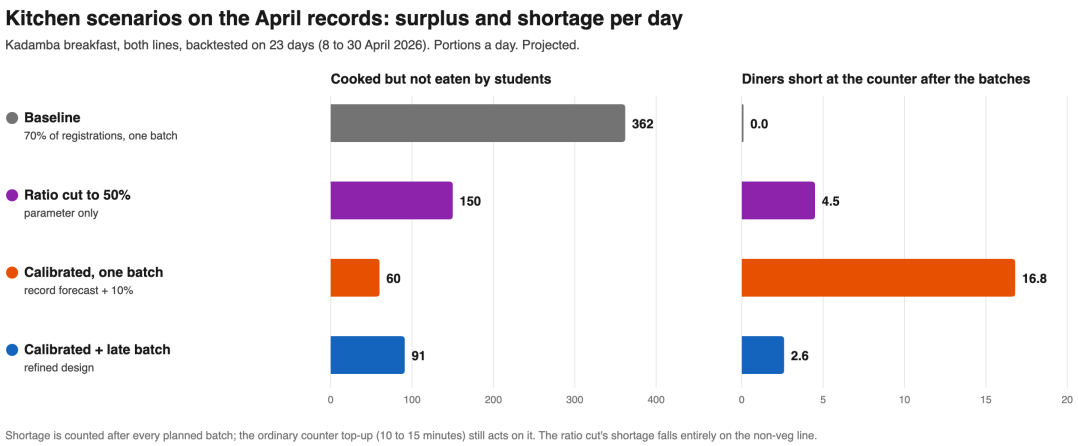

Figure 8.4. Kitchen rules backtested on 23 April days, portions a day, both lines. Source: Kadamba scan export, April 2026; model output (projected).

Projecting the day at 8:00 under-reads by 26.9% on average, since only 13.5 to 22.5% of diners are in by then, so the projection moved to 9:00. On three backtest Sundays the baseline cooked about 563 veg portions for about 190 eaten. The refined design would cook about 233, leaving about 45 surplus and about 2 short. On 19 April its batches track the diners' curve through the closing crest, while the baseline sits far above (Figure 8.5).

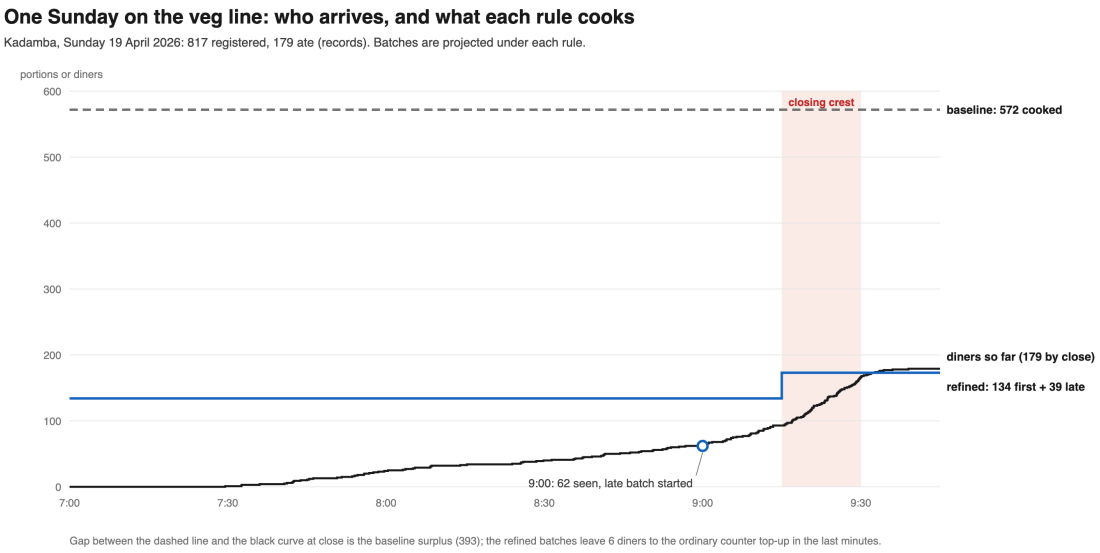

Figure 8.5. Kadamba veg line, Sunday 19 April 2026: recorded diners against the baseline and refined batches. Source: Kadamba scan export, April 2026 (confirmed); batches are model output (projected).

Under assumed shocks the design fails safe toward fewer eaters and loud toward more: from the lightest to the heaviest crest, line-days short (of 46) run 2, 7, 13 and 15. The late batch is load-bearing. If it fails, about 44 diners are short and 33 wait, the longest bars in Figure 8.6, so the raw reserve is mandatory and two failed Sundays running should stop the pilot.

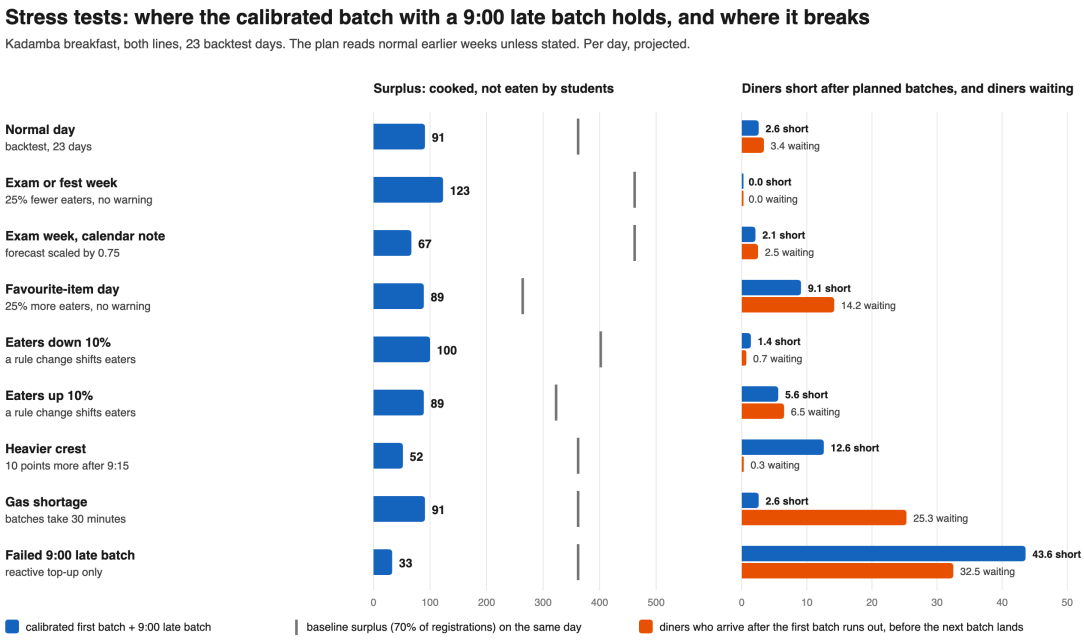

Figure 8.6. The refined kitchen design under assumed shocks on the 23 backtest days, beside the baseline surplus. Source: Kadamba scan export, April 2026; model output (projected).

Kitchen rules leave students' bills untouched. In April Kadamba students were charged for 20,221 breakfasts they did not eat (confirmed), Rs 10.58 lakh (projected). A Sunday skip at 25 to 50% uptake cuts that by Rs 0.40 to 0.80 lakh and an every-day skip by Rs 2.65 to 5.29 lakh (projected). A skip that returns nothing changes nothing, which is why Skip Meal fails today.

Simulated stakeholder responses

We simulated how twelve stakeholder groups would respond to A, B with C, D and H; no response comes from the actors. Opposition sits in three cells (Figure 8.7): Prism opposes D if paid on registered plates, the Bakul operator resists any run-

out risk, and outside diners and staff are harmed only if the allowance is unnamed. The calibrated batch saves Prism food cost on either plate basis, so the simulation argues against holding the Sunday proposal behind the kitchen.

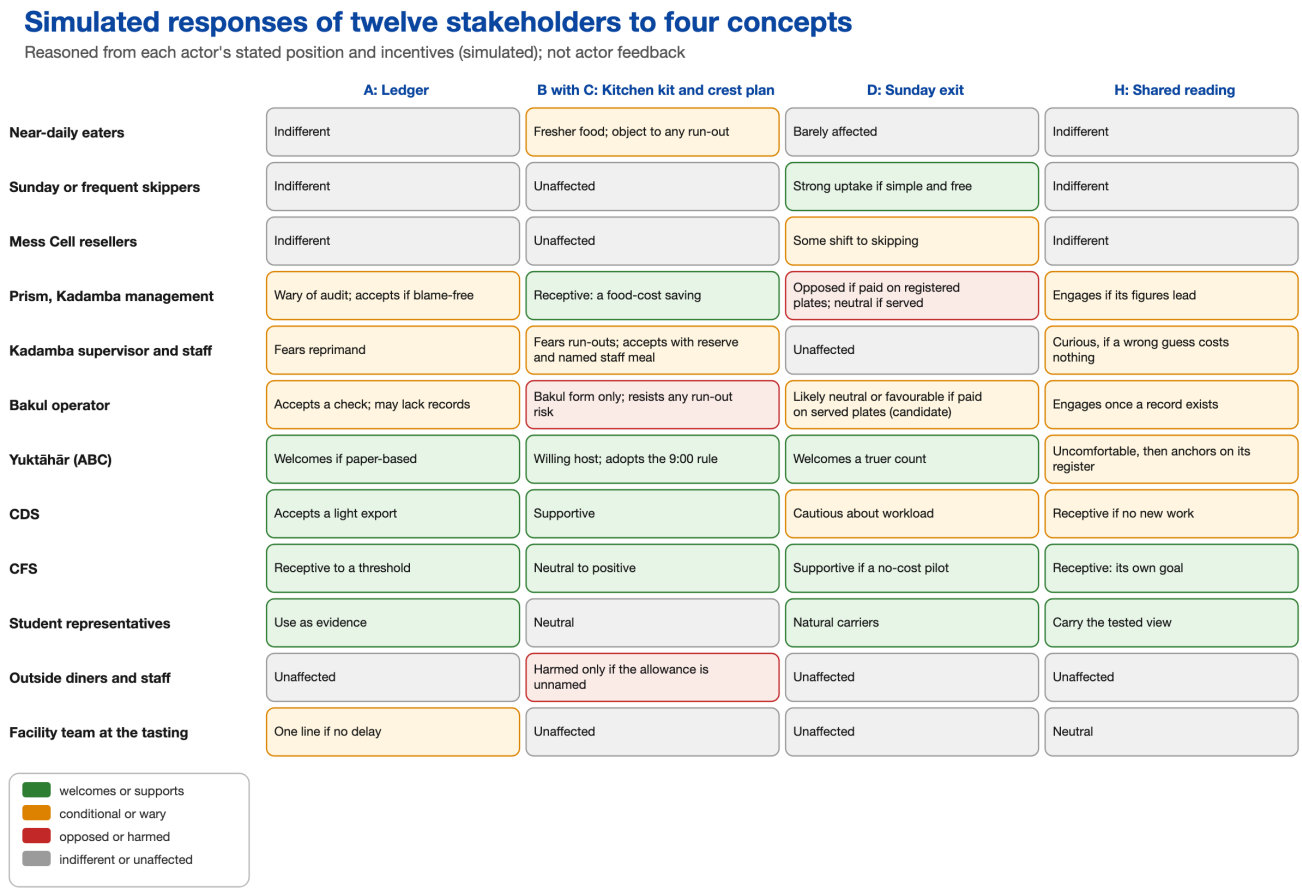

Figure 8.7. Simulated responses of twelve stakeholders to four concepts. Source: authors' reasoning from each actor's stated position (simulated; no actor feedback).

Gates and weighted scores

Four gates come before any score, and failing one makes a concept ineligible:

- G1 Absorbers kept: resale, reuse, outside-diner and staff meals, walk-ins and late service survive.
- G2 Safety kept: the food-safety sequence stays intact.
- G3 Aimed at structure: it neither asks students to behave better nor blames operators, and waits on no unknown that could reverse it.
- G4 Quick fix paired with a fundamental fix that lands on a date independent of kitchen performance.

Passing concepts are scored 0 to 4 on nine criteria weighted out of 400. Booked against eaten (O1), cooked against eaten (O2), crest service (O3) and fundamental capability (O4) weigh 15 each; fairness (O5), food quality (O6) and desirability (O9) 10; feasibility (O7) and reversibility (O8) 5. In revision food quality rose from 5 to 10 and desirability entered at 10.

Concept	Gates	Score	Main reason
A Breakfast Ledger	Pass	68% (first draft 74%)	Builds the route every lever needs; may read as audit
H Shared Reading	Pass	65% (new)	Acts on beliefs and goal; discomfort caps desirability
F Billed at consumption	Fails G3 (held)	63%	Operator effect depends on plate basis
C Crest Plan alone	Fails G4 alone	63%	Protects the crest; without B trims only slightly
D Sunday skip, ratio kept	Pass	60%	Rs 0.40 to 0.80 lakh a month; needs the rule owner
E Operator Menu Slot	Pass	60%	Cheap, safe, acts on L16 only
B Kitchen Learning Kit alone	Fails G4 alone	58% (first draft 65%)	Surplus 362 to 60, but 16.8 short a day

Concept	Gates	Score	Main reason
D every-day skip, ratio kept	Pass	50%	Non-veg short on 7 of 30 days; later step
D Sunday opt-in, ratio kept	Pass	50%	Crest 23 to 65 short each Sunday
G Ratio cut to 50%	Fails G1 and P4	48% (first draft 58%)	150 surplus portions remain; no allowance
First-draft bundle	Fails G4	79%	Sunday skip waited for the kitchen to prove itself
Re-sequenced bundle	Pass	79%	Same parts; skip and route on fixed dates

Read the gate column before the score bar: F and C outscore D and E, yet fail a gate (Figure 8.8). The two bundles share parts and score, because a weighted score cannot see sequencing. The first draft made the fundamental fix wait on the quick fix, a small case of shifting the burden, so only the re-sequenced bundle passes G4.

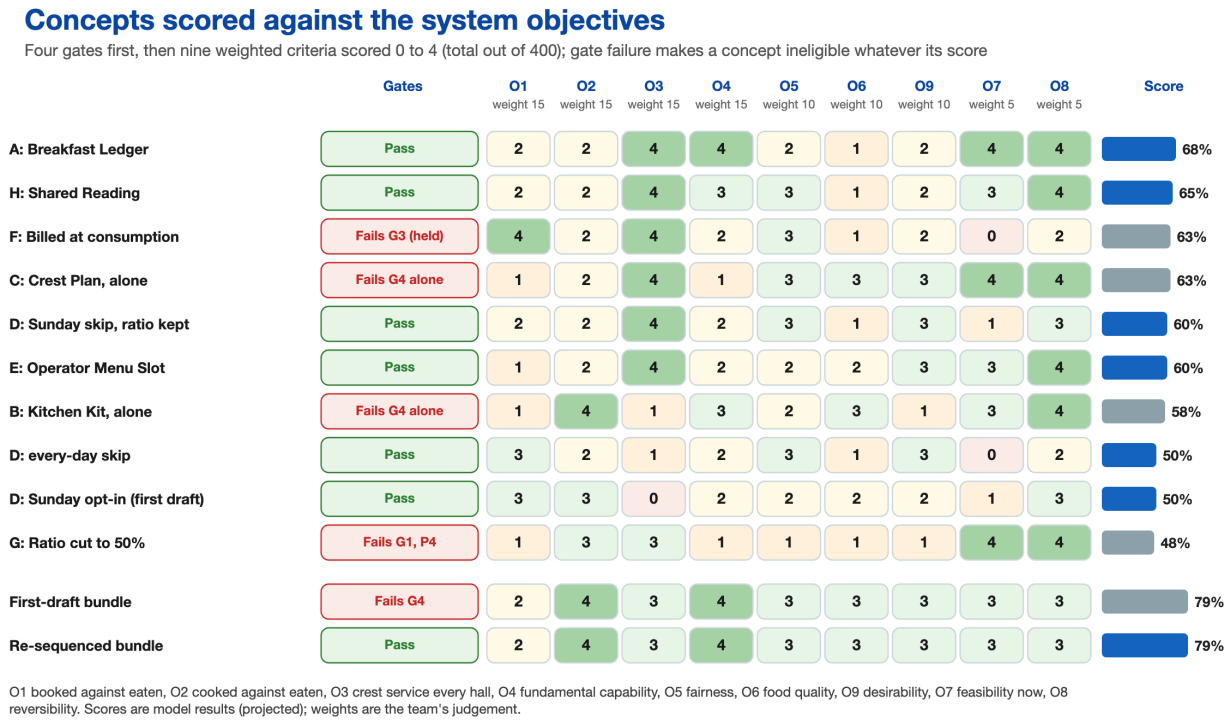

Figure 8.8. Concepts scored against the gates and nine weighted criteria, as percentages of 400. Source: authors' analysis; scores are model results (projected).

Unintended consequences

Several second-order effects are the loops pushing back. The likeliest is the hedge ratchet, a candidate loop (Figure 8.9): a run-out is loud and a surplus silent, so one complaint after a cut can return the batch to its old size. Judging results over months, recording every run-out, and keeping the late batch and raw reserve hold it down. A named allowance protects outside diners and staff, and a calendar note covers exams.

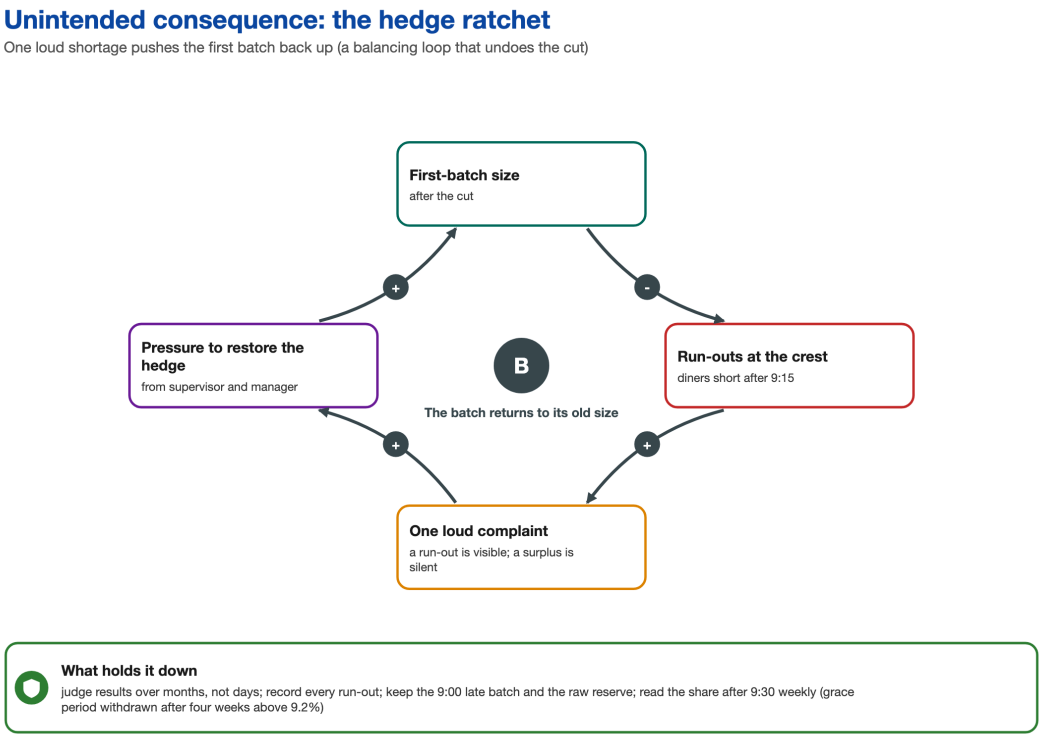

Figure 8.9. The hedge ratchet: one loud shortage restores the cut batch, and what holds it down. Source: authors' analysis; a candidate loop.

Refining the intervention

Of twelve refinements, sequencing changed the recommendation. The first-draft bundle proposed the Sunday skip only after the kitchen proved itself. The student-side scenarios showed the crest stays safe below about two-thirds uptake with the ratio kept, so the wait was needless, and it risked a better kitchen making the booking look unimportant. The skip was fixed at week 6.

Other changes followed the numbers. The projection moved to 9:00, where its error is 13% against 35% flat at 8:00, and a 1.5 cover cut diners left waiting from about 18 a day to 3.

The final recommendation

Route and read the gap together, open the Sunday exit on a fixed date, and calibrate the kitchen to buy time, labelled the quick fix wherever reported. The fundamental fix keeps its dates whatever the kitchen pilot shows (Figure 8.10, planned).

- Stage 1 (weeks 0 to 4): ledger with its charge line, co-design tests, joint reading at week 4, goal and indicators taken to CFS and CDS.
- Stage 2 (weeks 5 to 8): Sunday skip proposed at week 6 with the plate-basis question; Kadamba Sunday veg pilot of card, allowance, 9:00 batch and sheet.
- Stage 3 (weeks 9 to 20): skip in force from about week 10 if adopted, every-day decision at week 14, handover from week 16, end at week 20.

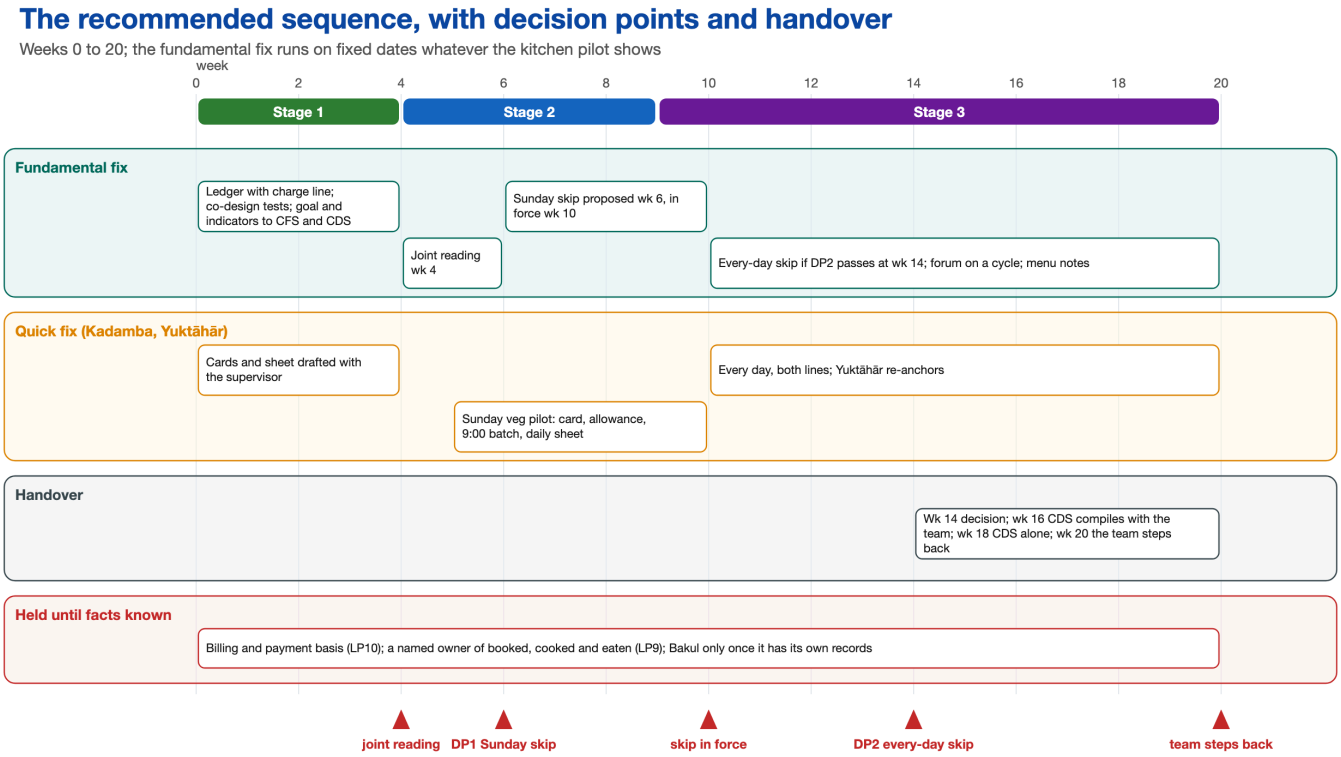

Figure 8.10. The recommended sequence, with decision points and handover. Source: authors' design (planned).

Bakul, a third regime with no records, joins only once it has scans, a measured top-up lead time, an item record, a kitchen lever that does not raise shortage risk and, for the Sunday exit, a confirmed payment basis.

The architecture path

The path moves in reversible steps toward a booking that matches intent. Thresholds are our judgement.

Decision point	When	Rule	If the evidence goes the other way
DP1 Sunday skip	Week 6, fixed; in force about week 10	None; goes on this date	If declined, the Sunday charge line goes to the escalation forum
DP2 Every-day skip	Week 14; from about week 16	Sunday uptake at least 25%; no Sunday over 10 short; eaten per remaining registration rising a few points; calibrated batch on both lines	Uptake under 10%: retest wording. Shortage: that line reads eaters first
DP3 Default matched to intent	After a term of DP2	Opt-in on lowest-turnout days; no automatic registration off campus; skip used by over half of no-shows; never costs a hall slot	Low uptake and stable charge: keep skip and default
DP4 Billing and payment	Once the plate basis is known	Billing at consumption, or payment on served plates; basis known operator by operator; resale reliance checked	If registered plates and no rate agreement, billing stays and the skip carries students

Theory of change and the iceberg

The bundle shifts the working goal (place 3) from the registration count to cooked against eaten, an aim already on the institution's own poster. Each root cause in Figure 8.11 has its own change and an expected response that is projected or candidate, never observed. The plan card and daily sheet would cut surplus from 254 to 471 portions a day to about 69 to 120, and the late batch would hold shortage to about 0.5 to 7.7 a day as the crest varies (projected).

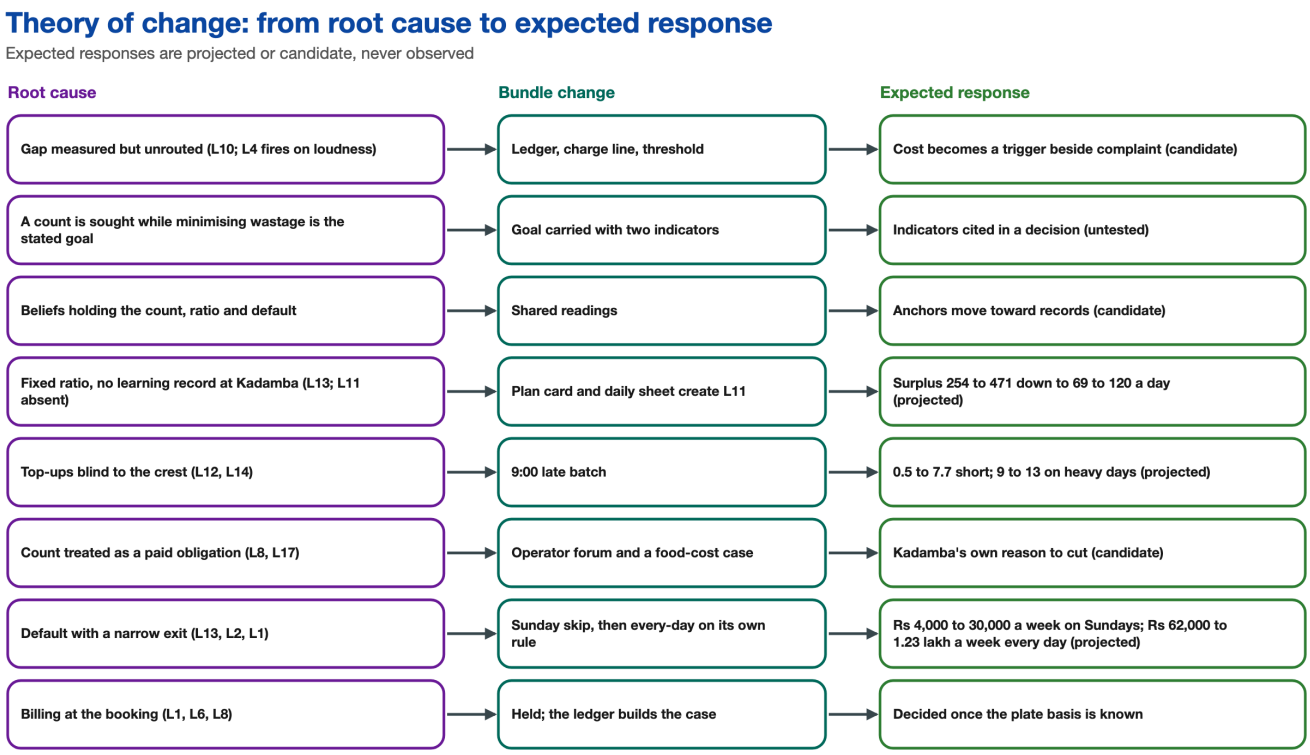

Figure 8.11. From root cause to bundle change to expected response. Source: authors' analysis; ranges are model output (projected) or reasoned (candidate).

Each level of the iceberg gets its own action: the daily sheet for events, the arrival card and weekly ledger lines for patterns, the ledger route, plan card, Sunday skip and stated goal for structures, and shared readings for mental models.

Pilot design and monitoring

The pilot runs on Kadamba's veg line at Sunday breakfast, judged at week 16, against Sunday non-veg, Saturday veg and earlier Sundays (planned). The Kadamba supervisor runs it; Prism's operations manager, ABC Hospitality Services, the facility team's parent body and the rule owner must agree.

Element	Specification (planned)
Result measures	Charges per week; first-batch surplus; run-outs and diners short after the late batch
Result measures (continued)	Grams, taste and texture at the tasting; holding time; share from a fresh or late batch; allowance meals served
Success	Sunday veg surplus from about 370 to under 100 (model about 45); no Sunday over 10 short; run-outs no more frequent than on comparison days
Success (continued)	Every allowance meal served; grams and taste held; every safety step recorded; no sheet-based reprimand; charge line reported weekly
Stop	Shortage beyond one top-up two Sundays running; late batch failing twice; any allowance meal missed; grams below standard; any figure used against staff; consent withdrawn

CDS compiles the ledger with us from week 16 and alone from week 18, so the team does not become what the system depends on. After handover each waste indicator is read beside the run-out indicator for the same hall and line, and an effective ratio drifting toward 70% for four weeks signals a fix that fails.

Who must act

Of thirteen actions, we can complete two alone (the ledger, cards and sheet, and the co-design tests), convene or support four, carry six as evidenced proposals, and must complete one, the handover by week 20. Prism's supervisor decides on the batches, and only the registration rule owner (Mess Office or CDS, unconfirmed) can open the Sunday exit.

Before and after

First-batch surplus falls from about 254 to 471 portions a day to about 69 to 120 over roughly four months. The weekly uneaten charge falls from about Rs 2.47 lakh to Rs 2.27 to 2.37 lakh with the Sunday skip, or Rs 1.23 to 1.85 lakh if the every-day skip follows (all projected). In Figure 8.12 surplus starts to fall with the Sunday pilot, while the charge moves only when the skip comes into force at week 10.

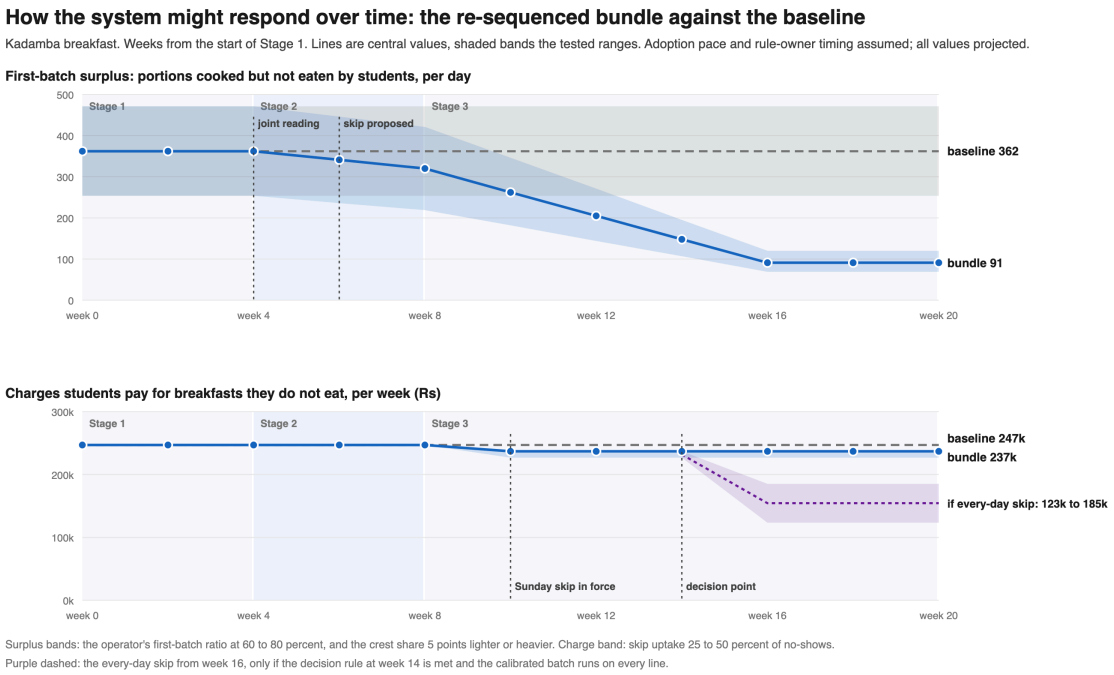

Figure 8.12. How Kadamba breakfast might respond over time under the bundle, against the baseline. Source: model output (projected); adoption pace and timing are assumed.

Outstanding evidence

Still open: what a plate counts for in each operator's monthly payment, the base of Kadamba's 5 to 10% waste figure, the real first batch per line and weekday, Bakul's scans and top-up lead time, and whether Skip Meal lowers the kitchen's count. No co-design test has run.

Decisions and revisions

- We kept G, a blunt cut we expected to fail, in every test, because testing only believed ideas risks every test passing.
- We let four gates overrule the weighted score, because a score cannot see sequencing: both bundles scored 79%, and only the re-sequenced one passes G4.
- We fixed the Sunday skip at week 6, independent of the kitchen pilot, because waiting would let a better kitchen make the booking look unimportant.
- We replaced the Sunday opt-in with an evening-before skip, because with the ratio kept the opt-in ran the crest 23 to 65 short and charged the forgetful Rs 81.
- We held F behind the plate basis, sized nothing for Bakul, and planned our own exit at week 20, because a team that stays becomes the thing the system depends on.

What it adds up to

The ten activities answered the nine research objectives in turn. Together they say that the gap comes from a default no one chose to keep, and that it is measured every day without reaching the people who set the rules.

Findings by objective

Objective	What we found
O1 Boundary and context	Registration is an automatic default with a narrow exit. About two thirds of breakfast registrations go unused, while lunch under the same rules mostly works.
O2 Participants	Twenty-four stakeholders, with the three operator firms at the centre. The body that controls the timetable has no channel to the people who run the mess.
O3 Power and leverage	Authority over the rules is split among offices that do not coordinate. Operators control quantity, students bear the cost, and the payment basis is unresolved.
O4 Process	No forecast sets the quantity. Each operator applies its own rule to the closed count, and the first batch can be topped up but never reduced.
O5 Behaviour over time	At Kadamba 62.9% of April registrations went unused, with a crest in the last quarter hour. Yuktāhār's breakfast turnout averaged about 28%, lunch 56 to 89%.
O6 Model and loops	Of eighteen suspected mechanisms, three close with every link evidenced, all inside a kitchen. No loop touches the registration.
O7 Root causes and reframe	Three structures (the default, billing at the registration, an unrouted gap) and two conditions hold the gap in place. The system is stable, not broken.
O8 Leverage and principles	Twelve leverage points, fourteen intervention points, seven principles. Beliefs and goals are beyond a student team; on Kadamba's vegetarian Sunday line three levers act together.
O9 Design and evaluation	Eight concepts were tested on paper. The recommendation builds the missing routes, sets a Sunday skip on a fixed date and tunes the kitchen as the quick fix.

How our beliefs changed

The first submissions of Phases 1 and 2 were written before most field evidence existed, and we changed our minds several times. Each correction was made in every deliverable in the same pass.

We first believed	What we found	What we did
About 30% of registrations go unused (two students' impressions)	62.9% at Kadamba, about 72% at Yuktāhār; 30% fits lunch	Replaced it everywhere and trusted records over recall
The vendors are one passive actor cooking to a forecast	Three operators, each with its own ratio: fixed at Kadamba, by item at Bakul, from books at Yuktāhār	Traced each caterer as its own system, from registration to bin
A forecast locked four days ahead is the bottleneck	The four-day close applies to registration; kitchens set quantities the day before	Named the cooked first batch, which can be topped up and never reduced, as the binding constraint
Composting absorbs the surplus	Kadamba has no composting; surplus goes to outside diners, staff, reheating and the bin	Removed composting and added outside diners and staff to the absorbers
Bakul cooks near the airport and raised its ratio from 70 to 80%	Its kitchen is at Hafizpet, about 8 km away; the share varies by item (single-sourced)	Withdrew the raise everywhere; Bakul became a third regime
Standing facts, such as the four-day close, are iceberg events	An event is one concrete occurrence (found on 13 September)	Rebuilt the reading from primary sources; of five loops re-tested, three became chains, one was disconfirmed, one was absent
Six loops were closed	Three close with every link evidenced (L4, L11, L12), all inside a kitchen	Adopted one scheme, L1 to L17 plus L3b, and a strict closure rule
Cuisine explains low uptake at the north Indian halls	It rested on one portal snapshot. Kadamba's vegetarian registrants attend at 32.4%, non-vegetarian at 48.5	Kept cuisine as one factor; the line and the item matter more
Friends pull each other to breakfast	Six respondents disconfirmed it	Dropped it as a driver of attendance

We first believed	What we found	What we did
The Sunday skip can wait until the kitchen proves itself	An improving kitchen can make the registration look unimportant, so the exit would never open (shifting the burden)	Made kitchen levers the quick fix; route, default, goal and belief levers run on fixed dates

The decisions that shaped the work

Each of these was a choice between options with costs.

Decision	What we set aside	Why
Treat the course's opening story as claims to test, link by link	Its single cause, the clock	Only its first link, late nights leading to skipped mornings, had testimony
Study breakfast only, with lunch and dinner as the control	Studying the whole mess	Lunch shares the portal, rules and caterer, so it tests the meal against the system
Label the confidence of every fact, and trust records over recall	Hedged wording	A reader needs each finding and how sure we are of it
Judge power one decision at a time	One power score per actor	An actor can hold strong authority over one decision and none over the next
Explain the gap by three root causes and two conditions	A single root cause	A root must plausibly stop the pattern if removed; the four-day lock failed that test
Never remove an absorber to create a signal	Making waste visible by cutting resale, reuse or outside diners' meals	Absorbers help real people, and the institution can measure the gap directly
Rank levers by depth, then feasibility; hold those that depend on the payment basis	Picking by depth or ease alone	Beliefs and goals are beyond a student team, and the plate basis decides which incentive points the right way
Keep a foil, the Ratio Cut, in every test, and apply gates before scores	Testing only promising concepts	Otherwise every test passes, and a weighted score cannot see that a concept removes an absorber
Open the Sunday skip on a fixed date, week 6, independent of the kitchen pilot (gate G4)	Letting the kitchen pilot set the date	A fixed date keeps both fixes moving
Freeze the evidence at a stated date	Waiting for more records	Data kept arriving in batches, and the open questions need contracts or records we cannot obtain

What it means for the institution

The gap is about equally large at Kadamba and Yuktāhār, which share the rules and differ in kitchen practice. Structure produces behaviour (Meadows, 2008): the registration explains the gap, and kitchen culture decides only how much of it becomes surplus. Every working loop sits inside a kitchen, so the system corrects locally and has no correction at the centre, and a quick fix in the kitchen can make the central problem harder to see, the shifting-the-burden risk (Senge, 1994).

The recommendation asks little of students and a good deal of the institution. The records that would calibrate every kitchen are already kept, so the first step costs a weekly sheet and a meeting. The second, a Sunday skip that removes the charge, is a decision for the rule owners, who today hold only fragments of the outcome.

No office is responsible for the whole chain from registration to bin. We hold back proposing a named owner until the payment basis is known, since it decides who has a reason to close the gap. The institution's posters give its aim as minimising food wastage through advance meal planning, and that planning happens at the registration, which is not where the quantity of food is decided.

What we would do differently

For several weeks we reasoned from the student side and the published rules, and the first honest look inside a kitchen changed more than any analysis had. With hindsight we would go to the kitchens first and ask for the records sooner, since they existed all along. We would also state how sure we were from the very first document, the habit that later let findings change cleanly when evidence arrived.

We also learned that measuring and routing are different things. The system measures its problem daily and still cannot see it, because the measurement is not delivered.

Limits and open questions

Limits of the evidence and the method

The evidence has five main limits.

- Six students were interviewed, one transcript had a labelling gap, and six people are not a population.
- Each kitchen was visited once, and the operators describe their own practice.
- Two sets of records cover two halls in two different months, and Bakul has none.
- The scenario model uses only Kadamba's scans, and its stress tests apply assumed shocks.
- Stakeholder responses are simulated from stated positions, no feedback has been collected, and every co-design test is planned, none run.

The method adds three more.

- The loop test is strict by design, so some real mechanisms stay candidates for want of one link. The count of three closed loops describes the evidence, not the system.
- The scoring weights are our judgement. Two bundles tied because a weighted score cannot see order, which is why the gates overrule the score.
- The three halls differ in five ways besides the planning rule. The comparison shows that a learning practice is possible under today's rules, without showing why it exists at only one hall.

Ethics

The study used interviews with students, staff and operators, photographs of kitchen records and an export of administrative data. People are named by role only, the data are used as aggregates, and no photograph shows a student identifier or face. No recorded figure may be used against the person who recorded it, a rule we carried into the design as a principle.

The intervention keeps every arrangement that feeds or protects someone: resale, reuse, the meals of outside diners and staff, and late service. The pilot's stop conditions include the withdrawal of consent and the use of any figure against a person. Our plan also includes our own withdrawal, so that the system does not come to depend on a student project.

The interviewers were students of the institution and the respondents were their peers and hall staff, a relationship that can shape what is said. We therefore read operator statements about their own practice as single-sourced until a record agreed.

Open questions

The plate basis matters most. It is one of three conflicts between sources that we deliberately left open, with whether Skip Meal reaches Kadamba's kitchen and who heads the dining services body.

Question	Who could answer it
What a plate counts for in each operator's monthly payment, registered or served; no contract has been seen	The institution's contracts or the dining services body
Whether Skip Meal lowers the count Kadamba cooks from (the institution says it does; the operator says it cooks on registrations alone)	The dining services body
How many students would take a Sunday skip (assumed 25 to 50% of no-shows, tested from 10 to 75)	The students
Whether a late batch cooked at 9:00 can land by about 9:15 without delaying service at the crest	The operators
What Kadamba's stated 5 to 10% waste figure is a percentage of	The operators
Who set the four-day close and the limit of five cancellations	The institution and its contracts
Who heads the dining services body (two different people are on record)	The dining services body
Which registration rule applied to Yuktāhār's June breakfasts	The institution and its contracts
Whether the April pattern holds in other months and at the halls without records	Scan exports not yet obtained

Conclusion

Conclusion and future work

The project began from the claim that students skip breakfast because early classes collide with late nights. It ended with a different account: registration is a default that no one chose to keep, the charge falls on the registration whether or not the meal is eaten, and the records that could correct each kitchen's guess never reach those who set the rules.

About 63% of Kadamba's breakfast registrations in April 2026 went unused, and about 72% at Yuktāhār in September, while lunch under the same rules worked. The paradox describes a stable system. Every actor behaves reasonably, and each consequence is absorbed before it reaches the people who could change the rules.

The response runs three parts together: routes built through a weekly ledger and joint readings, a Sunday skip on a fixed date, and kitchen tuning as the quick fix. A pilot on Kadamba's vegetarian Sunday line is designed with success and stop criteria. The model projects a large change in the kitchen and a small one for students until the rule moves. All of this is projected, simulated or planned, and none of it has been tested on the campus.

The next steps are to obtain the missing records, including an item-level record for Bakul, and to settle each operator's payment basis, which gates the held levers. After that come the planned co-design tests with a supervisor and a ledger keeper, the pilot with its stop conditions, a test of real uptake of the Sunday skip, and an extension of the analysis to lunch and dinner.

Deliverables produced

Each deliverable was written as a stand-alone document.

Phase	Activity	Deliverable
1	1 System context	System Context Brief
1	2 Stakeholders	Stakeholder Map
1	3 Power, interest, leverage	Stakeholder Power-Interest and Leverage Map; Power Analysis Brief
1	4 Existing process	Current-State Process Map and Process Trace
1	5 Behaviour over time	System Timeline and Behaviour-Over-Time Map
2	6 System map and loops	System Map
2	6 System map and loops	Causal Loop and Feedback Analysis
2	7 Patterns and root causes	Systemic Problem Analysis
2	8 Reframe	Reframed Systemic Problem Statement
2	8 Reframe	Design Opportunity Statement
3	9 Principles and intervention points	Leverage-Point and System Intervention Map
3	9 Principles and intervention points	Design Principles
4	10 Design, test, evaluate	Intervention Concepts and Prototypes
4	10 Design, test, evaluate	Scenario Model and Impact Evaluation
4	10 Design, test, evaluate	Final System Design Recommendation

> The full project document covers every activity, source and decision in detail. The project poster is a one-page summary.